Manual v.1.00.50813

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This is the first draft of a quite extensive manual to ICP-Base. I tried my best to explain everything so that it also makes (at least a little bit) of sense to users that are not deeply familiar with LA-ICP-MS. If you don't want to read the whole manual, every module chapter has a "Quick guide" at the end in which everything is explained in bullet points.

If you still have questions, are in dire need of a custom feature, found a bug, want to collaborate or have a LA-ICP-MS system for which the import is not yet implemented – **feel free to write me an email**! If you found a bug, please try to reproduce it and include that in the email, it will make the fix much easier. If you want your LA-ICP-MS data to be importable or you cannot import yours even though it should work, make sure to include those files in the email as well.

A good software lives of its users! If you don't tell me about a bug, I cannot fix it. If you don't tell me about a missing feature, I will not implement it (and so on, you get the idea).

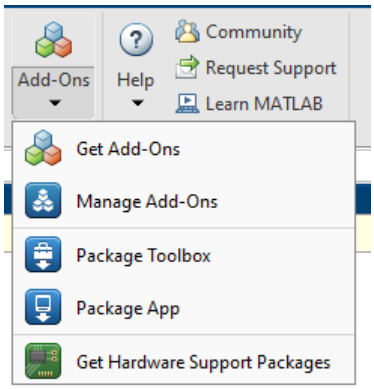
Enjoy and have fun!

1. How to get it running

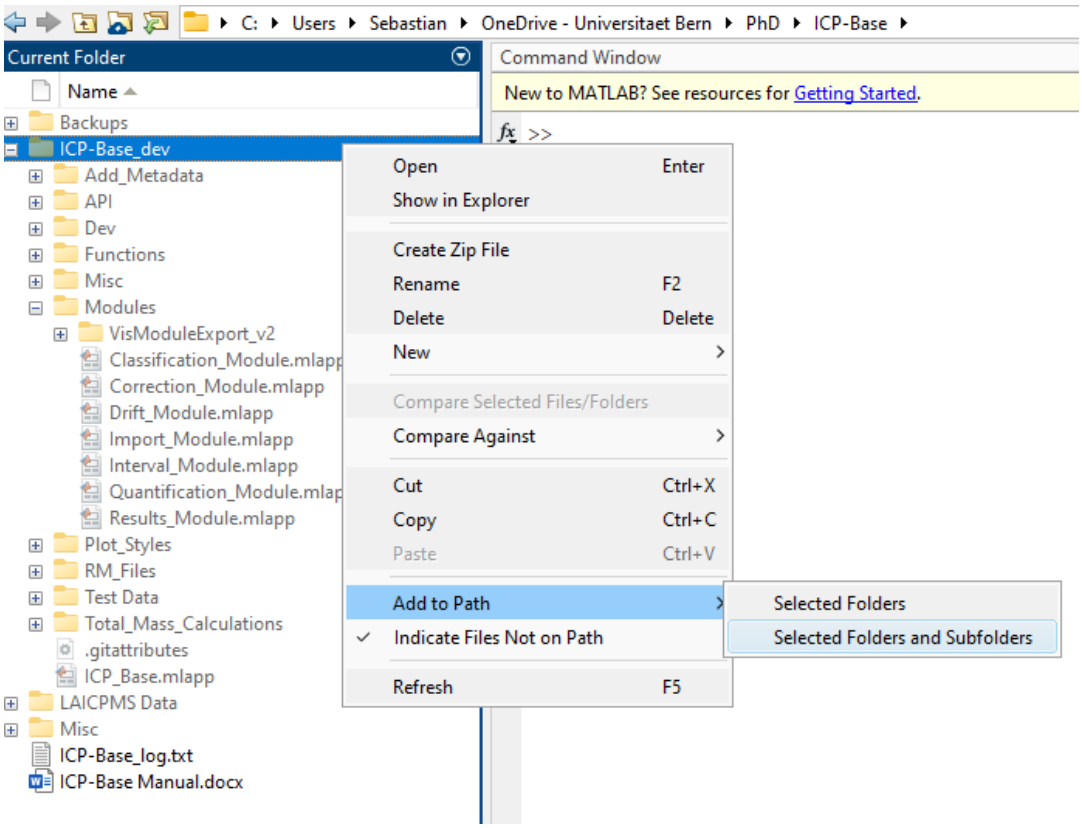
1.1. Uncompiled

If you choose to run it uncompiled, make sure to have MATLAB and the MATLAB App Designer with a valid license. When installing MATLAB make sure to add the Add-Ons “Image Processing Toolbox” and “Statistics and Machine Learning Toolbox”.

If your MATLAB is already set up but those Add-Ons are not installed you can get them using the “Add-Ons” Button in the “Home” Tab.



Once your MATLAB is running, navigate the path to the folder in which ICP-Base is stored. Then right-click the folder and selected “Add to Path” followed by “Select Folders and Subfolders”. The folders are then not greyed out anymore.



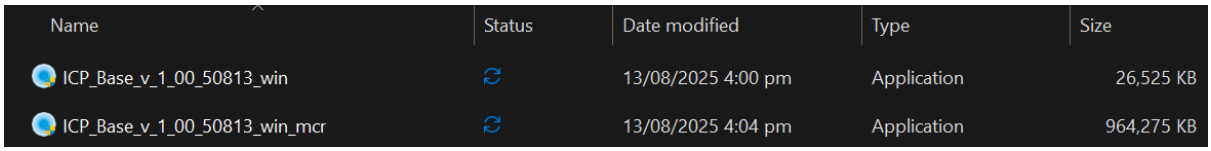
Next, either right-click on ICP_Base.mlapp and press “Run” to open it directly or double-click on it to open ICP-Base in the App Designer. If it is opened in the App Designer, go to the “Editor” or “Designer” Tab and then press the “Run” button.

1.2. Compiled

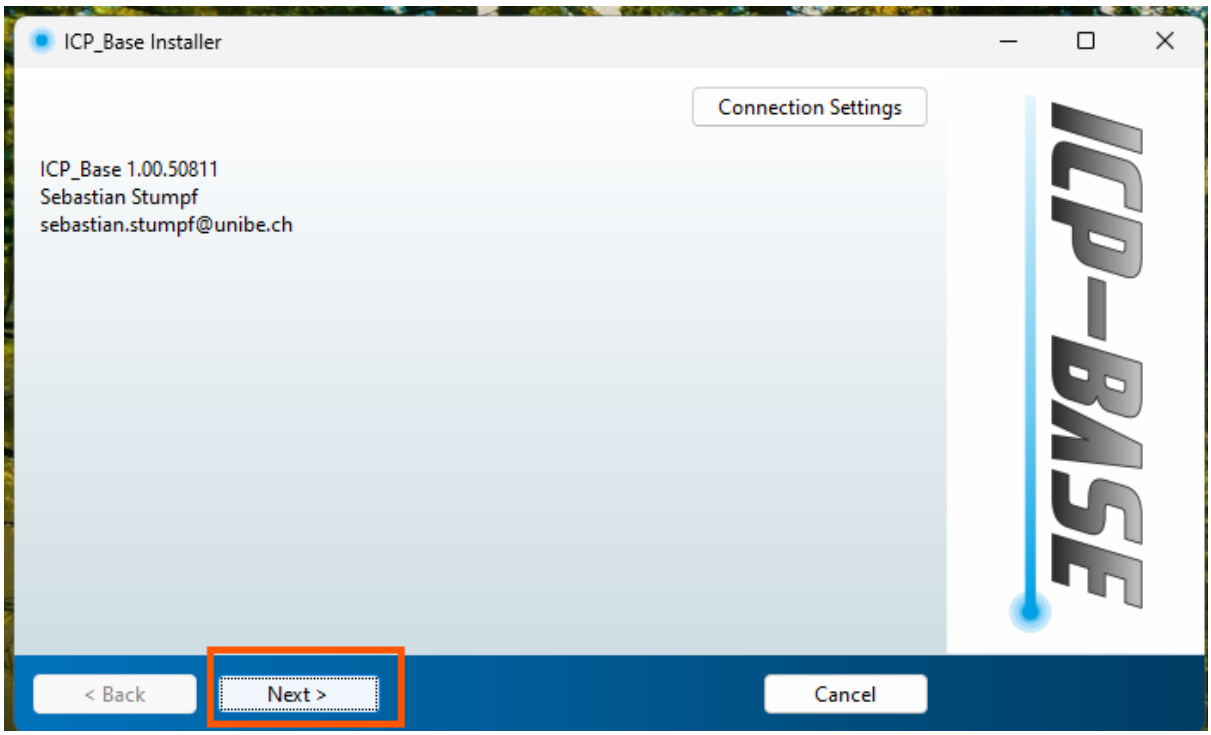
If the filename is ending NOT in “_mcr” you need internet do download the MATLAB Runtime library. So you need internet access to run ICP-Base.

If the filename is ending in “_mcr” you do the Runtime Library is already included and you can install ICP-Base on any computer, even without internet access. This file is therefore also larger in memory.

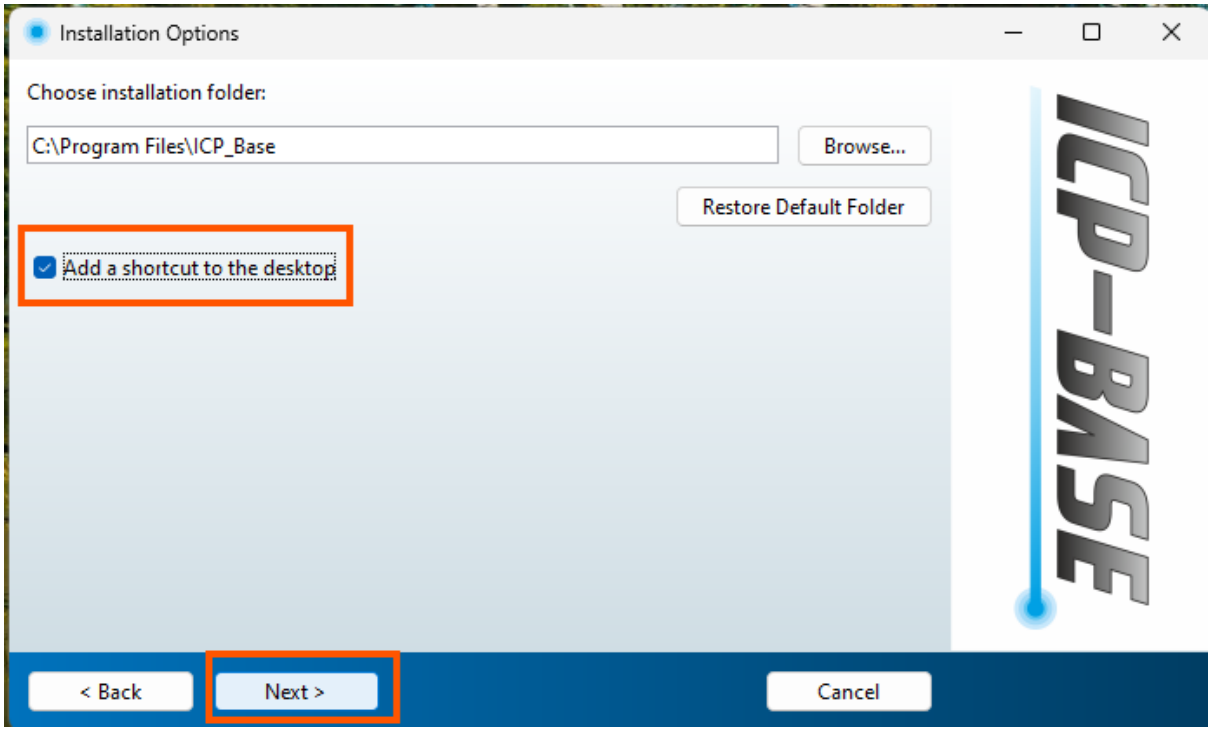
Simply double-click the .exe file and the installation process starts.



Name	Status	Date modified	Type	Size
ICP_Base_v_1_00_50813_win		13/08/2025 4:00 pm	Application	26,525 KB
ICP_Base_v_1_00_50813_win_mcr		13/08/2025 4:04 pm	Application	964,275 KB

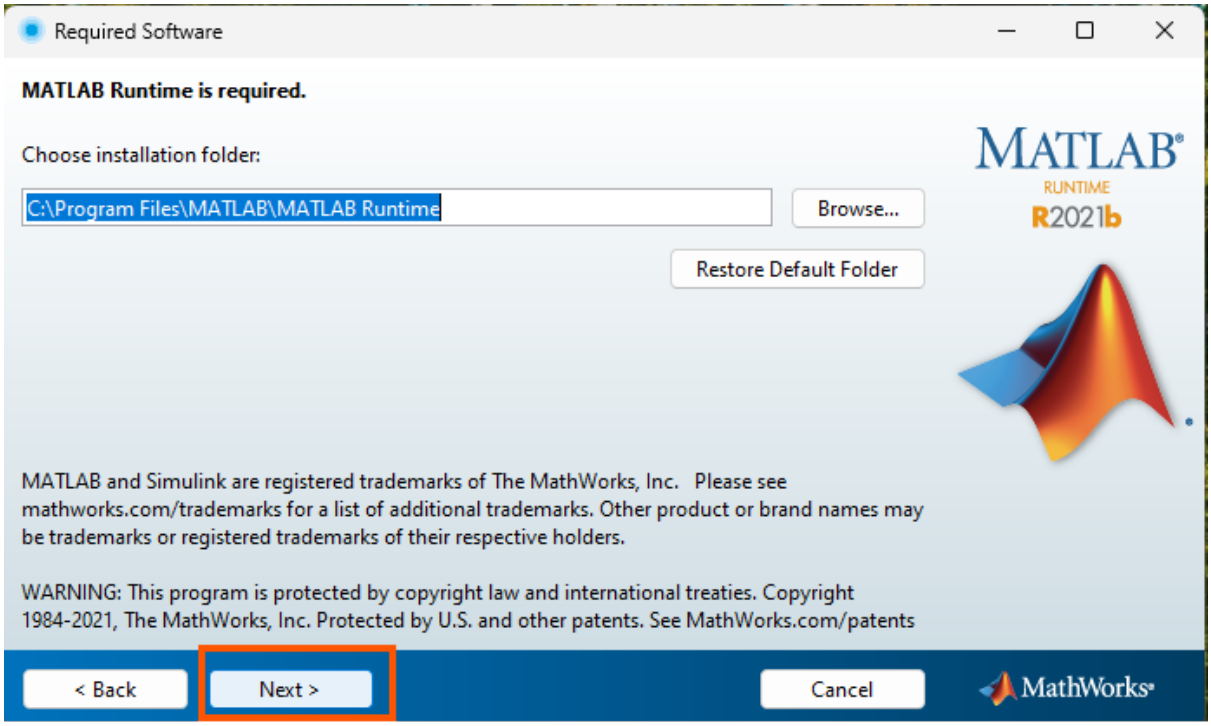


Click “Next >”.

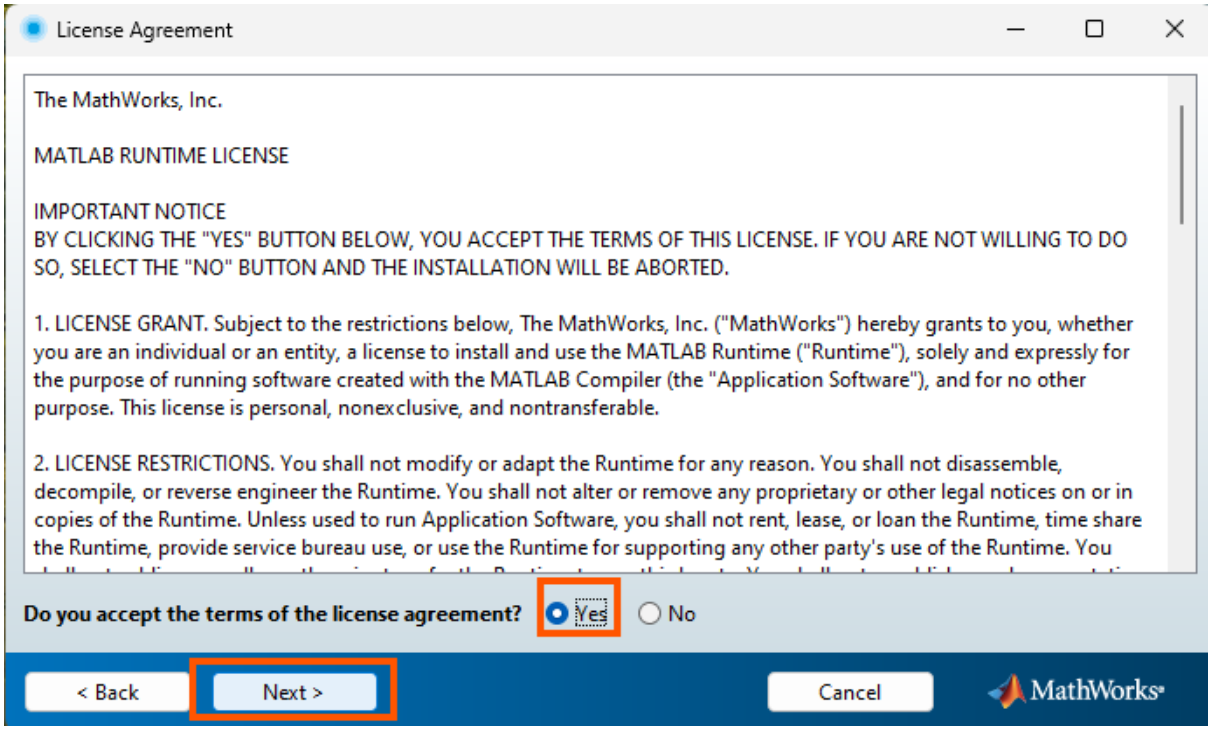


I would recommend to add a shortcut to the desktop. **Also make sure to remember where ICP-Base installation folder is located because you might want to access it later to customize your plots and reference material files!**

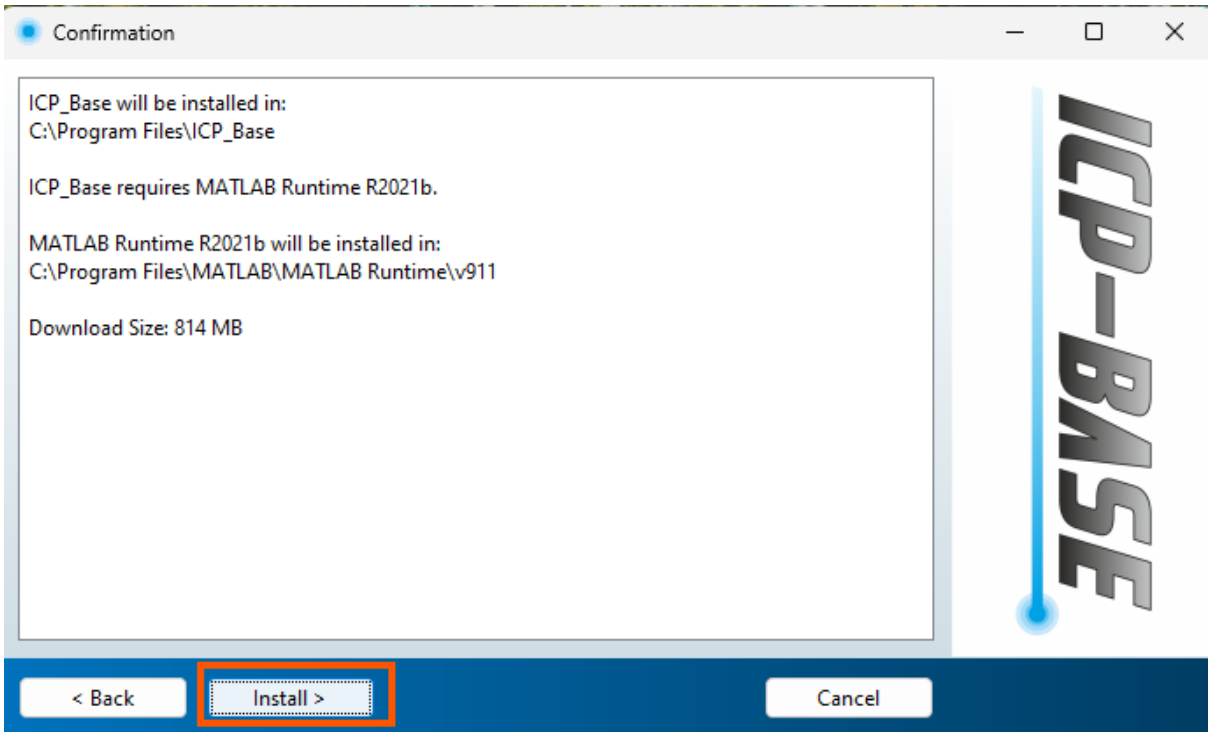
Then click “Next >”.



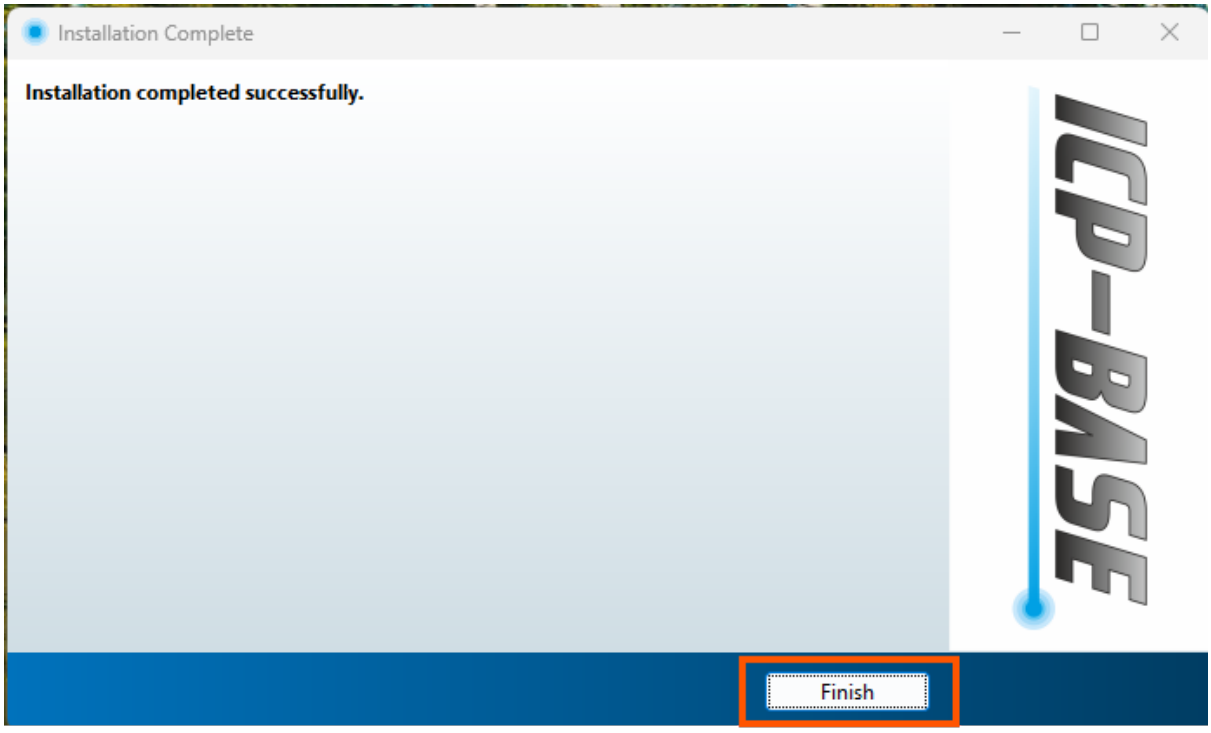
If you do not have the Runtime library already installed, MATLAB will do that for you. Just click “Next >”.



Agree and click “Next >”.

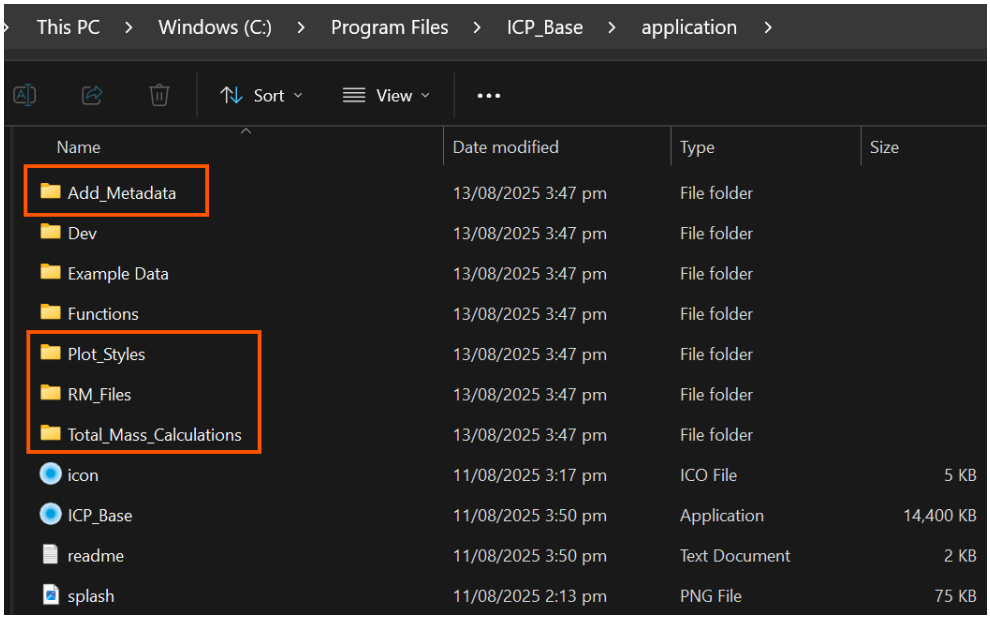


Click “Install”.



You did it! Click “Finish” and the icon should show up on your desktop.

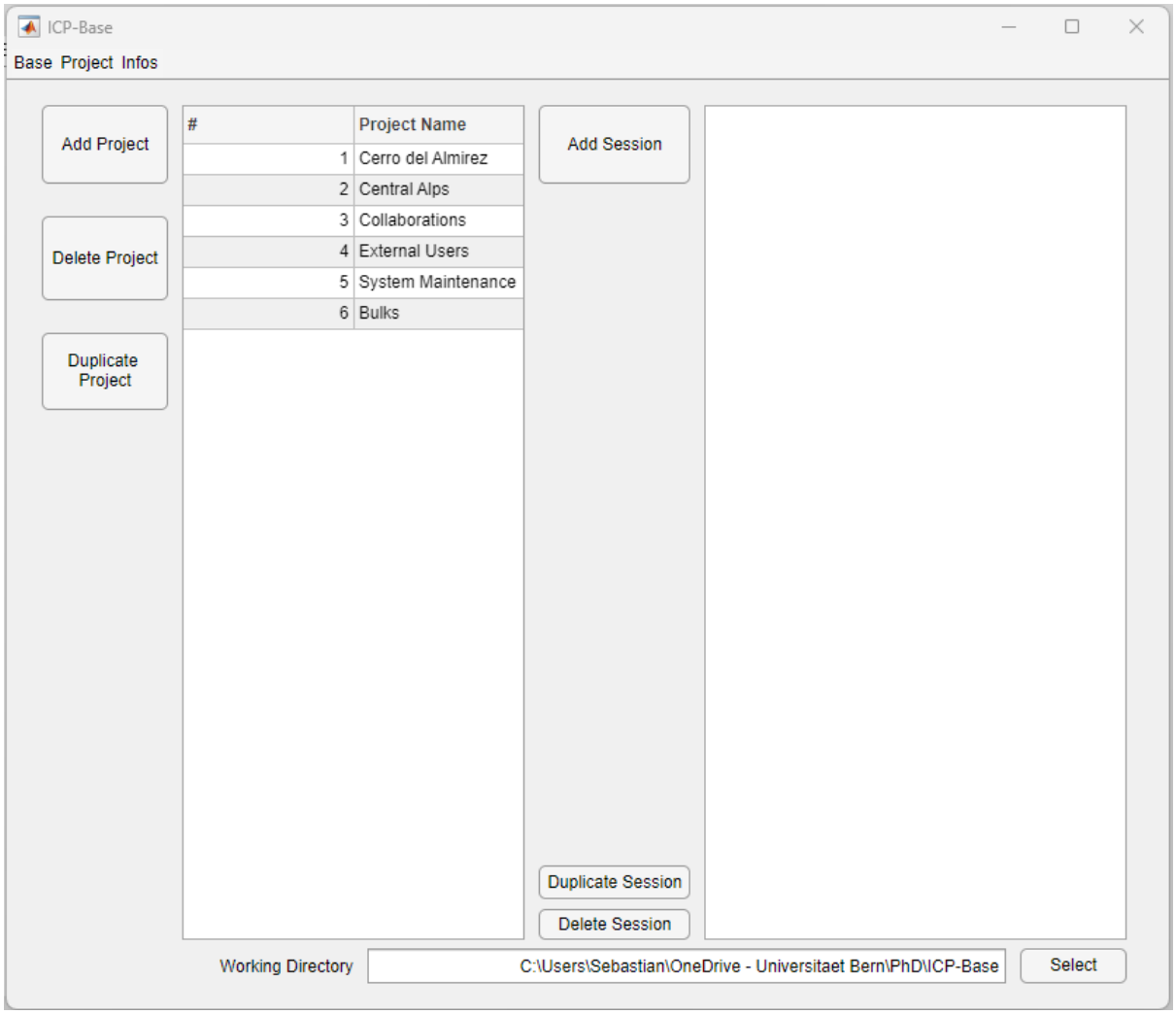
In the installation folder, you will find subfolders such as “RM_Files” and “Plot_Styles” which will become important later.



2. Base

The base, as the name suggests, is the basis of ICP-Base. Here, all your measurements, called “Sessions” are stored in “Projects”.

If this is your first time using it, just press the “Add Project” button and by double-clicking on the name, you can change it. The projects can also be deleted again and duplicated.



<u>By first selecting a project and then pressing the “Add Session" button, a measurement session will be added to the project and the data reduction starts.</u> There is no limit as to how many sessions can be added.

If you click the button and there is nothing happening for a few seconds – chill and wait, don’t stress it. Especially when starting the modules for the first time, MATLAB takes a while to initialize, but don’t worry, it will get quicker afterwards.

3. Import Module

Congrats, you successfully started the data reduction. This is a linear workflow where you will unlock one module after the other. Once you unlocked more of them, you can also jump between them using the colored buttons at the top.

There are multiple types of data that can be imported, not all of it is mandatory though. First, you need to figure out what types of data you have available and how the files look like.

There are three main types of data – Signal data, Laser metadata and ICP-MS metadata.

To import the correct data, you have to know which filetypes and datatypes you have. If you are not sure, check the example files that come with the software or ask the lab manager.

3.1. Signal data

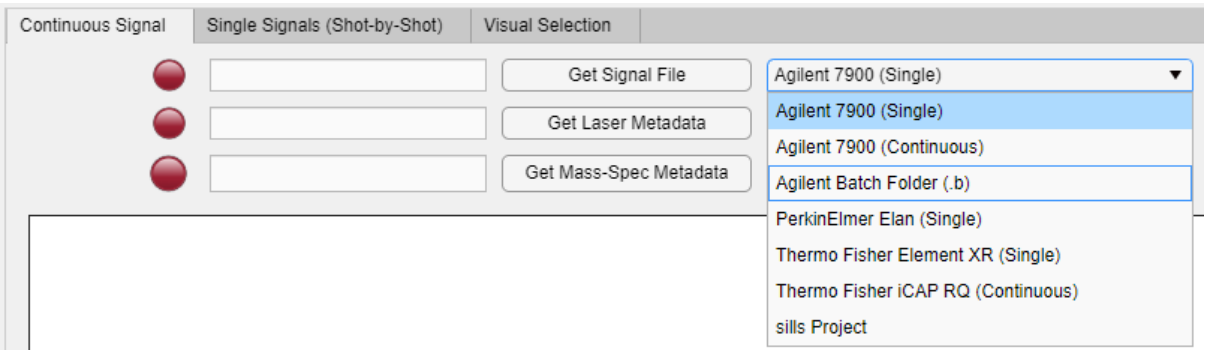
Signal data are the raw counts-per-second file(s) holding the data of the ICP-MS measurement. These files are specific to different mass spectrometer brands and also to some degree of how people like to measure ("Continuous" vs "Single").

First, check if you have one large file that holds multiple analyses/spots ("Continuous") or you have multiple files where every file holds one analysis/spot ("Single").

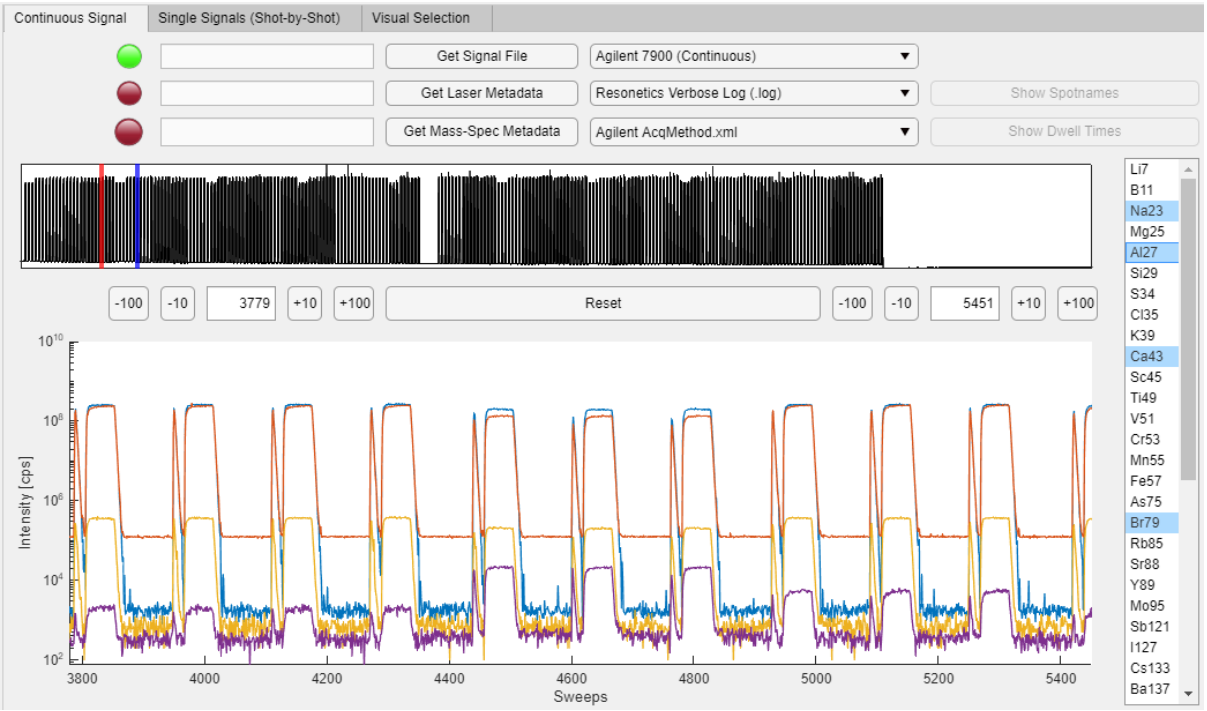
If you have measured with an Agilent make sure to also read chapter *3.1.3* before starting to import your data (but don't skip reading chapters 3.1.1 and 3.1.2).

3.1.1. Continuous

Navigate to the "Continuous Signal" Tab (default), choose the correct ICP-MS type and press the "Get Signal File" button. Then select the file that you want to import. For very large files, this can take a few seconds.



If the import was successful, the signal data will be visualized. By moving the red and blue lines, pressing the +/-10 and +/-100 buttons or by typing in a number in the fields it is possible to explore the data.



Now that the "Continuous" data was imported successfully, ICP-Base **requires** laser metadata to detect the signals. Continue reading chapter 3.2.

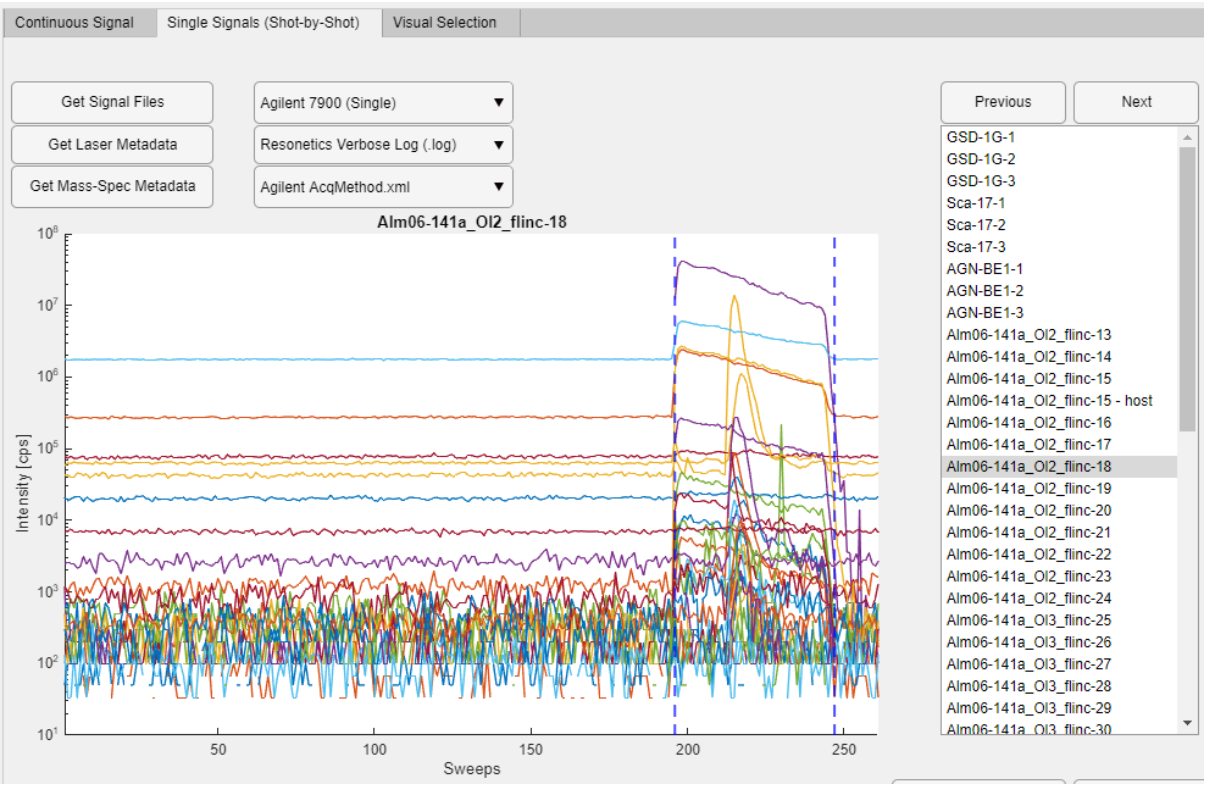
3.1.2. Single

Navigate to the "Single Signals (Shot-by-Shot" Tab, choose the correct ICP-MS type (same dropdown as for "Continuous", see above) and press the "Get Signal File" button. Then select the file(s) that you want to import. In ICP-Base, the signals will be ordered and visualized in the order that they lie in the folder. So if you have not sorted them correctly beforehand, they will also not be sorted magically in ICP-Base. For a high amount of files, this can take a few seconds.

If the import was successful, the signals are visualized in the plot, the blue dashed lines are the signal start and end that are automatically detected by ICP-Base. By default the spots are named based on the filename.

For Agilent and Element XR files, the acquisition time (necessary for the drift correction) is taken from the header of the file (it is written in there). For PerkinElmer .xl files, the acquisition time is taken from the file information. This means that the acquisition time corresponds to the time the file was saved on your computer. If you change the file and save it again, this time will change and will be wrong - so be careful.

For "Single" data it is not mandatory, but highly recommended, to import laser metadata (unlike for the "Continuous" data). If laser metadata is imported the filename changes to the name that the spot has in the laser metadata file. If you want to import laser metadata, continue with chapter 3.2, otherwise continue with chapter 3.3.



3.1.3. Agilent Batch Folder

If you have the Agilent batch folder (.b) you can select "Agilent Batch Folder (.b)" in the dropdown menu. Then after pressing "Get Signal File" you can select this folder and all the analyses (.d) will be imported. **It is not possible to select the .d folder directly! It has to be the batch folder (.b).** Make sure to not tinker with the folders beforehand.

If your data type is "Continuous", there will only be one .d folder within the batch. If this is the case, your data will be visualized in the "Continuous Signal" Tab. It is then required to also import laser metadata in order to detect the signals. See chapter 3.1.1.

If your data type is "Single" then there will be a lot of .d folders in the batch folder. In this case, your data will be visualized in the "Single Signals (Shot-by-Shot)" Tab. It is not required to import laser metadata. See chapter 3.1.2.

ICP-Base also automatically imports the AcqMethod.xml file and HardwareConfig.xml file (see chapter 3.3). Therefore, all the ICP-MS metadata is also imported already. If you are happy with the filenames and no laser metadata should be uploaded, you can immediately proceed to the next module (chapter 3.5).

3.2. Laser metadata

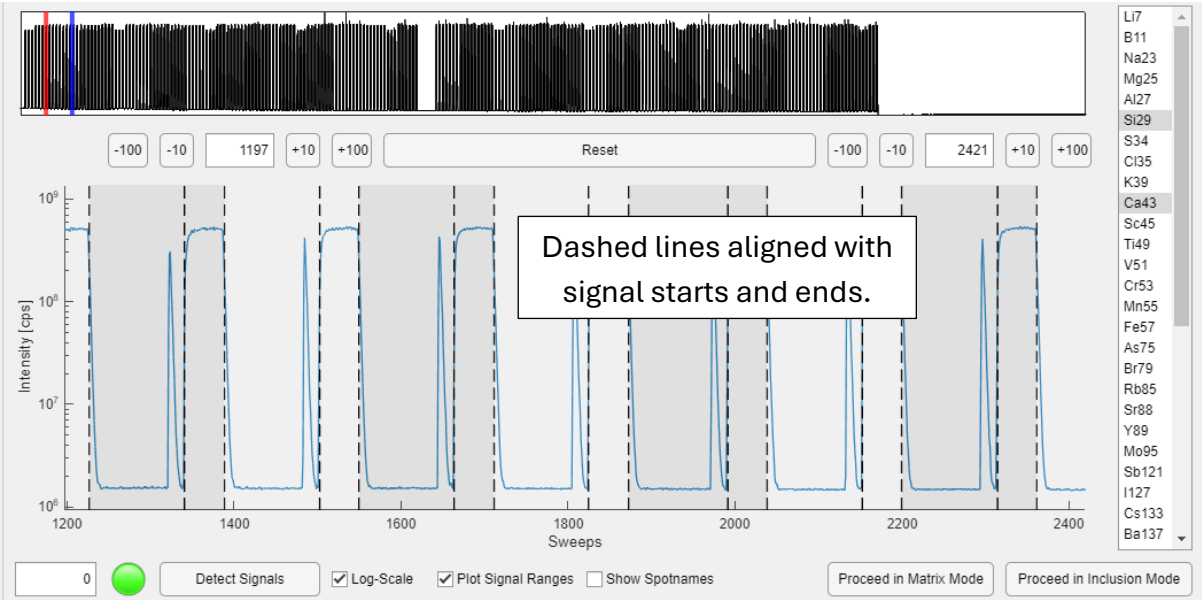
Laser metadata are logbook-type files that are output (mostly on request) by the software that runs the laser system. There are many different styles and filetypes but in general, they contain more or less the same information about the laser system: spot names, spot sizes, fluences, gas flows etc.

Press the “Get laser Metadata” button, select the type of your file and import it. Again, if you are unsure which type of file you have, check the example files and find a match.

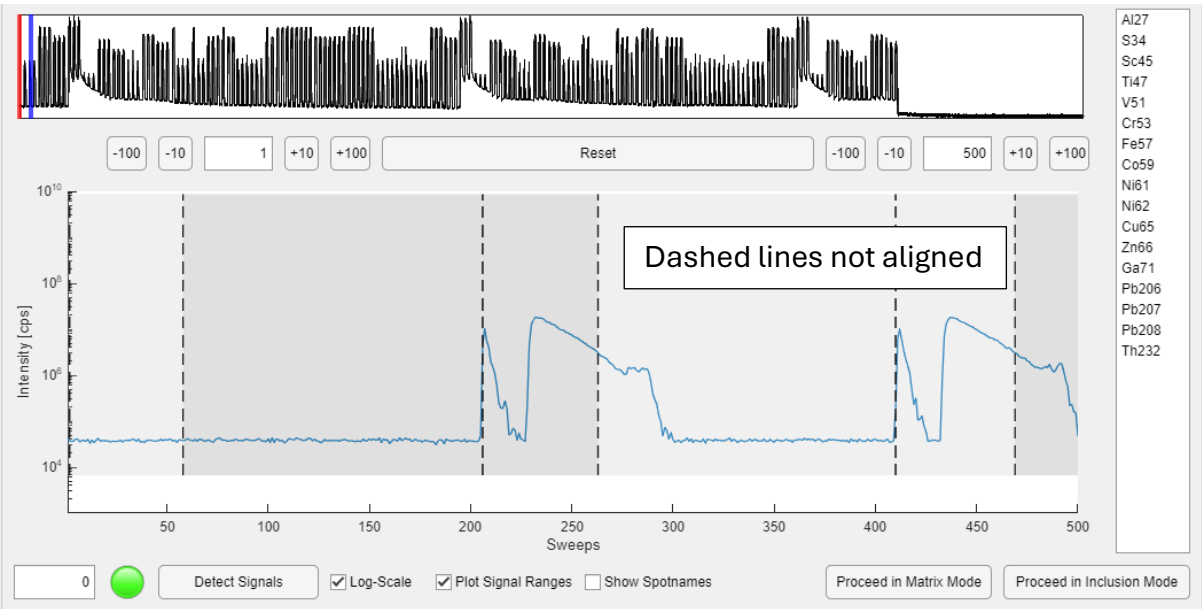
3.2.1. Continuous signal data

After successfully importing the signal data and the laser metadata, press the “Detect Signals” button. ICP-Base first uses a bisection method to detect the signal starts and ends based on the amounts of spots that were measured. Zoom in on the first signal all the way on the left of the plot to check if the detection was successful. If the dashed lines algin with the starts and ends, it worked.

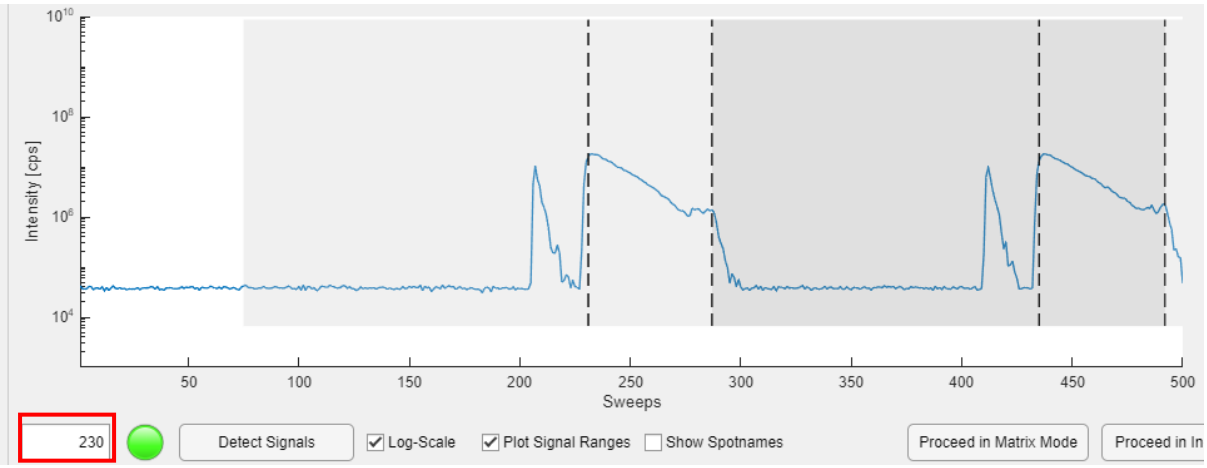
If it worked, it should look like this:



If the bisection method did not work, it looks like this:



In this case, ICP-Base uses the laser “On” and “Off” timestamps provided in the laser metadata file to make the signal starts and ends. You have to use the bottom left edit-field to move the starts and ends to the right place. To achieve this, type in the number of sweep (x-axis) of where the first signal starts and press “Enter” on your keyboard.

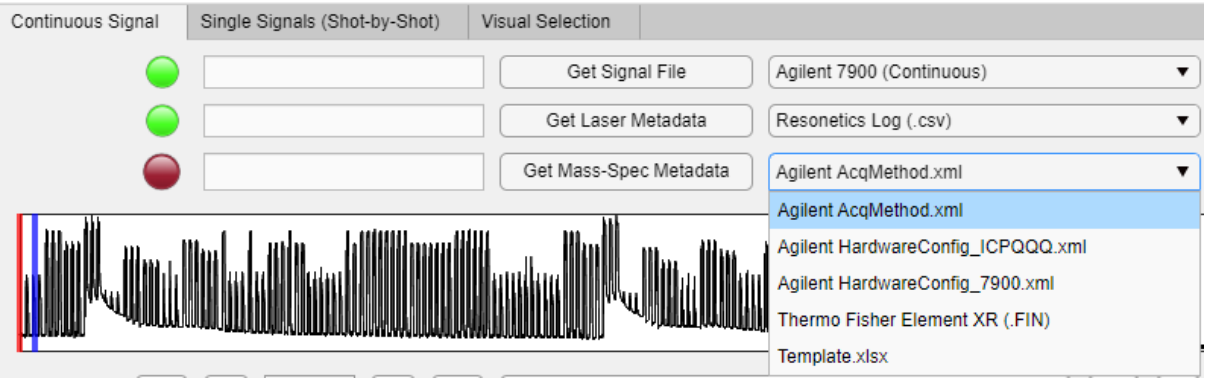


If this was successful, continue with the import of ICP-MS metadata.

3.3. ICP-MS metadata

ICP-MS metadata are also file(s) that are output by the ICP-MS systems. These file(s) can hold crucial metadata like the dwell times which are used for the calculation of the limits of detection or machine specific parameters used for long term monitoring.

For importing this metadata, you do not need to worry about “Continuous” or “Single”. Just import the chosen metadata once to cover the whole session.



3.3.1. Dwell times

These values are important to correctly calculate the limits of detection.

If the dwell times are unknown or were not provided by the lab manager, you can ignore this import step. ICP-Base will assume dwell times of 10 ms for all analytes, which is the most pessimistic approach that gives the highest limits of detection. Later, the dwell

times can be changed manually in the Quantification Module and you can recalculate the results, to get the right detection limits (see chapter 7.4).

3.3.1.1. Agilent

Agilent provides the dwell times of a measurement session in the AcqMethod.xml file that is by default stored in the batch folder.

3.3.1.2. Thermo Fisher Element XR

Thermo Fisher provides the dwell times in the .FIN file.

3.3.1.3. Thermo Fisher iCAP RQ

The iCAP RQ provides the dwell times in the signal file. If you imported the signal data, the dwell times already have been imported too.

3.3.1.4. Template .xlsx

This template is a provisory solution to import any dwell times by just making the first row of the sheet hold the exact same analyte names as the signal file and the first row holding the dwell times in seconds. This has become somewhat redundant since ICP-Base has the defaults at 10 ms.

3.3.2. Machine specific parameters

These parameters are mass spectrometer tuning and machine settings like RF Power, RF matching, Extracts, Lenses, Gas flows etc.

3.3.2.1. Agilent

Agilent provides the AcqMethod.xml and the HardwareConfig_XXX.xml files.

3.3.2.2. Thermo Fisher iCAP RQ

The iCAP RQ output files also holds all the parameters. If you imported the signal data, the parameters already have been imported too.

3.4. Sills projects

ICP-Base can import sills projects. Up to this point, it imports the (spike corrected) data, the dwell times and the intervals (background, host, inclusion). If there is a demand, it would also be partially possible to import the internal standards.

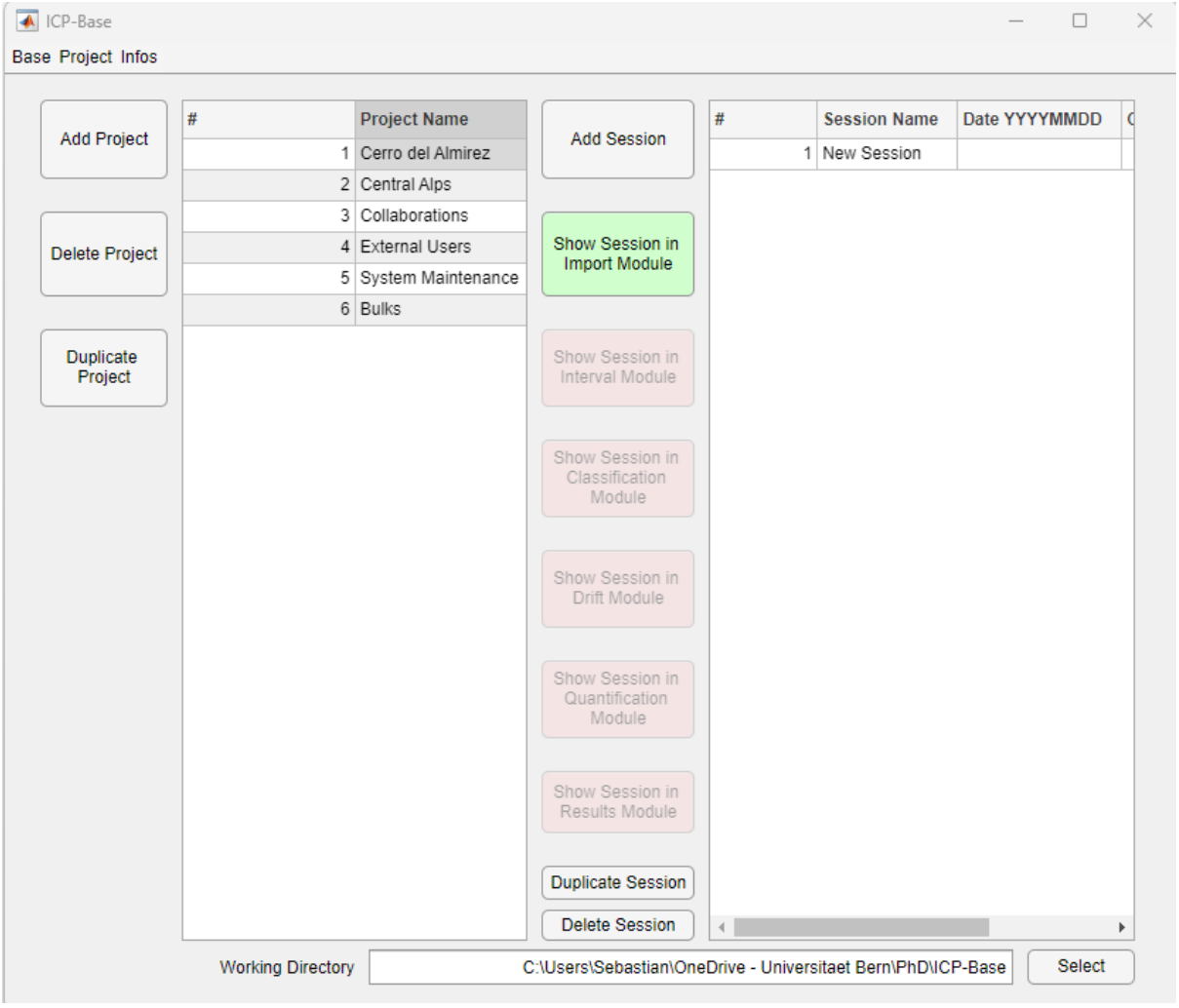
3.5. Proceeding to next module – Matrix vs Inclusion Mode

After importing all the data needed, it is possible to proceed in “Matrix Mode” and “Inclusion Mode” with the bottom right buttons. Both lead to the same modules and data reduction workflow, they just show/hide different buttons/functions/tables.

“Matrix Mode” should be chosen if “normal” spot analyses want to be quantified. This mode facilitates quick interval setting and hides a few inclusion related buttons to make the GUI easier to read.

“Inclusion Mode” should be chosen for inclusion quantification. This mode facilitates a slower but more thorough way of setting the intervals.

ICP-Base will ask if you want to save the data – press “Save” to proceed. This means that your imported data will be saved back to the base. The “New Session” will now be visible within the project in the base. This does NOT save the data permanently on your computer though! For this you have to save the whole base (see chapter 8.5)



3.6. Quick guide

- Import signal data files (raw counts-per-seconds data)
    - “Continuous” = One file for multiple spot analyses
        - Needs laser metadata to detect signals
    - “Single” = One file per spot analysis
        - Does not need laser metadata
        - Multiselect possible
        - For PerkinElmer .xl files the acquisition time is taken from the file information, if the file is changed and saved it again, this is wrong.
    - Agilent batch folder
        - Also imports ICP-MS metadata
        - “Continuous” = One .d folder in batch folder
            - Needs laser metadata to detect signals
        - “Single” = Multiple .d folders in batch folder
            - Does not need laser metadata
- Import laser metadata (“Logfiles”)
    - If “Single” the spot names get overwritten by the names in the logfile
    - If “Continuous” press the detect signals button on bottom left
        - Zoom in on lower plot to check if dashed lines align with signal starts and ends
            - If not type a number in the field on bottom left and press “Enter” on the keyboard to shift the dashed lines to signal starts and ends
- Import ICP-MS metadata
    - Optional
    - Dwell times are stored in:
        - Agilent AcqMethod.xml
        - Thermo Fisher Element XR (.FIN)
        - Thermo Fisher iCAP they are already imported with the signal data.
    - If no dwell times are imported, ICP-Base assumes 10 ms
        - This can be changed later in the Quantification Module
    - Machine specific parameters
        - Agilent AcqMethod.xml and HardwareConfig_XXX.xml
        - Thermo Fisher iCAP they are already imported with the signal data.
- Proceed in:
    - “Matrix Mode” for “normal” spot analysis
    - “Inclusion Mode” for inclusion analysis

4. Interval Module

From here on, it does not matter anymore if the datatype is “Continuous” or “Single”, all the data is treated and visualized in the same way. There are some small differences though if “Matrix Mode” or “Inclusion Mode” have been chosen.

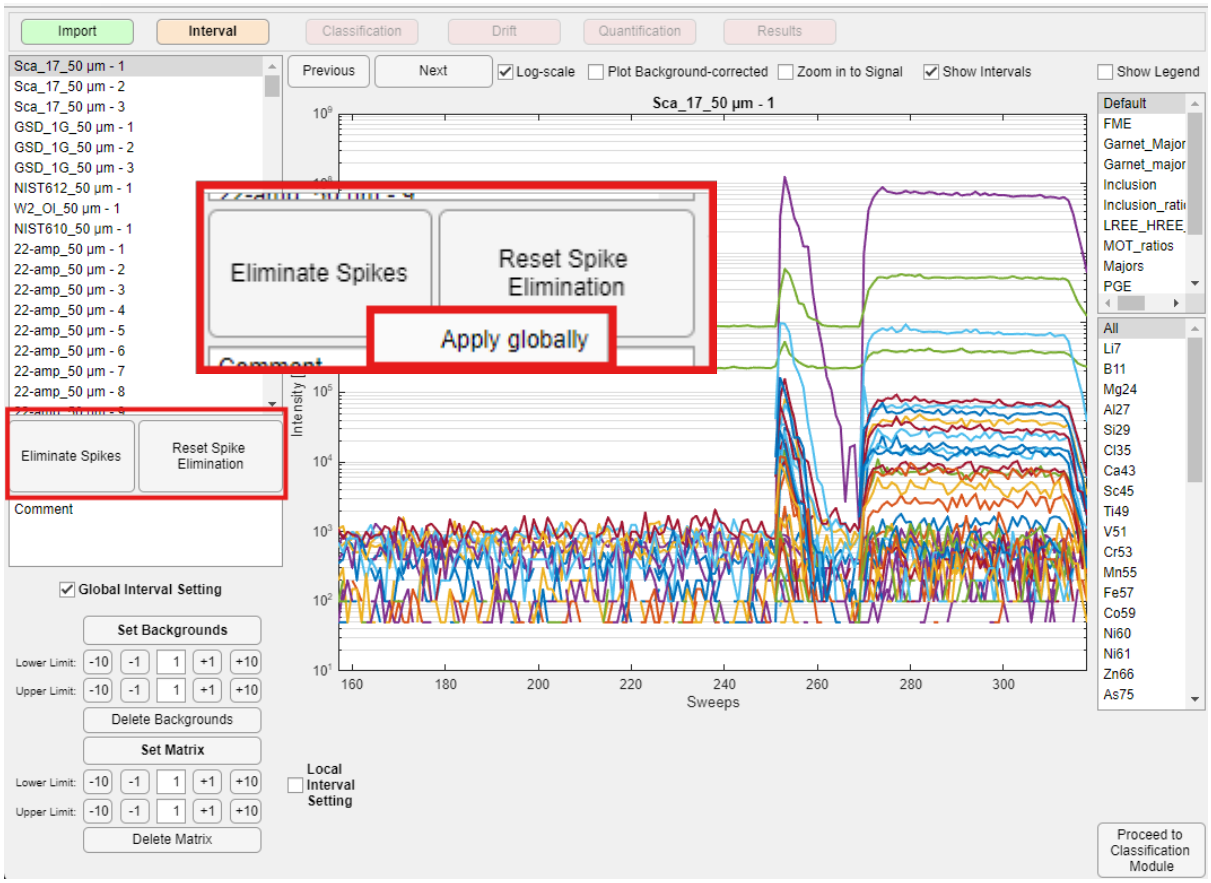
4.1. Spike elimination

4.1.1. Matrix Mode

I would recommend to start with a spike elimination so that possible spikes do not interfere with the interval setting. ICP-Base uses a Grubbs’s Test to check if a sweep for a given analyte is an outlier by comparing the value of that sweep with the values of four sweeps before and after. Every sweep below 600 counts per second is not considered. This threshold can be changed in the code if necessary. The spike elimination only works once per spot (so clicking it more often does not eliminate more spikes).

To eliminate the spikes for a given spot, press the “Eliminate Spikes” button and to bring them back the “Reset Spike Elimination” button.

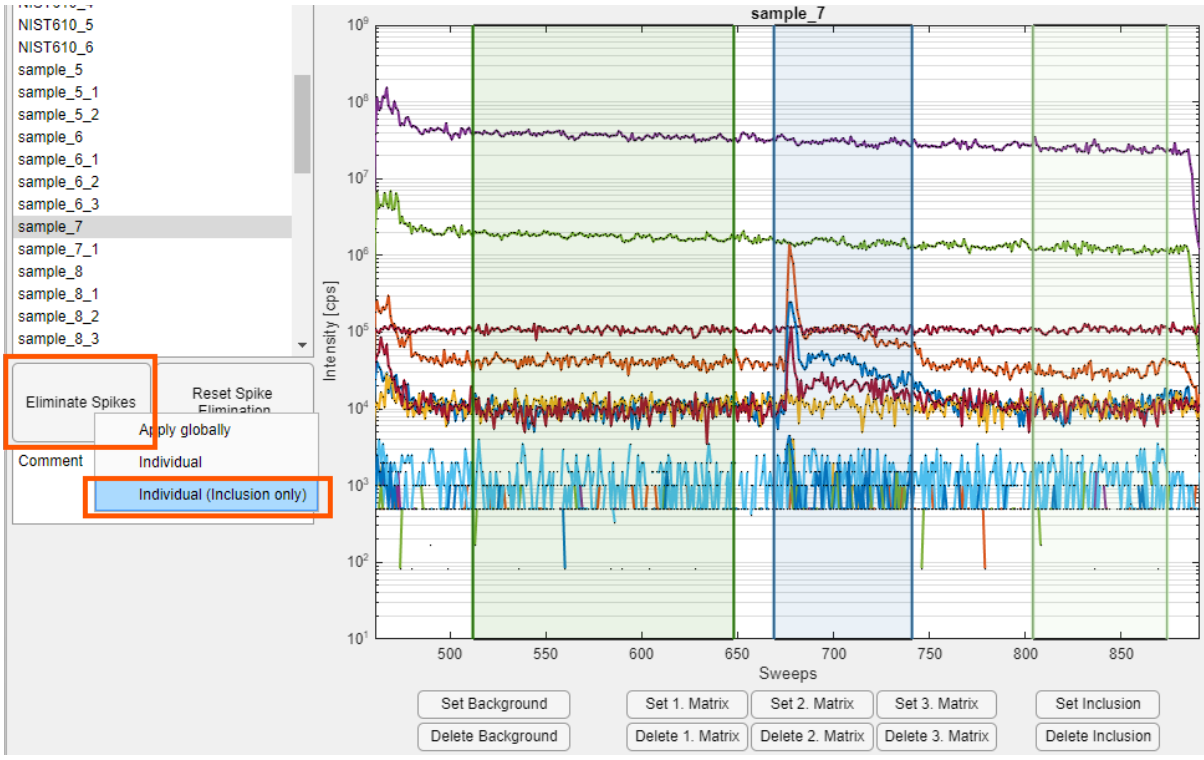
If you want to eliminate the spikes in every spot, you can just right-click the “Eliminate Spikes” button and then click “Apply globally”. The same is possible with the “Reset Spike Elimination” button. Regarding the “Individual” and “Individual (Inclusion only)” continue reading the next chapter.



4.1.2. Inclusion Mode

It generally works the same as for the “Matrix Mode” with the difference, that ICP-Base ignores all the data within the “Inclusion” Interval. So if you set your inclusion intervals first and at the end press the “Apply globally” button (see above) all the spikes will be cleared except for the data within the inclusion intervals, which will remain unchanged.

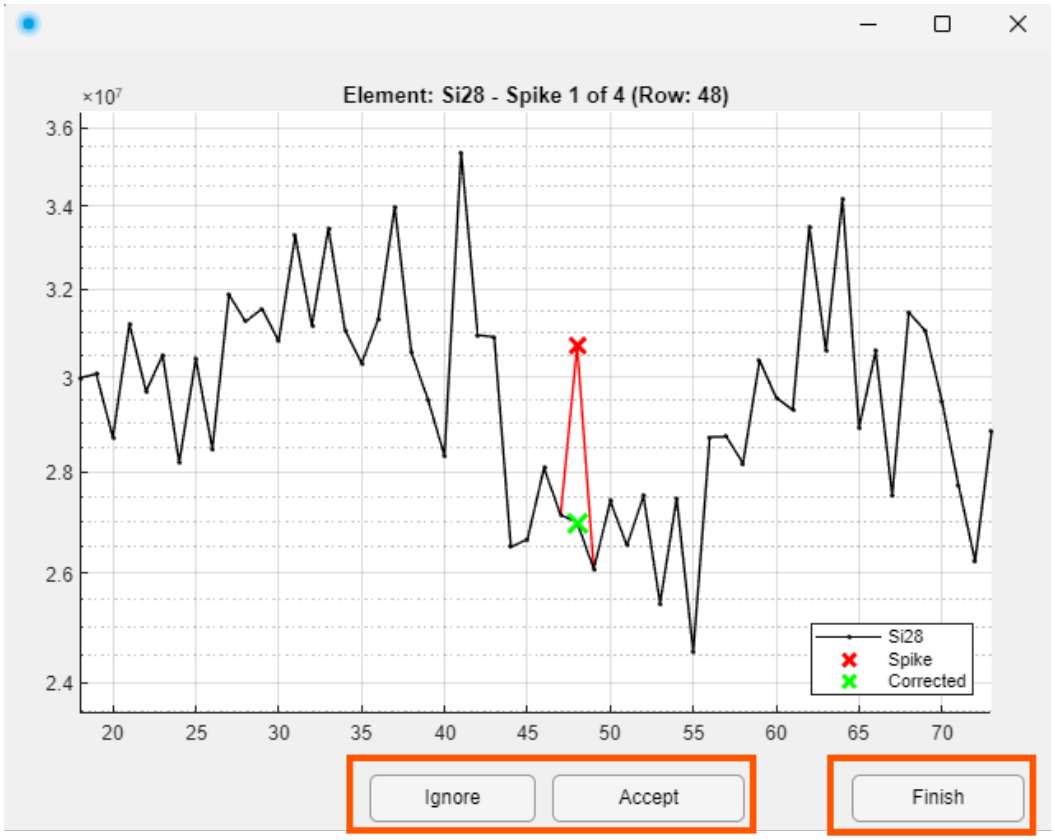
Next, you can right-click the “Eliminate Spikes” button and select “Individual (Inclusion only)” which will let you manually ignore or accept spikes just in the inclusion interval.



A the Spike Submodule will then pop-up showing the spikes in red and the suggested corrected value in green. If you accept, the data will be corrected. If you ignore, the spike will not be corrected. When you are done clicking through all the spikes, click “Finish” and continue in the Interval Module.

If you just select “Individual” you can correct all spikes manually, also those that are not in the inclusion intervals.

A good practice is to first set all the inclusion intervals, then do the spike elimination globally which will ignore the inclusion data. Next, do the “Individual (Inclusion only)” spike elimination to carefully ignore and accept the spikes in the inclusion interval.

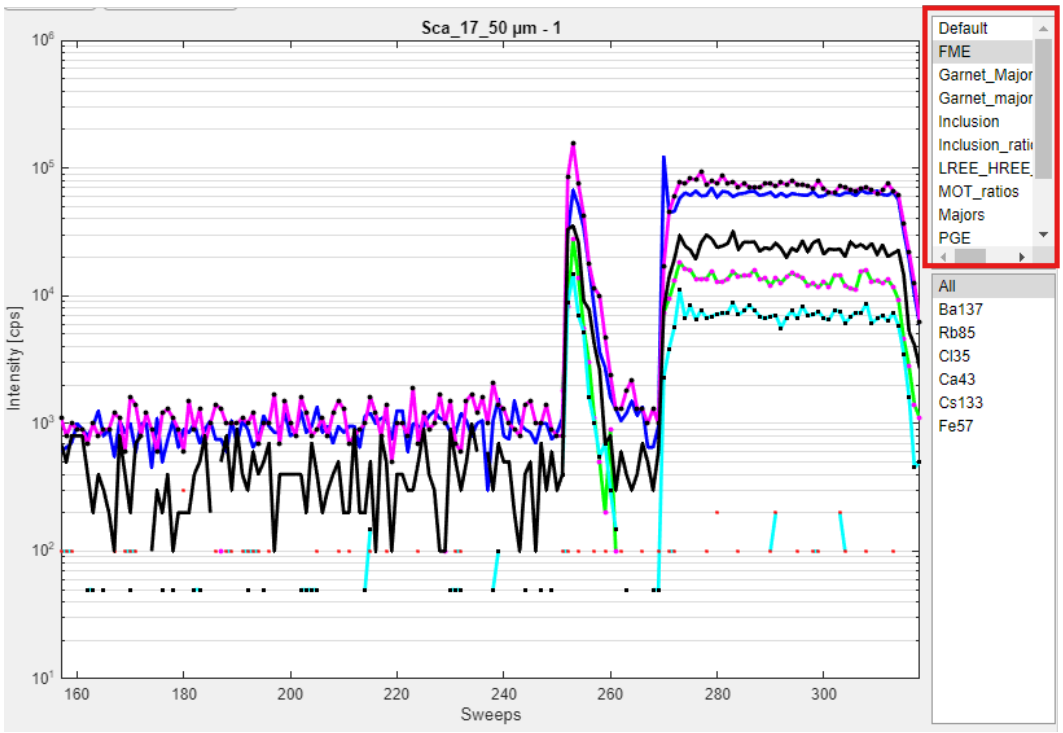


4.2. Plot styles

Either you use the default visualization, or you can choose plot styles to visualize and highlight analytes that are of interest and plot analyte ratios. If the second case is what you want, you first have to think about what you want to show.

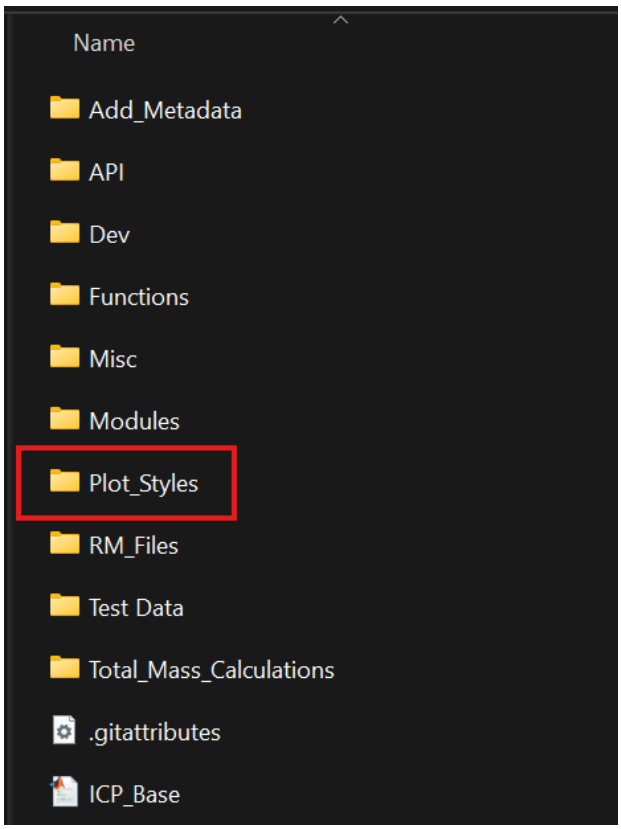
Are there certain analytes that are more important than others? Are there certain analytes that are commonly found in unwanted mineral inclusions or impurities that should be avoided?

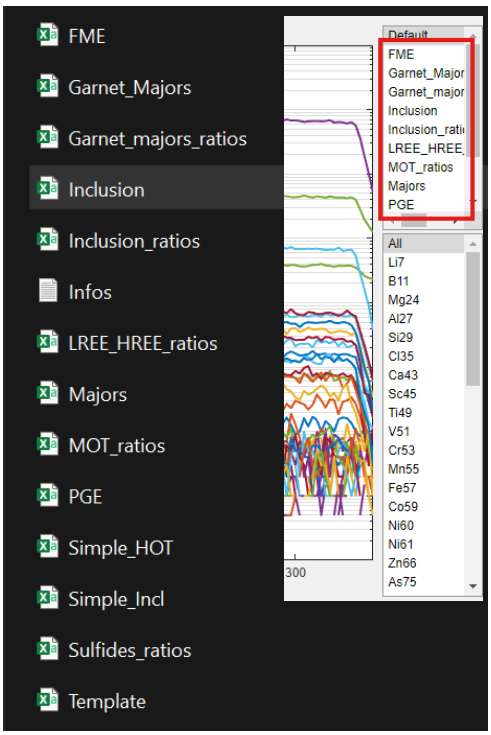
Once you know what you want to show, you can generate your very own plot style. They are shown in the top right corner of the Interval Module.



4.2.1. *How to get make them*

In the ICP-Base folder you will find a subfolder called Plot_Styles. Within that folder you will find a few .csv sheets that contain the plot styles.





You can now simply copy an existing style and modify it or make a completely new one based on the Template.csv that is automatically containing all the analytes that were measured this session.

To make these, simply fill in the plot properties and save the file in the same folder. **<u>When naming the file, never use – and avoid whitespaces and other special characters!</u>** If you do use them, you will probably not be able to load the plot style.

4.2.2. <u>"Normal" plots</u>

These plot properties are partly restricted by MATLAB itself. E.g. there are the colors "black", "blue" and "cyan" that can be used like this because they are default colors in MATLAB. This does not work for most other colors though. If you want a specific colour, just use the hex code (e.g. #FFB442), these always work and are easy to get.

If no marker is wanted, put Marker, MarkerFaceColor and MarkerEdgeColor to "none" and MarkerSize to 0.

Here is an example and some possible line and marker styles:

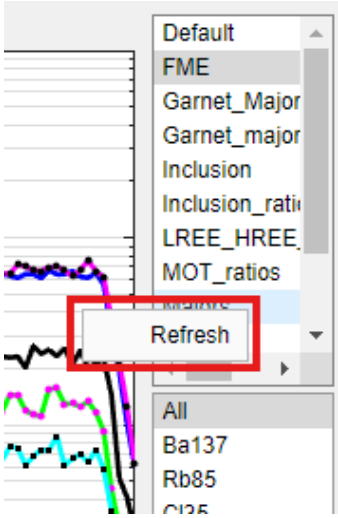
	A	B	C	D	E	F	G	H
1	Analytes	Color	LineWidth	Linestyle	Marker	MarkerSize	MarkerFaceColor	MarkerEdgeColor
2	Mg25	black	2	--	none	0	none	none
3	Si29	black	2	-.	none	0	none	none
4	Ca43	blue	2	-	o	2	magenta	magenta
5	Fe57	black	2	:	none	0	none	none
6	Ba137	green	2	-	none	0	none	none
7	Cl35	cyan	2	-	*	2	magenta	magenta
8	S34	#FFB442	2	-	none	0	none	none
9	K39	blue	2	-	s	2	blue	black
10	Na23	blue	2	-	d	2	green	green
11	Br79	cyan	2	-	*	2	black	black

Line Style	Description	Resulting Line
"-"	Solid line	————
"--"	Dashed line	- — — — —
":"	Dotted line	.....
"-."	Dash-dotted line	—.—.—.—.-

Marker	Description	Resulting Marker
"o"	Circle	○
"+"	Plus sign	+
"*"	Asterisk	✳
"."	Point	•
"x"	Cross	×
"_"	Horizontal line	—
"|"	Vertical line	|
"square"	Square	□
"diamond"	Diamond	◇
"^"	Upward-pointing triangle	△
"v"	Downward-pointing triangle	▽
">"	Right-pointing triangle	▷
"<"	Left-pointing triangle	◁
"pentagram"	Pentagram	☆
"hexagram"	Hexagram	✡

If there is an analyte in the plot style that was not measured this session, it will simply not be shown in the Interval Module. Also if your input was nonsensical or wrong it will also not show up in the Interval Module.

Once you have made your plot style, right-click the list box and press “Refresh”. The new plot style should now show up. If it does not, you might have named it wrong (using special characters or – or whitespaces, see above) or saved it to the wrong place.



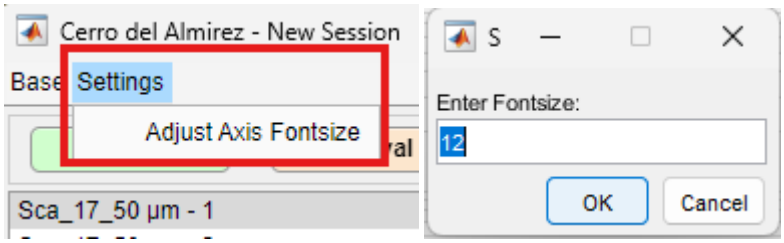
4.2.3. *“Ratio” plots*

In the same manner as the “normal” plots, you can also visualize analyte ratios by establishing the nominator and denominator. The rest of the visualization properties are the same as for the “normal” plots (see above).

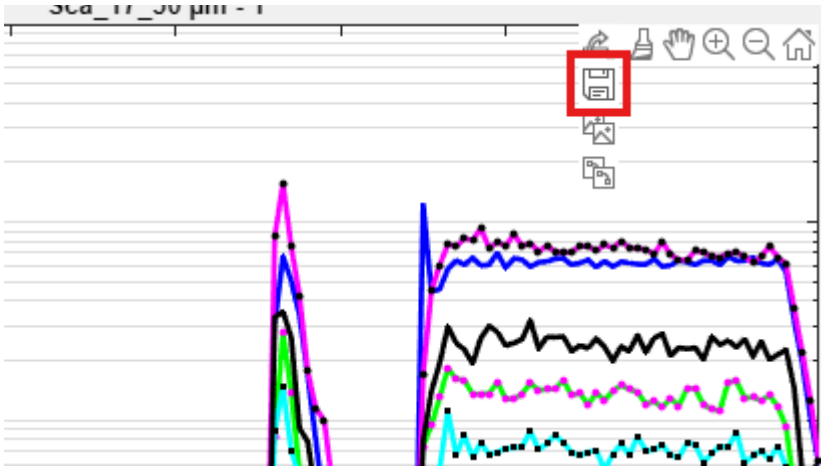
	A	B	C	D	E	F	G	H	I	J
1	Nominator	Denominator	Color	LineWidth	Linestyle	Marker	MarkerSiz	MarkerFa	MarkerEdgeColor	
2	Ca43	Al27	blue	2	-	*	4	cyan	black	
3	Ba137	Fe57	cyan	2	-	o	4	green	green	
4	Mg25	Si29	green	2	-	s	4	red	red	
5	Si29	Al27	magenta	2	-	*	4	green	green	
6	Mg25	Al27	black	2	-	o	4	red	black	
7	Ba137	Lu175	red	2	-	none	0	none	none	

4.2.4. *Saving signal figures*

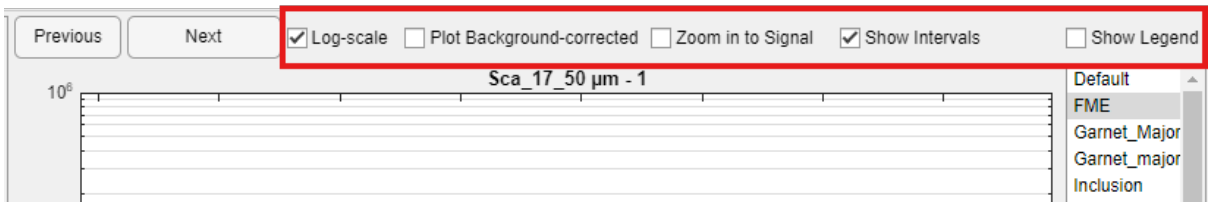
If you want to save the figure of the signal for publication, first go to “Settings” an then “Adjust Axis Fontsize”. This will trigger a pop-up in which you can change the font size of the plot. For publication-ready figures a font size of 20-30 is good, 12 is the default.



Once that is done, you can navigate to the top right corner of the plot, select the leftmost icon and save the figure in the desired format.



4.3. General visualization features



4.3.1. Plot Background-corrected

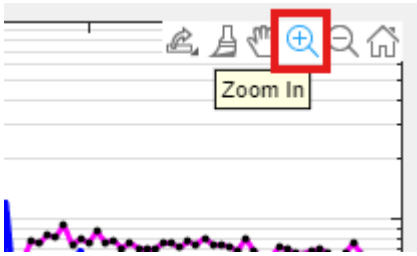
If a background interval is already set, you can check the "Plot-background corrected" checkbox to, who would have though, plot the signal background-corrected. ICP-Base subtracts the mean of the background counts-per-second of each analyte from the whole signal. This means that for sweeps that are higher than the mean there will still be some counts left, even in the designated background. Don't worry about this, just accept and ignore it.

4.3.2. Zoom in to Signal

If you check the "Zoom in to Signal" checkbox ICP-Base will automatically use the signal start and end (as detected in the Import Module) to be the x-axis limits. If the starts and ends were not correctly detected, this will also not correctly display anything.

4.3.3. Zooming in and out manually

It is also possible to use the built-in "Zoom In" and "Zoom Out" function of MATLAB. If you use these you will be able to zoom in and out where ever you want. If this tool is selected, you are not able to move the interval lines anymore. To be able to move them again, click on the selected tool (e.g. "Zoom in") once more to deselect it. The interval lines are only movable if you have **not** selected any tool!



4.3.4. Show Intervals

The "Show Intervals" checkbox can be used to hide or show the intervals (if there were any set). This can be desired if the intervals are already set but the figure should be exported without the intervals.

4.3.5. Show Legend & Log-scale

"Show Legend" shows the legend and "Log-scale" changes the plot from log to linear scale and vice versa.

4.4. Commenting a spot

For every spot, it is possible to write a comment. So e.g. if hit an apatite inclusion by accident, you can write “Ca and P inclusion” and later in the results, you instantly know why your Ca and P are way too high with respect to the other comparable spots.

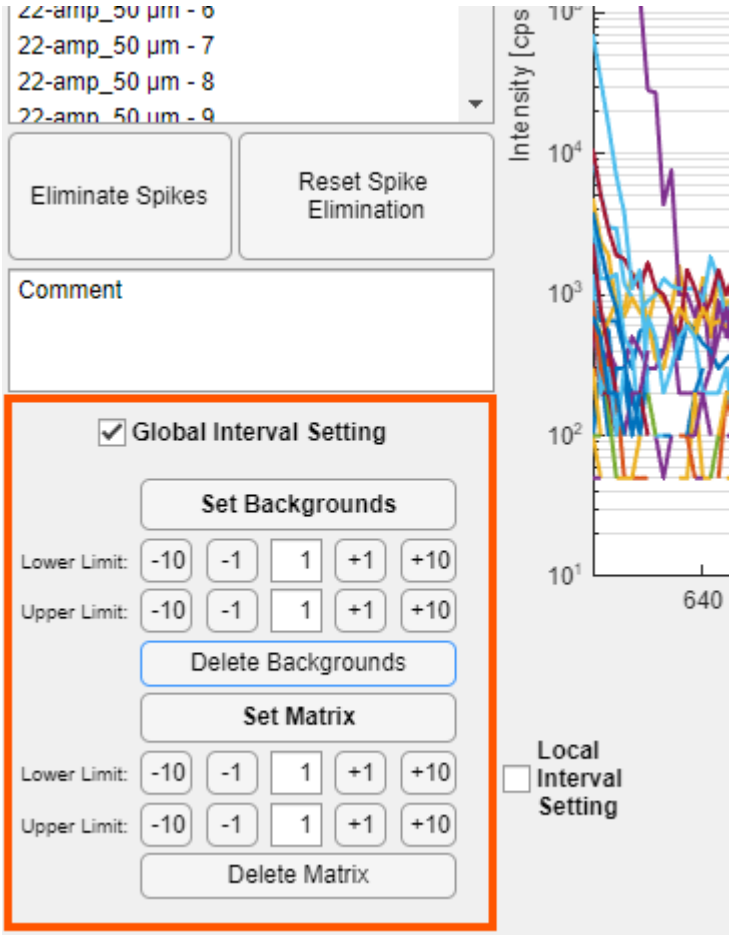
4.5. Interval setting

Disclaimer: If your computer can be referred to as a “potato”, meaning that it is old and slow (sorry), the dragging of the intervals can be quite slow and lagging behind. Right now there is nothing I can do about that, I might try to optimize this in the future (if that is even possible).

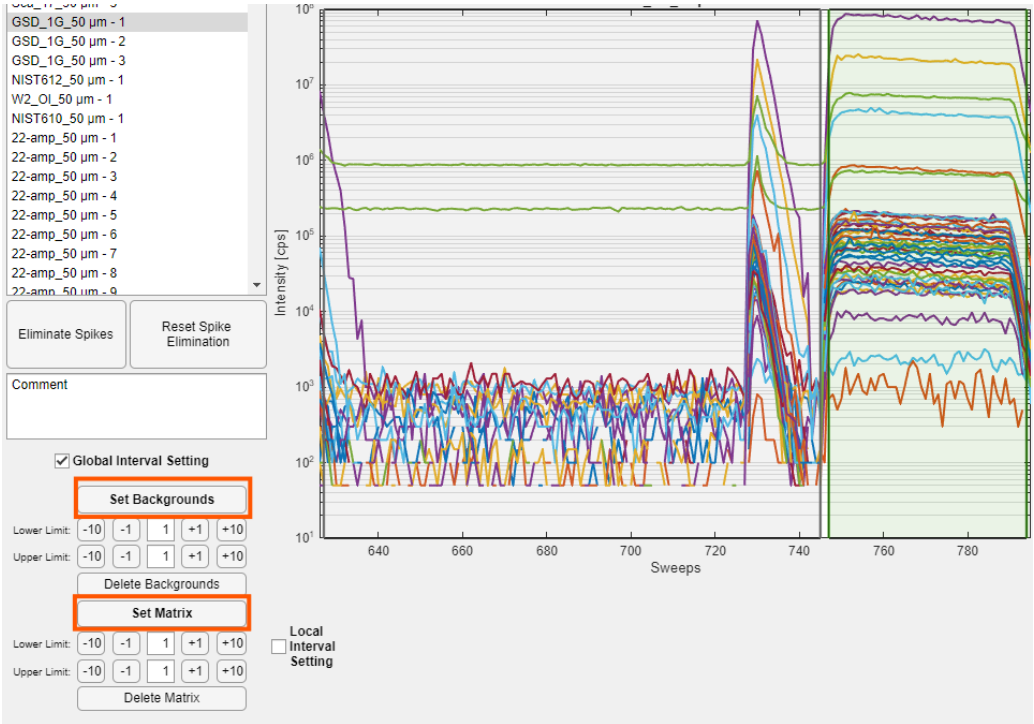
4.5.1. Matrix Mode

4.5.1.1. Global Interval Setting

By default, the “Global Interval Setting” panel is visible. Like the name suggests, it lets you set the background and matrix intervals for all spots at the same time. This is based on the signal start and end of each spot as they were detected in the Import Module.



First, press the “Set Backgrounds” and “Set Matrix” buttons. This will set the backgrounds (grey interval) and the matrix intervals (green) for all spots.



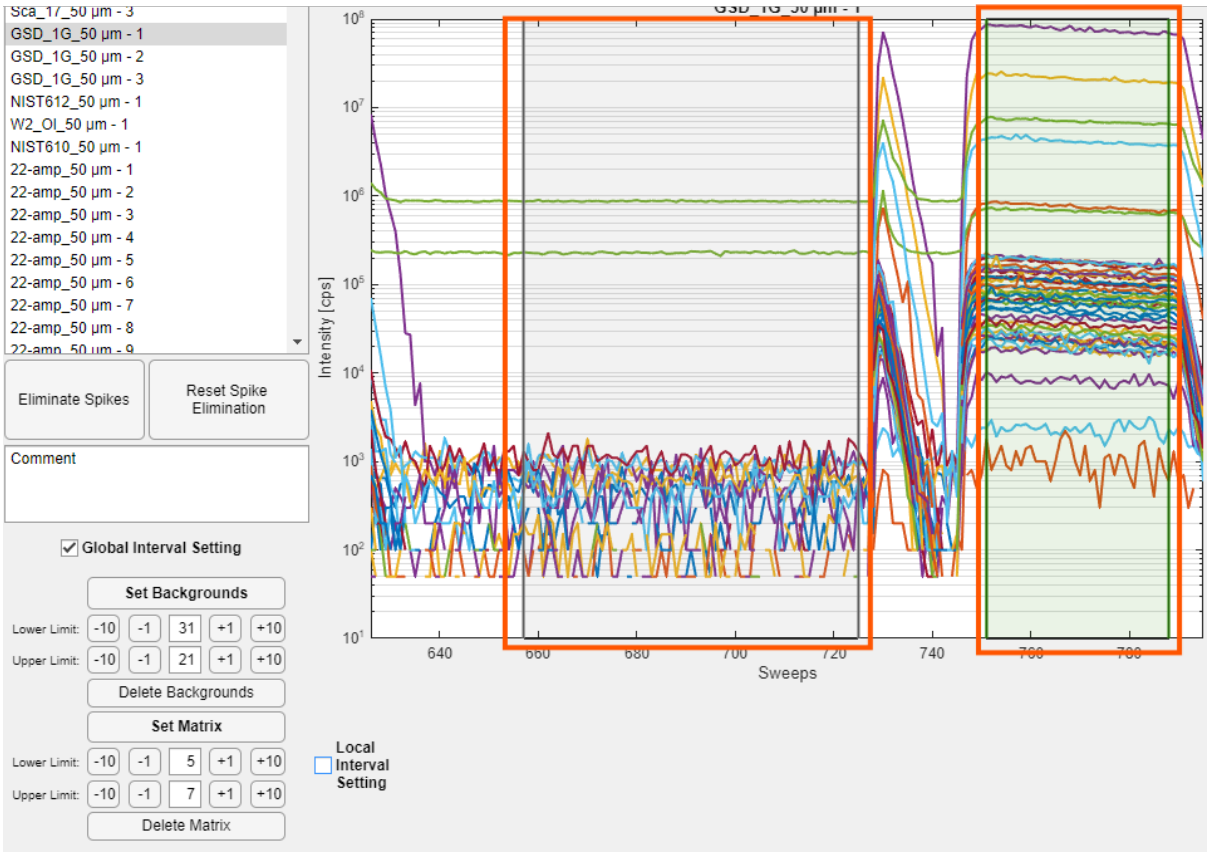
The numbers of ±10, ±1 and the number in the field (1 by default) correspond to the number of sweeps that will be intend from the signal start and end (as detected in the Import Module).

So for the background “Lower Limit”, if you press +10 it will move 10 sweeps to the right of the signal end of the spot beforehand. If you press -10 on the “Upper Limit”, it will move 10 sweeps to the left of the signal start of this spot. For the first spot where there is no preceding signal end, ICP-Base will assume this based on the signal starts and ends of all other spots.

For the matrix “Lower Limit”, if you press +10 it will move 10 sweeps to the right of the current signal start and if -10 is pressed it will move 10 sweeps to the left of the current signal end.

Once they are set, it is possible to move the whole intervals by clicking, holding and dragging the middle of the interval and the lower and upper limit by clicking, holding and dragging the interval borders. When you move them like this, make sure to not overlap them.

If you click “Delete Backgrounds” or “Delete Matrix” they will be deleted globally! The same is true for the shifting of the lower and upper limits – all the intervals will be moved accordingly. Make sure to not press these buttons if you already spent some time manually adjusting the intervals.



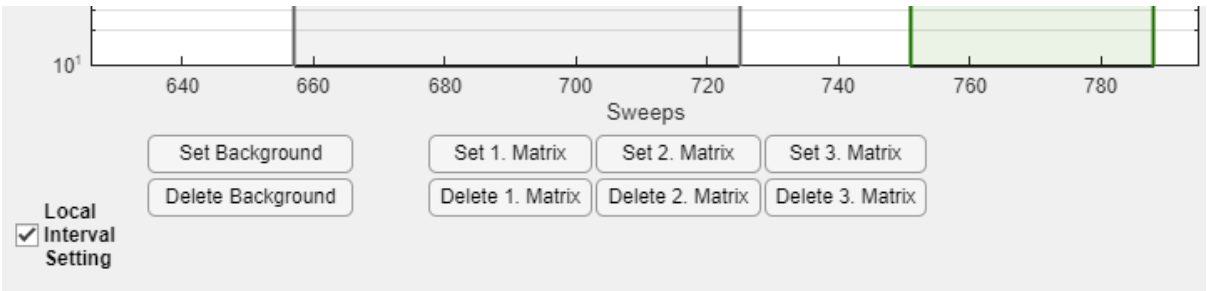
4.5.1.2. Local Interval Setting

By checking the “Local Interval Setting” checkbox a few buttons show up that will only affect the current spot.

By clicking the “Set XY” buttons, you can set a given interval by dragging and dropping the cursor over the plot.

By clicking the “Delete XY” buttons, you can delete an interval just for this spot.

For the matrix, it is possible to set three different intervals which makes it possible to avoid inclusions. The counts-per-second of the different matrix intervals will be averaged (weighted) in the Quantification Module.



4.5.1.3. How to do this efficiently

First, navigate to one of the first spots (but not the first one as it has no preceding spot). Then press the “Set Backgrounds” and “Set Matrix” button and adjust the intervals using

the ±10 and ±1 buttons. Check these settings for a few of the spots at the start, middle and end of the measurement session. Once you are happy and the intervals fit the spots, uncheck the "Global Interval Setting" to hide the buttons as to not press them by accident. Next, manually click through all of them (yes, really) to check for inclusions and impurities that should be avoided. Adjust the intervals manually by dragging them and delete or add more using the "Local Interval Setting".

4.5.2. Inclusion Mode

In the "Inclusion Mode", the "Global Interval Setting" is hidden to make sure users actually look at all their spots. All buttons will only affect the current spot.

By clicking the "Set XY" buttons, you can set a given interval by dragging and dropping the cursor over the plot.

By clicking the "Delete XY" buttons, you can delete an interval.

Additionally to the three matrix intervals, the user can also set an "Inclusion" interval (blue).

Once you set your background, host (matrix) and inclusion intervals, continue reading chapter "4.6 Signal Unmixing" to get more information about how to deconvolve the signals.

4.5.3. Deleting a spot

If a spot should be disregarded completely, delete **all** intervals of that spot using the "Local Interval Setting". It will still show up in the next modules but nothing will be quantified. There is no option to actually delete the data itself.

4.6. Signal Unmixing

When analyzing fully enclosed inclusions of any sort (fluid/solid/single-phase/poly-phase), the raw signal data will always be the sum of the counts-per-second (cps) derived from ablation of the inclusion and the surrounding host (since both are ablated at the same time).

This implies, that the data within the inclusion interval that you set is not just the cps of the inclusion but is the cps of a host-inclusion-mix. In order to quantify the inclusion, we need to deconvolve this host-inclusion-mix to pure host and pure inclusion contributions.

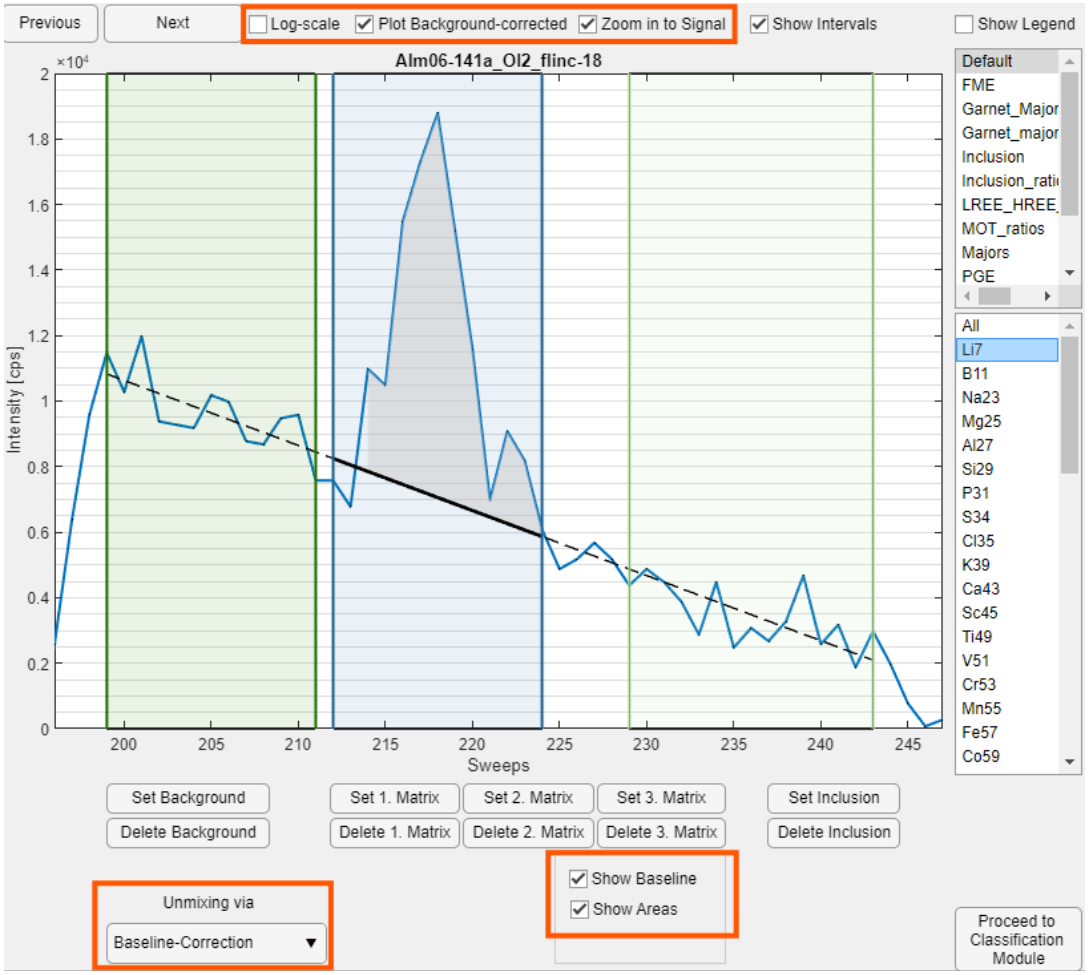
There are two unmixing modes which can be changed in the "Unmixing via" dropdown menu located below the "Delete Background" button (see figures below).

4.6.1. Baseline correction

This is the default unmixing mode of ICP-Base since it is the easiest and most straightforward solution. Actually, when you set a background, a host (matrix) and an inclusion interval, the unmixing already happened.

A baseline is a straight line fitted to all cps datapoints of a given analyte in the available host intervals. This means that the unmixing process of every analyte is just dependent on itself. The baselines are extended to the inclusion interval where they estimate the pure host contribution to the mix for each analyte. If the cps of a given sweep is higher than the baseline, the correction will subtract the estimated pure host contribution from that sweep. This then yields the pure inclusion counts-per-second for each sweep which will be averaged and quantified later.

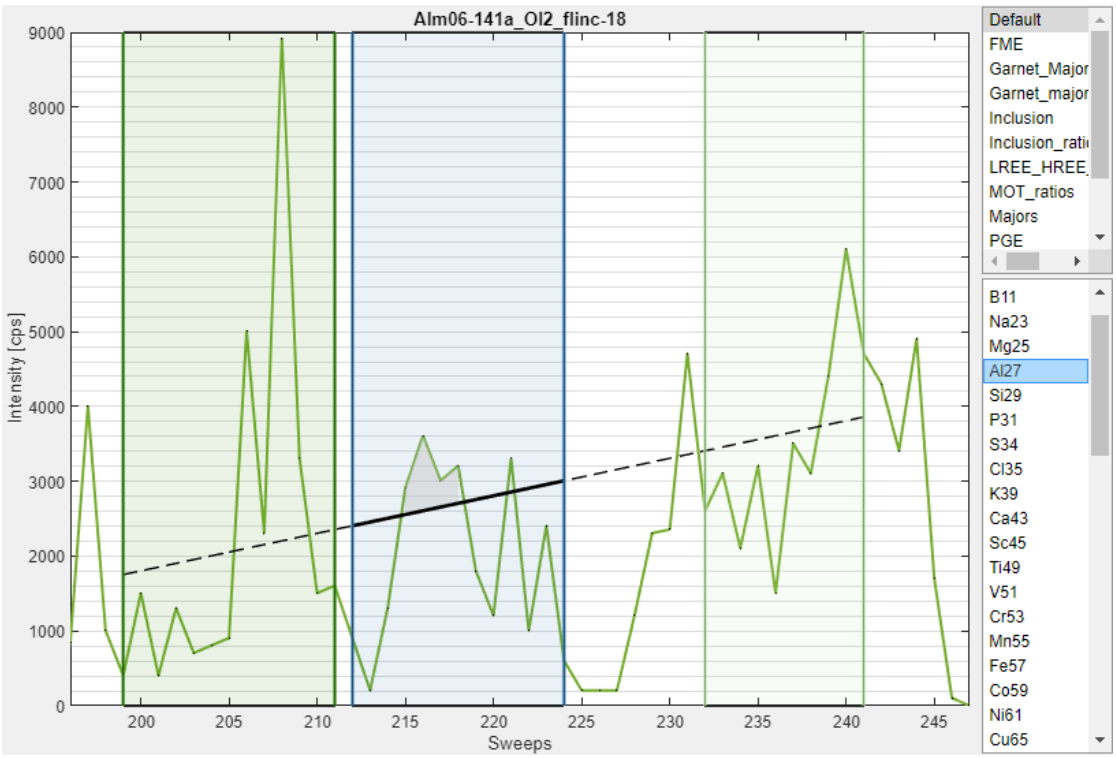
You can visualize the baselines by checking the "Plot Background-corrected" & "Show Baseline" checkboxes. This is necessary because the baselines are fitted to the background-corrected cps data. You can also show the area under the inclusion by checking the "Show Areas" checkbox. This is not really good for anything except that I like it. Sometimes, looking at the baselines works better if linear-scale is used.



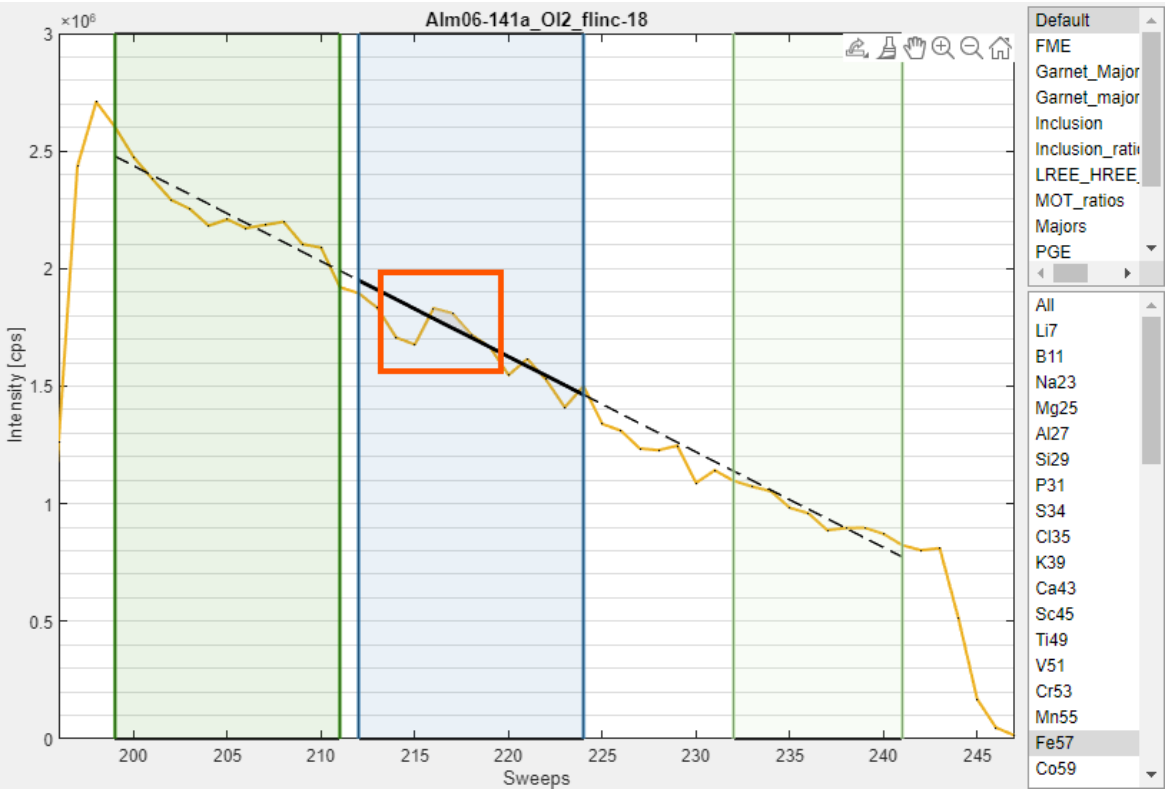
Since every analyte has its own baseline, I recommend looking at the analytes of interest/importance to see how well the fit is. By moving the host and inclusion intervals the baseline can be adjusted on-the-go. In the example above, the baseline is as good as it gets.

There are a few cases though, where the unmixing takes place even though the pure host and pure inclusion contributions obtained this way are quantitively non-sensical.

In this example below, the unmixing process will ascribe a few cps to the inclusion even though Al is not actually part of the inclusion, it is just scatter of the data in the host and inclusion intervals.



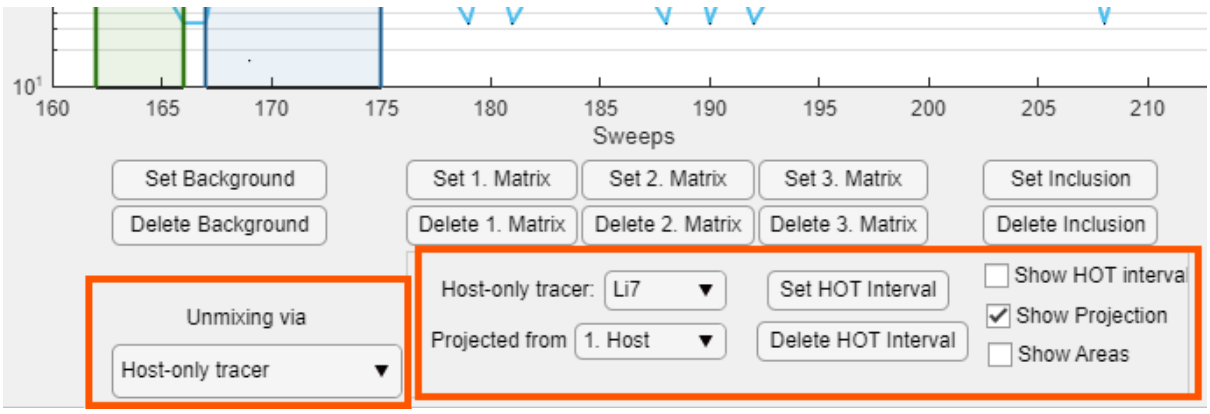
The next example is from an analysis of an inclusion in olivine, which contains a lot of Mg and Fe. So the inclusion itself is actually nowhere visible. But still, a few sweeps of Fe land above the baseline just due to the scatter of the data. Those sweeps will be considered as part of the inclusion (which they are not). The cps of Fe (a major in the host) are way to high (10^6) for the small inclusion cps to be a substantial part of the host-inclusion-mix.



<u>Since the software cannot know what makes sense and what does not, it is on you to check this</u>! ICP-Base will quantify and sell such non-sensical inclusion data to you as if it were correct!

4.6.2. <u>Host only tracer (HOT)</u>

This unmixing technique is a refined version of the "Matrix-only tracer" first established in sills. I recommend using this mode if the baseline-correction just is not good enough e.g. when the host intervals are very short. This mode only requires one host (matrix) interval and sometimes its enough if it is only two sweeps long.

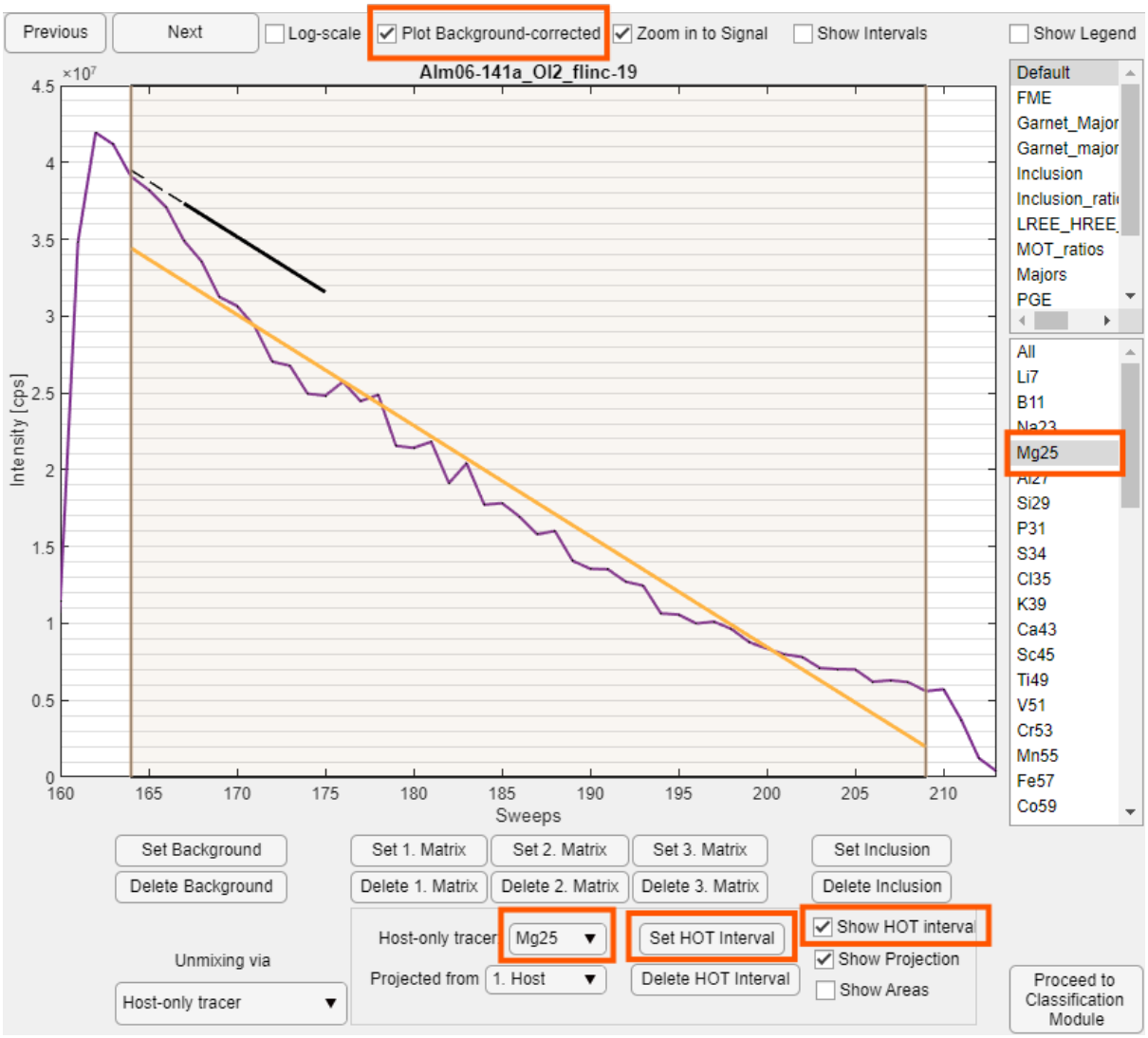


The “Host-only tracer” is an analyte, that is abundant in the host matrix and **not** in the inclusion. This is typically the case for major elements that have so much counts they don’t “feel” the inclusion or an analyte that is just very incompatible.

So first step is to click through your analytes and find one that you like and that shows no sign of the inclusion. Then select this analyte as your host-only tracer (HOT) in the dropdown menu (This can be changed later without problems). Next, press the “Set HOT Interval” button which hides the existing intervals and lets you draw a new interval by clicking, holding and dragging on the plot. You can now set this interval over the whole length of the signal.

ICP-Base can always either show the matrix and inclusion intervals or the HOT interval as they are fully independent of each other and can overlap. You can change what should be shown by checking “Show Intervals” or “Show HOT interval”.

Next, check the “Plot Background-corrected” checkbox to visualize the fitted lines.

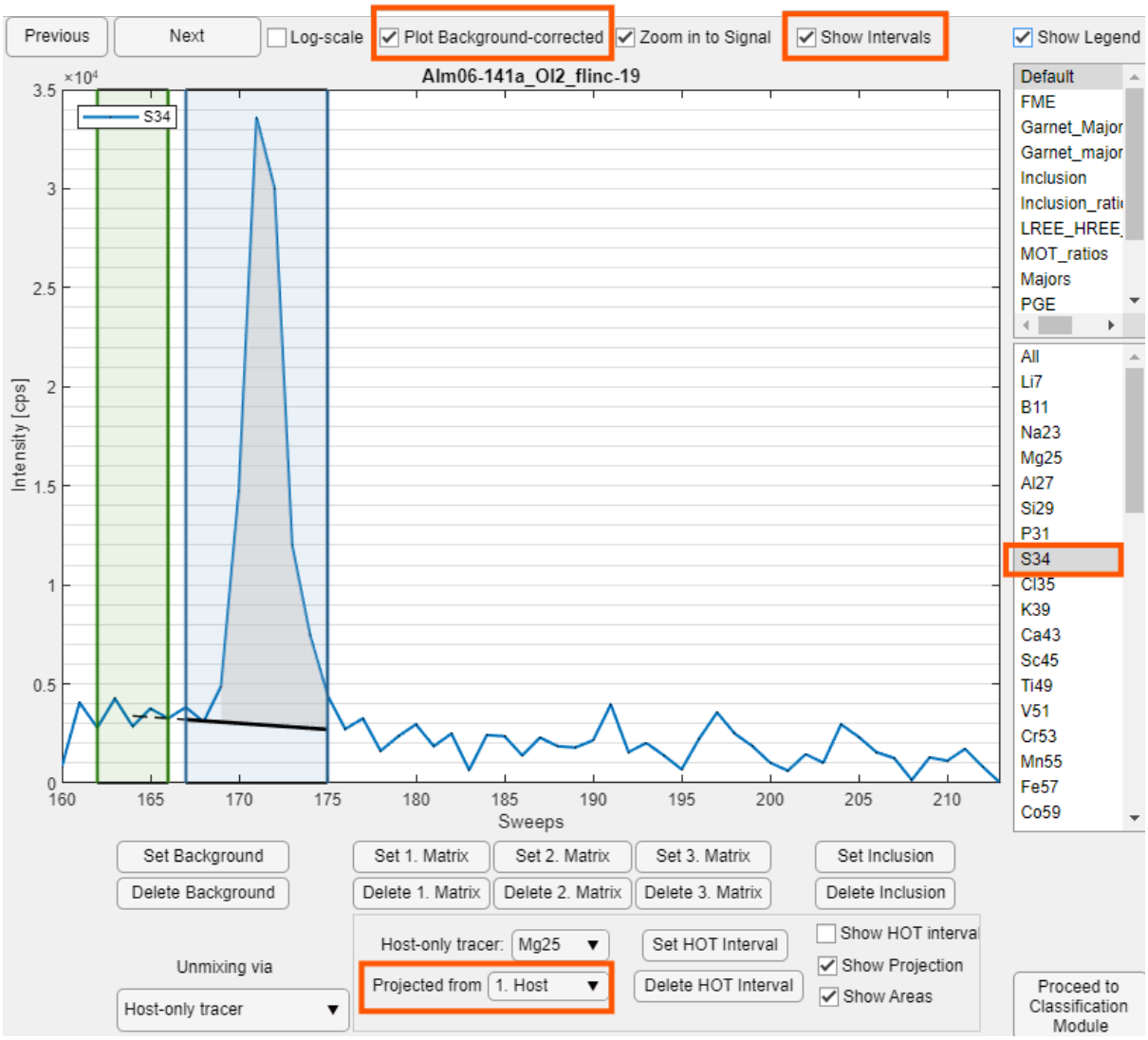


In this example, I used Mg25 which is a major element in the host olivine as the HOT. I then set the HOT interval over the whole length of the signal, completely disregarding the matrix and inclusion intervals as they don't matter for the HOT. The orange line visualizes a linear fit of the HOT cps over the whole interval.

The slope of this fit is then projected from the average of every given analyte within the chosen Host (Matrix) interval. This is already visualized as the dashed and thick black line. Of course, for Mg25 this makes no sense and the projection is useless (as it should be).

If we now look at e.g. S34 which is abundant in the inclusion (example below), we can see that the unmixing worked quite well! This means that the slope of Mg25 cps over time has approximately the same slope as S34 cps over time.

The unmixing itself works in exactly the same way as described in the baseline-correction above. It is just that the origin of the thick black line, which depicts the estimated host contribution to the mix, is based on the slope of the HOT and therefore is **<u>not</u>** independent of other analytes (in contrast to the baseline-correction).



It is still possible to change the host-only tracer by simply selecting a different analyte in the dropdown menu. You can also adjust the HOT interval to better fit your downhole slope.

Also for this unmixing technique, ICP-Base will unmix everything even if it is non-sensical. Therefore, I shall copy and paste the warning from the Baseline-correction chapter.

Since the software cannot know what makes sense and what does not, it is on you to check this! ICP-Base will quantify and sell such non-sensical inclusion data to you as if it were correct! **If you don't know what I am talking about right now, go and read the Baseline-correction chapter immediately and fire up the good ol' brain of yours!**

4.7. Quick guide

- Spike elimination
  - Works spot-wise
  - Press "Spike Elimination" button
  - Right-click on button and press "Apply globally"
    - Same is possible to reset it
  - No spikes are eliminating in the inclusion intervals by the "Spike Elimination" button or the "Apply globally" option
    - Spikes in the inclusion intervals can be manually accepted or ignored via right-click on the "Spike elimination" button and then selecting "Individual (Inclusion only)".
  - To manually accept or ignore all spikes select "Individual" after right-clicking the "Spike Elimination" button.
- Interval setting
  - Global Interval Setting
    - Changes the intervals globally
    - Depends on the signal start and end as detected in the Import Module
    - The numbers and ± 1, ±10 are the number of sweeps that are intend from the starts and ends
    - Also moves the intervals that were manually adjusted so watch out
  - Local Interval Setting
    - Set intervals by dragging and dropping the cursor over the plot
    - In Inclusion Mode there is also the "Inclusion" interval
  - Manually moving intervals
    - Click, hold, drag and drop the middle or borders of the intervals to adjust them
    - This just changes the one interval that was moved
- Signal unmixing
  - Only in Inclusion Mode

    - Only visible when plotting background-corrected
    - Baseline-correction
        - Default
        - Depends only on the analyte itself
    - Host-only tracer (HOT)
        - Depends on an analyte that is abundant in the host and not in the inclusion
        - Set a new interval (HOT interval)
            - If you don't understand where to put it, read the proper instructions
        - Choose HOT analyte
        - Choose which Host (Matrix) to project from
- Plot styles
    - Make them yourself
        - Navigate to ICP-Base folder
        - Get subfolder "Plot_Styles"
        - Based on .csv files
            - Open/Delete/Change the plot styles
            - Also able to plot ratios
    - Right-click on the top right list box
        - Press "Refresh" to load in new or changed plot style
- General visualization features
    - If the built-in zoom function is used (top-right icons in the plot) make sure to deselect the zoom function by clicking on it again. Otherwise you cannot move the intervals anymore.
    - Other features are self-explanatory

4.8. Proceeding to next module

After doing the spike correction, setting the intervals and doing the host-inclusion-unmixing you are good to go to the next module. Make sure to not have standalone background intervals (which have no matrix interval) and vice versa.

5. Classification Module

In this module, all you have to do is to tell ICP-Base which spots are what.

5.1. Types

There are four types that a spot can be: “Unknown”, “In-House Reference Material”, “Secondary Reference Material” and “Primary Reference Material”.

All spots are set to “Unknown” (UNK) by default. These are the spots about which nothing is known (as the name suggests).

An “In-House Reference Material” (IHRM) is basically an “Unknown”, just that it gets measured systematically so that the analyst can compare the results of this measurement session to the results of a past session. The “real” composition of an IHRM is not known, it is just used to be compared to itself to see relative differences.

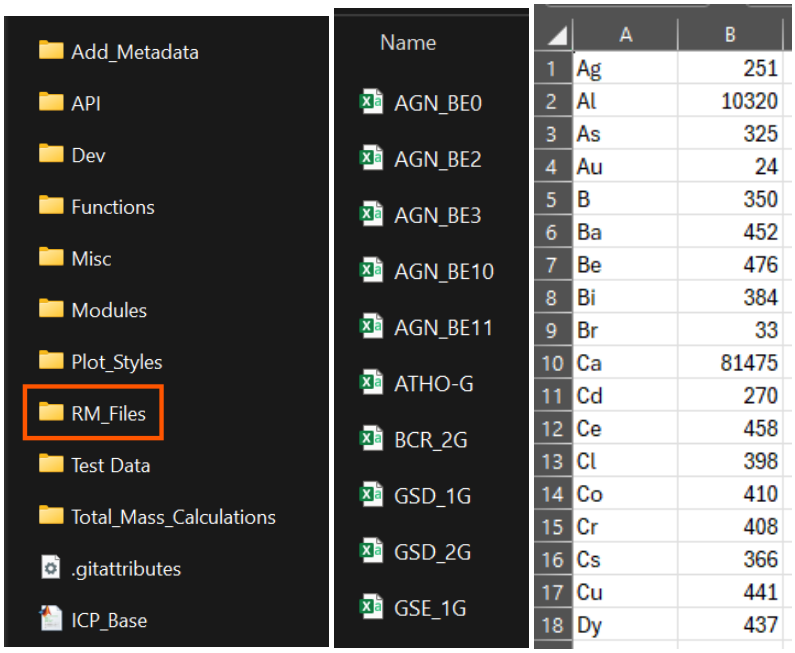
A “Secondary Reference Material” (SRM) is a material for which the “real” composition is known. Those are solely used for quality control and have no influence whatsoever on the actual quantification of the other spots. They are treated like “Unknowns” during the whole quantification process, just at the end you can compare the results with the “real” composition. I know that SRM can also stand for “Standard Reference Material” but this term is just wrong so I use it for “Secondary Reference Material” out of spite.

A ”Primary Reference Material” (PRM) can also be referred to as the “external standard”. The quantification of all other spots relies on them as they are used to calculate the sensitivity.

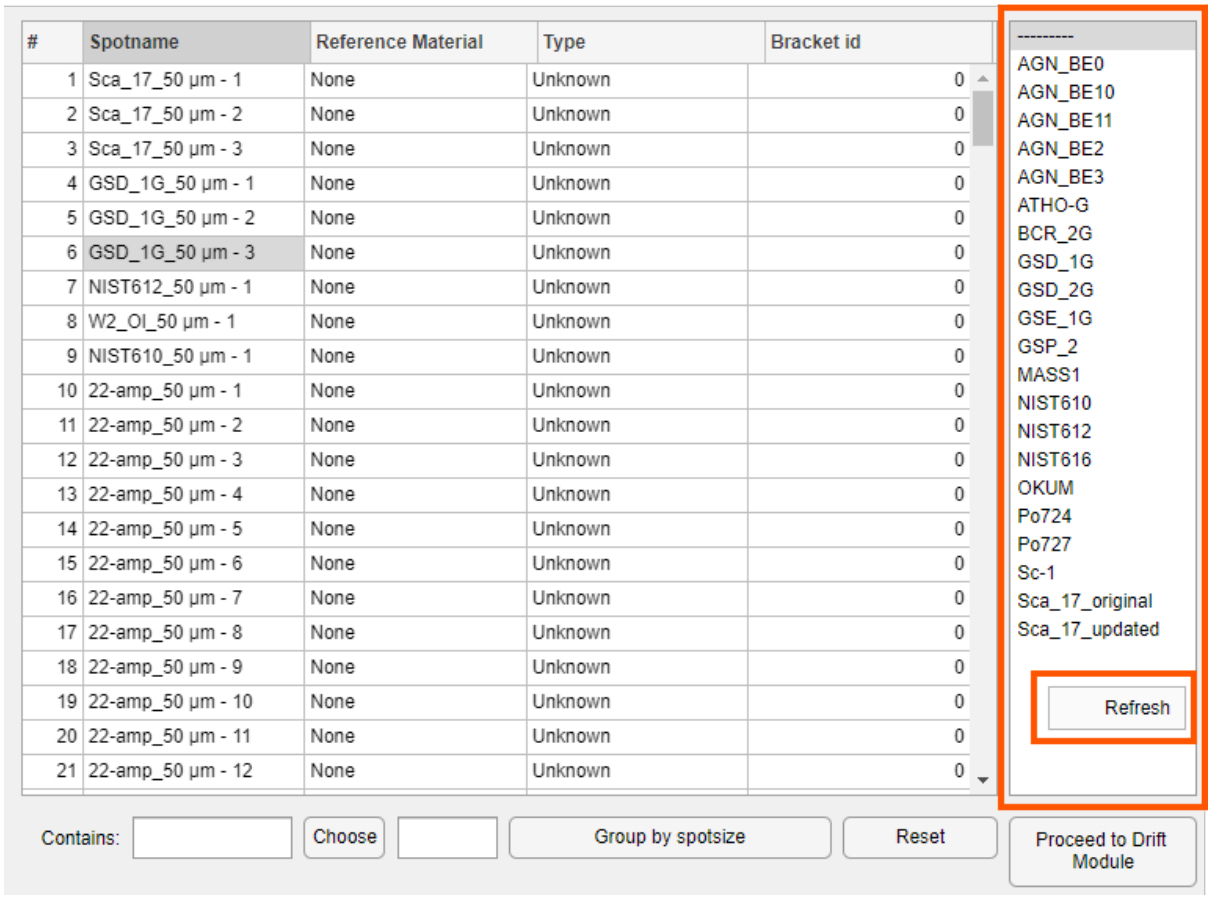
5.2. Reference materials

For UNK and IHRM this is of no matter since we don’t know the “real” composition. On the other hand, every spot classified as SRM and PRM needs to have an assigned reference material.

In the ICP-Base folder you will find a subfolder called RM_files which holds a bunch of .csv files that contain the composition of a given reference material in µg/g. You can add new ones or alter the existing ones as you like, just stay true to the format and don’t name the files with special characters, - or whitespaces.



These reference material files will then be imported to ICP-Base automatically when the Classification Module is launched. If you add a file later or alter one, right-click the list box and click “Refresh”.



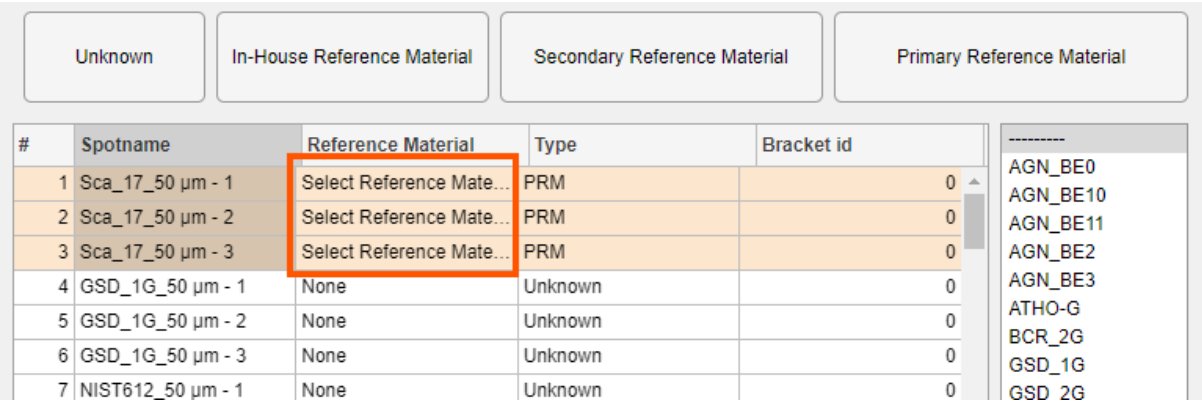
5.3. How to do the classification

5.3.1. Manually

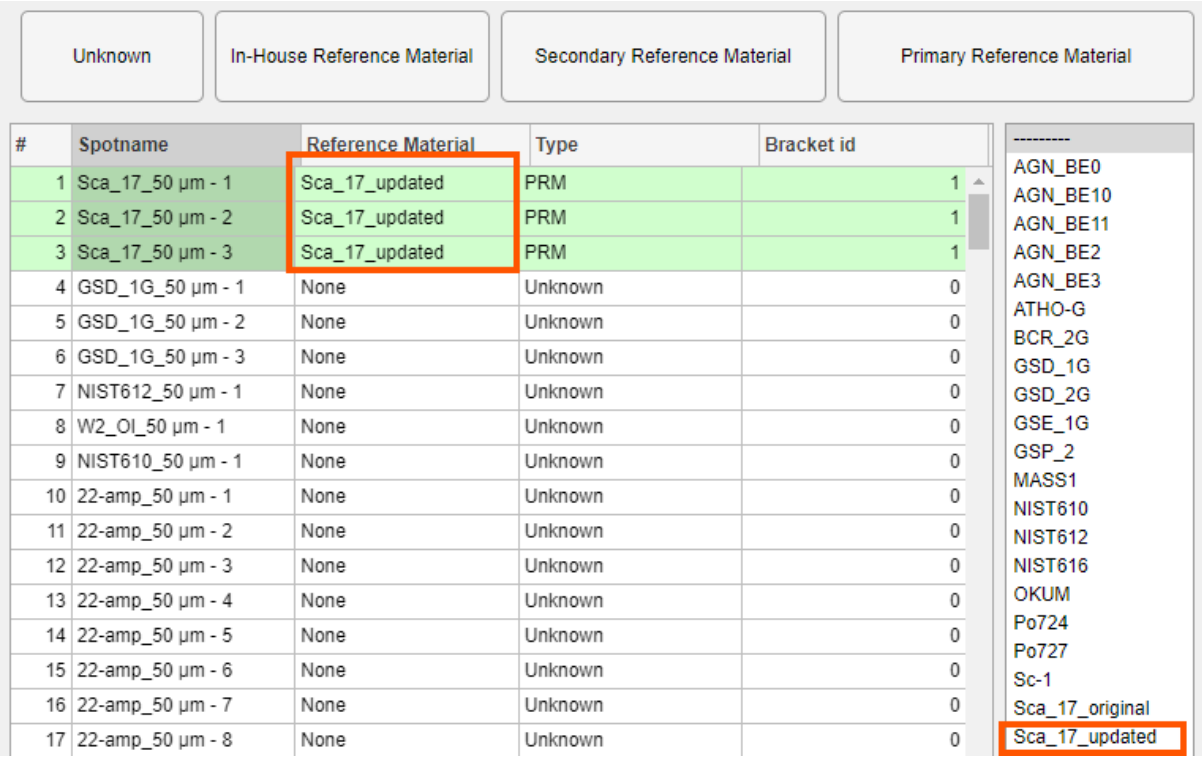
You can simply mark one or more cells of the table (any column) by clicking, holding and dragging the cursor over the table. Once you have selected those spots, press one of the buttons on top to select a type.

For UNK and IHRM you are done, there is no further information needed.

For SRM and PRM the rows will turn orange and you also have to assign a reference material.

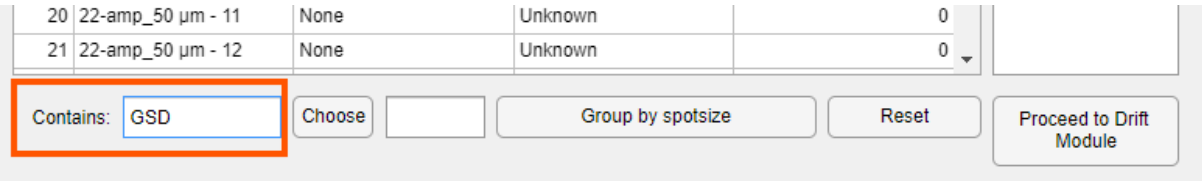


Simply click on the reference material that you want to allocate to those spots.

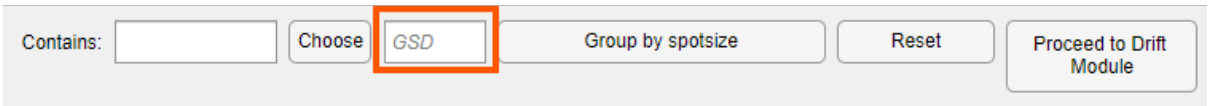


5.3.2. *Using the spot names*

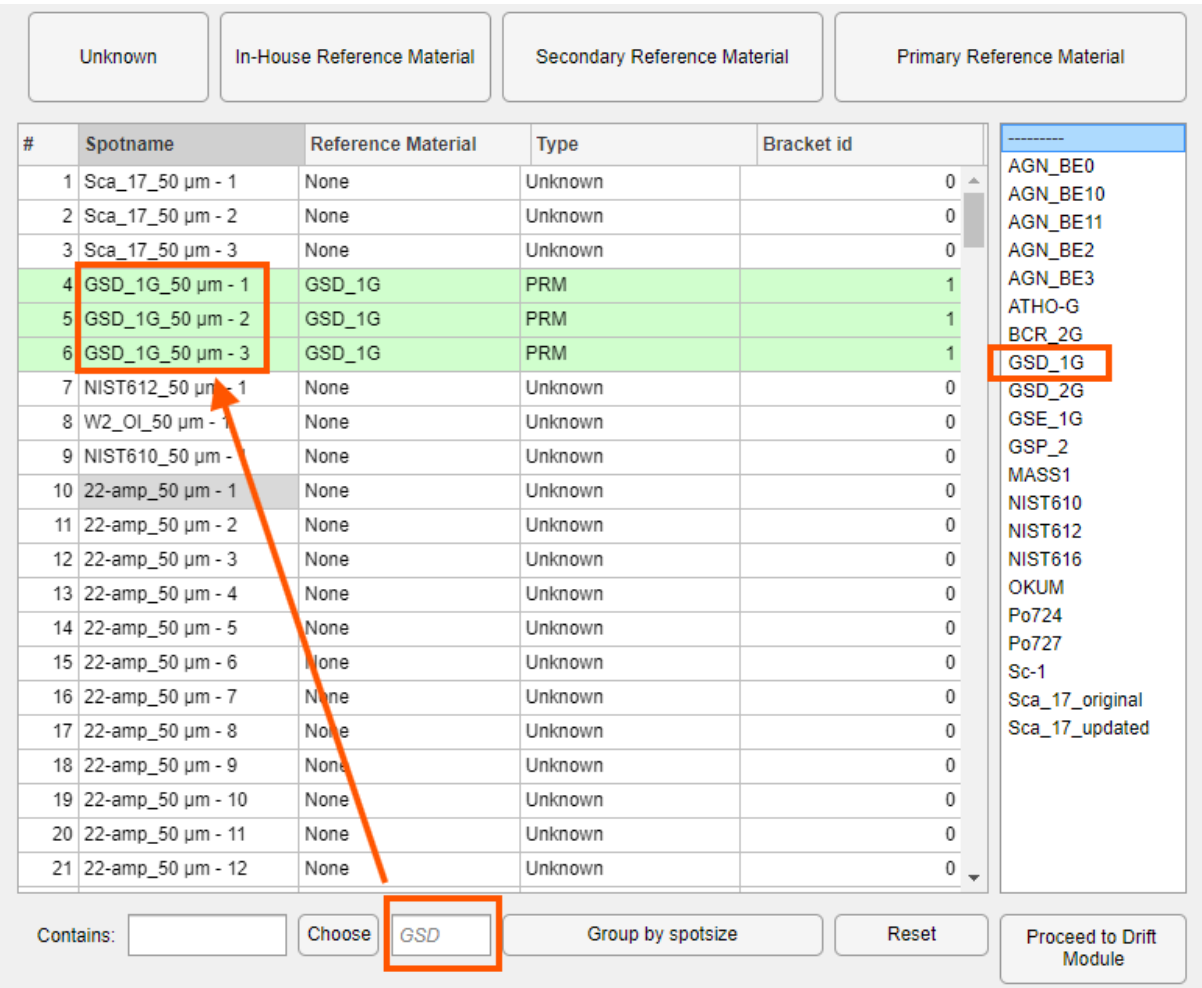
A more efficient way of allocating a high amount of spots quickly is by using the spot names. Type a string in the field on the bottom left and press the “Choose” button.



This will “lock” the string in



Now all spots that contain that string are selected (even though, unfortunately, it is not visible).



If you click anywhere in the table, the selection will be changed to the selection in the table and the grey string will vanish.

Below find an example of how it should look like. Also note the column “Bracket id” that we will discuss next.

#	Spotname	Reference Material	Type	Bracket id
1	Sca_17_50 µm - 1	Sca_17_updated	PRM	1
2	Sca_17_50 µm - 2	Sca_17_updated	PRM	1
3	Sca_17_50 µm - 3	Sca_17_updated	PRM	1
4	GSD_1G_50 µm - 1	GSD_1G	PRM	1
5	GSD_1G_50 µm - 2	GSD_1G	PRM	1
6	GSD_1G_50 µm - 3	GSD_1G	PRM	1
7	NIST612_50 µm - 1	NIST612	SRM	0
8	W2_OI_50 µm - 1	None	IHRM	0
9	NIST610_50 µm - 1	NIST610	SRM	0
10	22-amp_50 µm - 1	None	Unknown	0
11	22-amp_50 µm - 2	None	Unknown	0
12	22-amp_50 µm - 3	None	Unknown	0
13	22-amp_50 µm - 4	None	Unknown	0
14	22-amp_50 µm - 5	None	Unknown	0
15	22-amp_50 µm - 6	None	Unknown	0
16	22-amp_50 µm - 7	None	Unknown	0
17	22-amp_50 µm - 8	None	Unknown	0

5.4. Bracket id

First of all: Bracket ids are only important for PRMs, all other types will and should always have a bracket id of 0.

Secondly, what even is a bracket? A bracket is (loosely) defined as two or more PRM spots that enclose spots of all other types. All the UNK, IHRM and SRM type spots within the bracket will be quantified based on the sensitivity of the enclosing PRM spots. ICP-Base will take adjacent PRM spots of the same reference material and the same bracket id and average them together for drift correction. Why this is the case and what that means I explain in more detail in chapter 6. Drift Module.

However, if PRM spots of the same reference material were measured with different settings that leads to different cps signal data (e.g. different fluence or spot size) they have to be treated separately. If you ablate a e.g. NIST SRM 612 with 80 µm spot size will give you vastly more cps than ablating it with a 20 µm spot size. This will result in vastly different sensitivities. Averaging them together would be wrong.

In the example of my measurement session, the session starts with measuring all spots with 50 µm. Later in the session, smaller spots (38 µm) were analyzed and the PRMs were correspondingly also set to 38 µm. It is important that the PRM spots that were measured with 50 µm are not treated together with the PRM spots with 38 µm spot size.

#	Spotname	Reference Material	Type	Bracket id
167	19-amp_50 µm - 8	None	Unknown	0
168	19-amp_50 µm - 9	None	Unknown	0
169	19-amp_50 µm - 10	None	Unknown	0
170	Sca_17_50 µm - 19	Sca_17_updated	PRM	1
171	Sca_17_50 µm - 20	Sca_17_updated	PRM	1
172	Sca_17_50 µm - 21	Sca_17_updated	PRM	1
173	GSD_1G_50 µm - 19	GSD_1G	PRM	1
174	GSD_1G_50 µm - 20	GSD_1G	PRM	1
175	GSD_1G_50 µm - 21	GSD_1G	PRM	1
176	Sca_17_38 µm - 1	Sca_17_updated	PRM	1
177	Sca_17_38 µm - 2	Sca_17_updated	PRM	1
178	Sca_17_38 µm - 3	Sca_17_updated	PRM	1
179	GSD_1G_38 µm - 1	GSD_1G	PRM	1
180	GSD_1G_38 µm - 2	GSD_1G	PRM	1
181	GSD_1G_38 µm - 3	GSD_1G	PRM	1
182	22-amp_38 µm - 1	None	Unknown	0
183	22-amp_38 µm - 2	None	Unknown	0
184	22-amp_38 µm - 3	None	Unknown	0
185	22-amp_38 µm - 4	None	Unknown	0
186	22-amp_38 µm - 5	None	Unknown	0
187	22-amp_38 µm - 6	None	Unknown	0

So what we can and should do now, is to assign a different bracket id to the 38 µm PRM spots. This be can be done by manually selecting the cells and typing in e.g. “2” and pressing “Enter” on your keyboard. If laser metadata holding the spot sizes was imported, you can also press the “Group by spotsize” button.

#	Spotname	Reference Material	Type	Bracket id
167	19-amp_50 µm - 8	None	Unknown	0
168	19-amp_50 µm - 9	None	Unknown	0
169	19-amp_50 µm - 10	None	Unknown	0
170	Sca_17_50 µm - 19	Sca_17_updated	PRM	1
171	Sca_17_50 µm - 20	Sca_17_updated	PRM	1
172	Sca_17_50 µm - 21	Sca_17_updated	PRM	1
173	GSD_1G_50 µm - 19	GSD_1G	PRM	1
174	GSD_1G_50 µm - 20	GSD_1G	PRM	1
175	GSD_1G_50 µm - 21	GSD_1G	PRM	1
176	Sca_17_38 µm - 1	Sca_17_updated	PRM	2
177	Sca_17_38 µm - 2	Sca_17_updated	PRM	2
178	Sca_17_38 µm - 3	Sca_17_updated	PRM	2
179	GSD_1G_38 µm - 1	GSD_1G	PRM	2
180	GSD_1G_38 µm - 2	GSD_1G	PRM	2
181	GSD_1G_38 µm - 3	GSD_1G	PRM	2
182	22-amp_38 µm - 1	None	Unknown	0
183	22-amp_38 µm - 2	None	Unknown	0
184	22-amp_38 µm - 3	None	Unknown	0
185	22-amp_38 µm - 4	None	Unknown	0
186	22-amp_38 µm - 5	None	Unknown	0
187	22-amp_38 µm - 6	None	Unknown	0

Be aware that the Sca_17 spots and the GSD_1G spots can have the same bracket id because they are different reference materials, so there is no need to split them up using the id. **<u>Bracket ids should only be used to separate PRMs that are the same reference materials. They should not be used to separate different reference materials from each other!</u>** Since they are different reference materials, they are separated by this already.

If you measured PRMs with different spot sizes but you did not sort them, don't worry it does not matter for ICP-Base. Simply assign them the bracket ids as above. In this case, the sensitivity (see next chapter 6. Drift Module) will be calculated for all spots within that bracket. This means that the 80 µm spots could also be quantified using the 50 µm PRMs and vice versa. The UNK, IHRM and SRM always have bracket id 0, they are not explicitly assigned to a PRM. I explain this in more depth in chapter 7.1.1.

#	Spotname	Reference Material	Type	Bracket id
1	MASS1_50 - 1	MASS1	PRM	2
2	MASS1_50 - 2	MASS1	PRM	2
3	MASS1_80 - 1	MASS1	PRM	1
4	MASS1_80 - 2	MASS1	PRM	1
5	NIST610_50 - 1	NIST610	PRM	2
6	NIST610_80 - 1	NIST610	PRM	1
7	NIST612_50 - 1	NIST612	SRM	0
8	NIST612_80 - 1	NIST612	SRM	0
9	Po724_50 - 1	Po724	SRM	0
10	Po724_80 - 1	Po724	SRM	0
11	Po727_50 - 1	Po727	PRM	2
12	Po727_80 - 1	Po727	PRM	1
13	PS66-238-002_Sulf_80 - 1	None	Unknown	0
14	PS66-238-002_Sulf_50 - 1	None	Unknown	0
15	PS66-238-002_Sulf_50 - 1	None	Unknown	0
16	PS66-238-002_Sulf_50 - 2	None	Unknown	0
17	PS66-238-002_Sulf_80 - 1	None	Unknown	0
18	PS66-238-002_Sulf_80 - 1	None	Unknown	0
19	PS66-238-004_Sulf_50 - 1	None	Unknown	0
20	PS66-238-004_Sulf_50 - 1	None	Unknown	0
21	PS66-238-004_Sulf_50 - 2	None	Unknown	0
22	PS66-238-004_Sulf_80 - 1	None	Unknown	0

5.5. Quick guide

- Types
  - "Unknown" is an unknown (yes, really)

    - “In-House Reference Material” (IHRM) is an unknown that gets measured repeatedly to be compare to itself.
    - “Secondary Reference Material” (SRM) is treated like an unknown but has a known composition to which the results can be compared to.
    - “Primary Reference Material” (PRM) is actually used for the quantification of the other types.
- Reference Materials
    - Their compositions are stored in files in the subfolder RM_Files in the ICP-Base folder
    - Add/Change/Delete them as you like
        - Right-click on the listbox and press “Refresh” to load the new ones in.
- Assign types and reference materials to the spots by selecting them directly in the table or via string search
- Assign bracket ids if you measured the same reference material with e.g. different spot sizes to separate them.
    - Different reference materials do not have to be separated by the different bracket ids. They are already separated by the fact that they are different reference materials
    - Bracket ids only should be used to separate spots of the same reference material
    - If this is unclear to you, go to chapter 5.4. I tried my hardest there to explain it well

5.6. Proceeding to the next module

After setting all your types, reference materials and assigning the bracket ids, you can proceed to the next module.

6. Drift Module

In this module, the user cannot give any input in the software. The drift is calculated for you in a fixed manner. All you have to do is look and think!

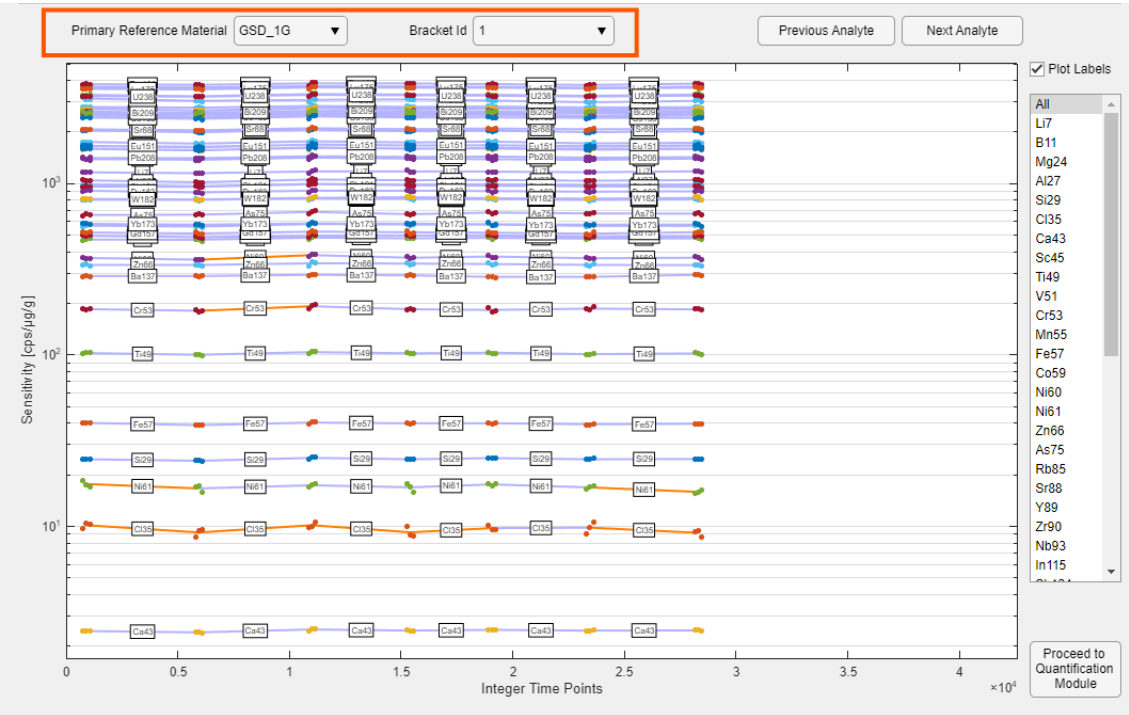
The PRMs are used to calculate the sensitivity of every analyte that was measured and is in the reference material file. e.g. for a NIST SRM 612, which contains 44 µg/g of Ti you get 1000 cps. The sensitivity is then 1000 cps divided by 44 µg/g equals a sensitivity of 22.72 counts-per-second-per-microgram-per-gram (cps/µg/g). ICP-Base will do this for every spot that is classified as a PRM. If the spots are adjacent to each other **and** of the same reference material **and** of the same bracket id, they will get averaged together.

ICP-Base will then fit a line from each average to the next average. In other words - for each bracket, a line will be fit from the bracket start to the bracket end.

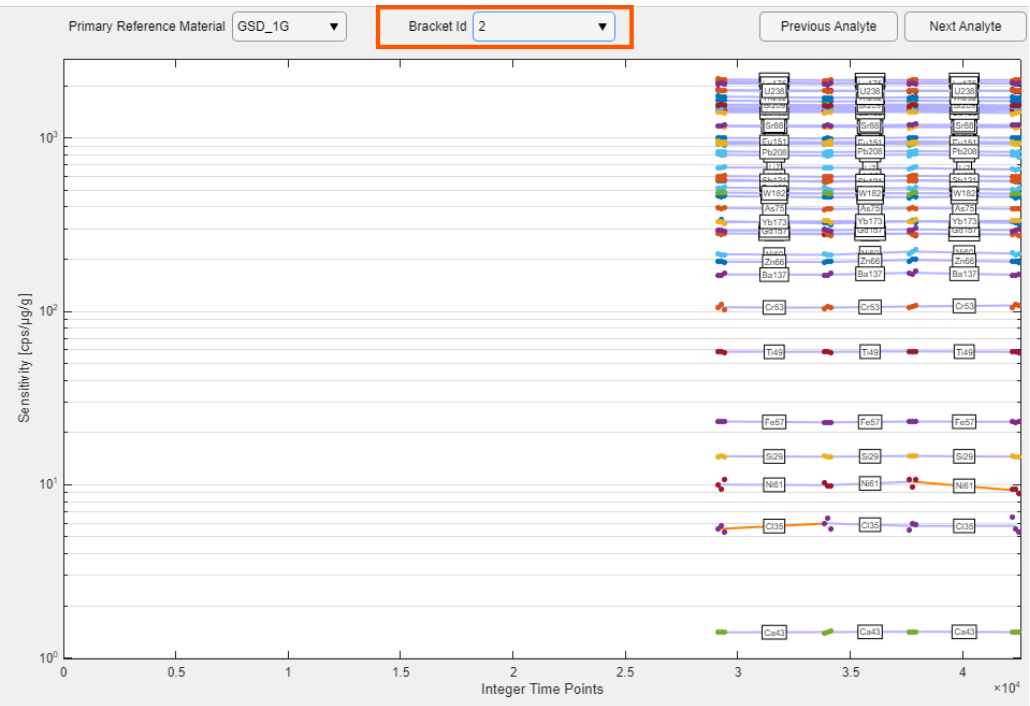
Time (horizontal axis) is expressed as "Integer Time Points" because they are much easier to handle during programming. The higher the integer time point, the more time has passed during your measurement session.

ICP-Base visualizes the changes of sensitivity of a reference material over the time of the whole measurement session. The small dots represent the actual measured and calculated sensitivities for the selected PRM and bracket id. The lines show a linear fit between the averages.

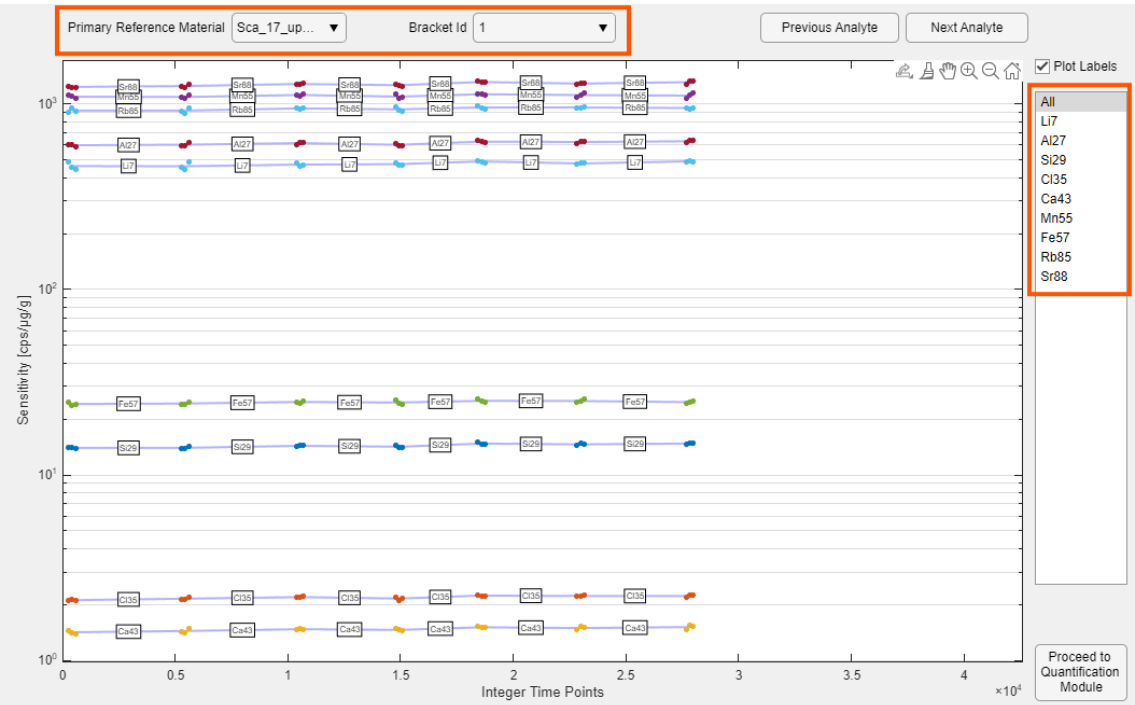
A blue line means that the sensitivity of a given analyte changed less than 5 % from the start of the bracket to the end of the bracket. Orange indicates a change of 5 to 10 % and red (not shown in the example) shows a change of more than 10 %.



In this example, we see the sensitivities of the GSD-1G reference material of bracket id “1” which corresponds to the 50 µm spot size. By changing the “Bracket id” dropdown to “2” we see the 38 µm spots. Of course, those are further back in time since they were analyzed at the end of the session. Note the drop of the sensitivity of one magnitude!



The same can be done for the PRMs. In the example below, we now visualized the Sca-17 scapolite reference material of 50 µm spot size. This reference material only contains nine analytes in its reference material file. Therefore, we can only use it to calculate the sensitivity of those nine analytes.



The UNK, IHRM and SRM spots lie between the PRM spots in time. ICP-Base gets the integer time points of them, fits them to the line and derives the sensitivity from them. This process is commonly referred to as “Drift correction”. In other words - we are correcting the drift of sensitivity for each UNK, IHRM and SRM over time based on a linear regression of the sensitivities of the PRMs.

6.1. Disregarding a PRM

If you now see that a certain PRM is just throwing off the sensitivity and you want to disregard it, you can go back to the Interval Module and delete all Intervals set for that PRM. See chapter 4.5.3.

6.2. Quick guide

- The plot visualizes the drift of the sensitivities of all analytes of a given primary reference material and bracket id pair
    - The dots are the sensitivities (vertical axis)
    - Horizontal axis is time
- Using the dropdown menus at the top you can navigate through the primary reference material and bracket id pairs.
- Line colors
    - A blue line means that the sensitivity of a given analyte changed less than 5 % from the start of the bracket to the end of the bracket.
    - Orange indicates a change of 5 to 10 %.
    - Red shows a change of more than 10 %.
- Just look at it and think, there is nothing to input

6.3. Proceeding to next module

Once you have looked at your drift, you can immediately proceed to the next module. There is nothing to do for you except look and think.

7. Quantification Module

In this module, the user is asked to input the internal standards. The internal standards play an important role in the quantification process and can be the deciding factor between "My data is sh*t" and "I have the best data ever".

ICP-Base features a few practical and novel quantification techniques due to the way it manages reference materials and the bracket ids.

7.1.1. Implications of using multiple PRMs and bracket ids

To explain the implication in its simplest case, lets focus on the first bracket of my example. The first six spots are all PRMs, spots 7 to 32 are UNK, IHRM and SRM, spots 33 to 38 are again PRMs. The PRMs are of the same bracket id but two different reference materials (Sca-17 and GSD-1G). This means that for spots 7 to 32, ICP-Base calculates the sensitivity twice – once for each reference material – bracket id pair (GSD-1G – id 1, Sca-17 – id 1)**.** This means that you can quantify these spots twice – once for each reference material – bracket id pair. In this example, the bracket ids don't matter because all PRMs in the first bracket have the same bracket id.

But if you were to not only use two different reference materials but also e.g. two different bracket ids for a given bracket, you could quantify the spots within that bracket four times! Once for each reference material – bracket id pair e.g. GSD-1G – id **1**, GSD-1G – id 2, Sca-17 – id **1**, Sca-17 – id **2**.

This implies, that a given spot is not strictly assigned to one reference material but that it can be quantified using multiple PRMs in parallel. This allows for direct comparison of the results of multiple PRMs (and bracket ids) without having to redo the data reduction.

Basically, we are splitting up the data reduction into multiple strands that run in parallel. These strands can also be run in sequence – and that is where it gets real spicy. But first, lets talk about the "normal" parallel quantification.

7.2. Internal standardization

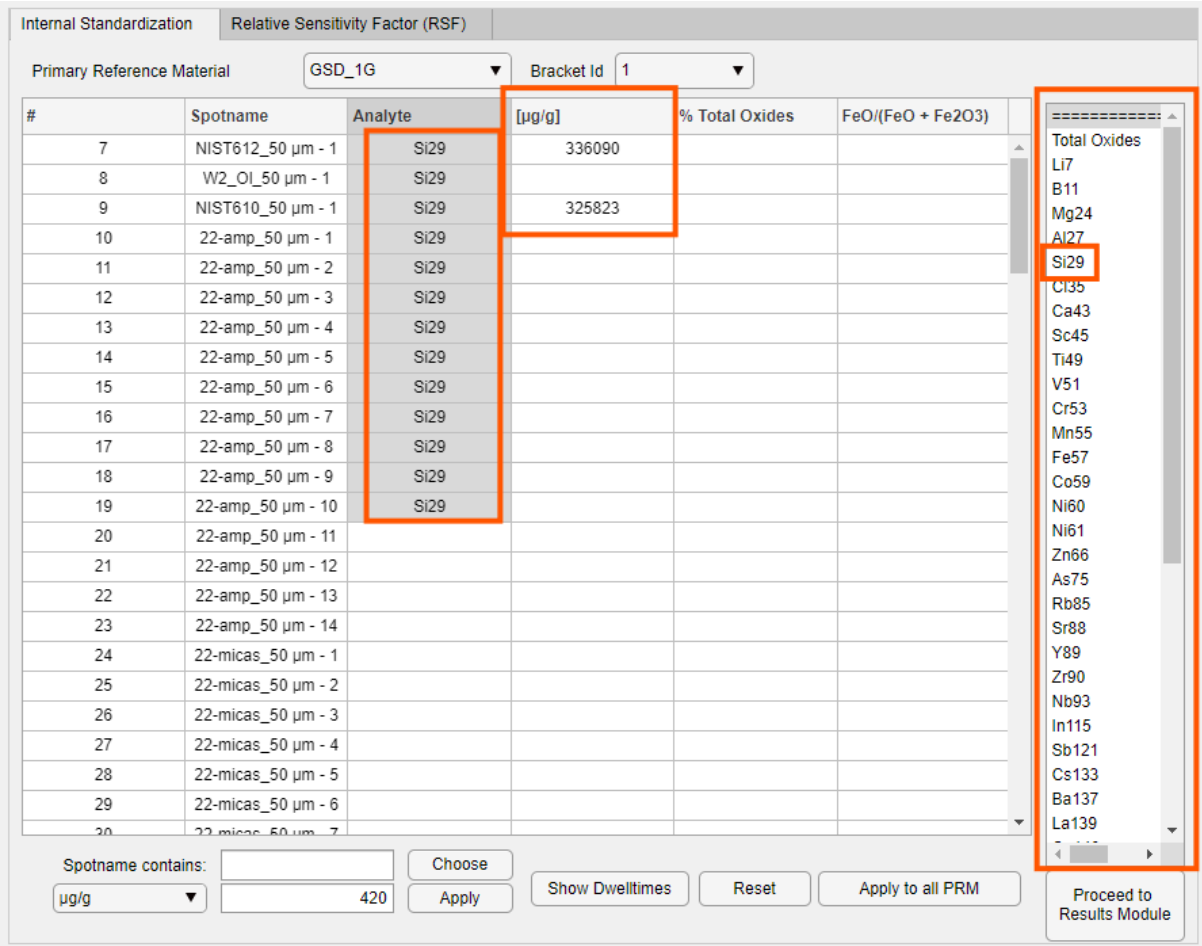
For quantification, ICP-Base lets you choose two types of internal standards – Analyte mass fractions in µg/g and a normalization to a given amount of total oxides. If you come from a different software and only have NaCl eq. data, don't worry it will also work (see chapter 7.2.4.1).

Generally explained, if you use an internal standard µg/g value of 10, you will get some results. If you change it to 20 µg/g, you will get exactly double of what you got with 10 and so on. So the internal standard just "shifts" the data to the right place. If you double the internal standard, you will "shift" it double and get double the µg/g.

7.2.1. Analyte [µg/g]

If you want to use an analyte as internal standard, click in the “Analyte” column. This makes the list of possible internal standards for a given reference material (GSD-1G in this case) pop up on the right hand side. Then click on the analyte of choice in this list box.

For the NIST612 and NIST610 spots, which are classified as SRM, ICP-Base knows the value for Si from the reference material file and automatically fills it in.



For the other spots, simply mark the cells of µg/g and type in the number and press “Enter” on your keyboard.

Primary Reference Material: GSD_1G Bracket Id: 1

#	Spotname	Analyte	[µg/g]	% Total Oxid
7	NIST612_50 µm - 1	Si29	336090	
8	W2_OI_50 µm - 1	Si29		
9	NIST610_50 µm - 1	Si29	325823	
10	22-amp_50 µm - 1	Si29	66666	
11	22-amp_50 µm - 2	Si29	66666	
12	22-amp_50 µm - 3	Si29	66666	
13	22-amp_50 µm - 4	Si29	66666	
14	22-amp_50 µm - 5	Si29	66666	
15	22-amp_50 µm - 6	Si29	66666	
16	22-amp_50 µm - 7	Si29	66666	
17	22-amp_50 µm - 8	Si29	66666	
18	22-amp_50 µm - 9	Si29	66666	
19	22-amp_50 µm - 10	Si29	66666	

If the Sca-17 PRM is chosen, the list contains only the nine analytes that are in the reference material file. Only these can be used to quantify the spots with Sca-17.

Internal Standardization | Relative Sensitivity Factor (RSF)

Primary Reference Material: Sca_17_updated Bracket Id: 1

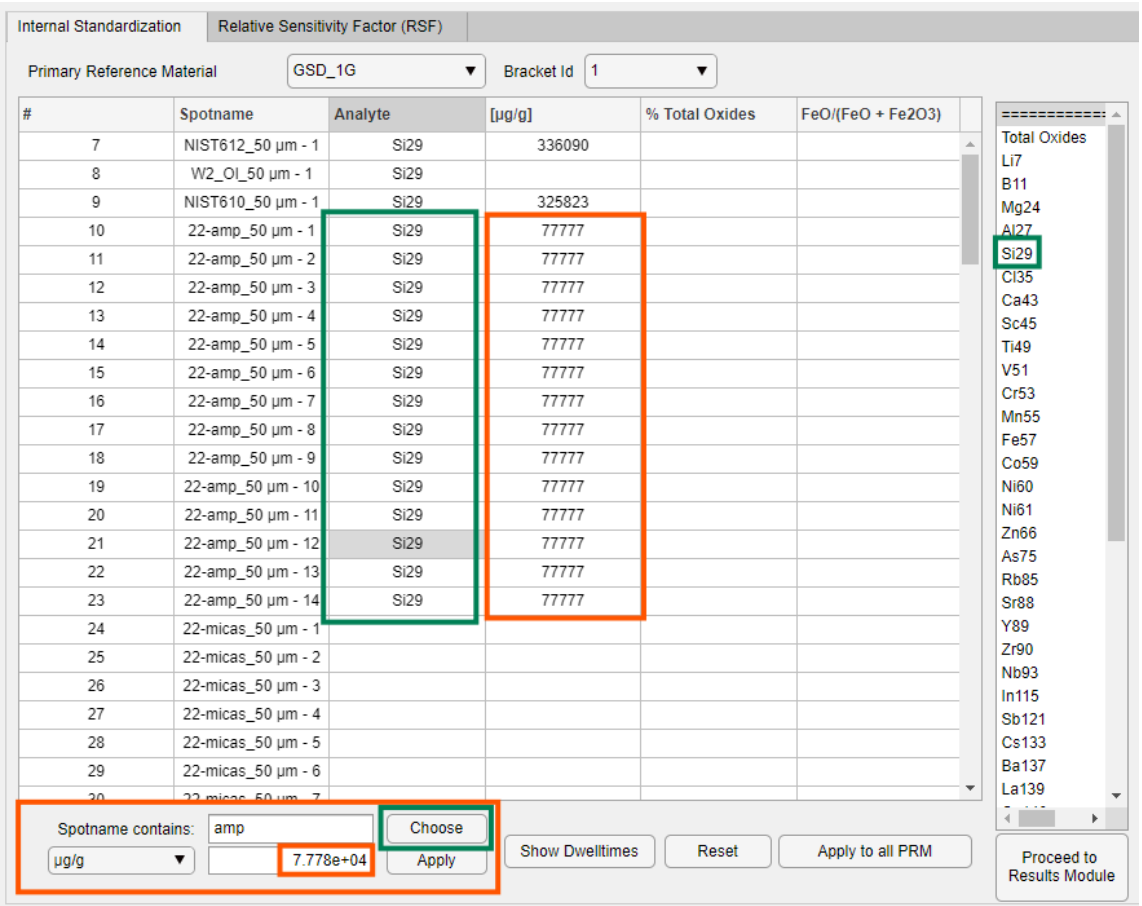
#	Spotname	Analyte	[µg/g]	% Total Oxides	FeO/(FeO + Fe2O3)
7	NIST612_50 µm - 1				
8	W2_OI_50 µm - 1				
9	NIST610_50 µm - 1				
10	22-amp_50 µm - 1				
11	22-amp_50 µm - 2				
12	22-amp_50 µm - 3				
13	22-amp_50 µm - 4				
14	22-amp_50 µm - 5				
15	22-amp_50 µm - 6				

==========
Total Oxides
Li7
Al27
Si29
Cl35
Ca43
Mn55
Fe57
Rb85
Sr88

You can also use a string search similar to the one in the Classification Module.

First, click in the “Analyte” column to make the list box holding the analytes pop up. Next, type in the string and press the “Choose” button. Now all spots containing that string are chosen. Then click on an analyte in the list box to select the internal standard element. It is also possible to use && (and) and || (or) for the string search. For example you could type “amp && micas” (which would select all spots containing “amp” and “micas”. Similarly it would work with “amp || micas” which selects all spots containing “amp” or “micas”. This is it though, no parentheses or other fancy stuff. Just two strings with either a && or a || in between them.

Next, select the µg/g option in the drop-down menu, type in the number and press the “Apply” button.

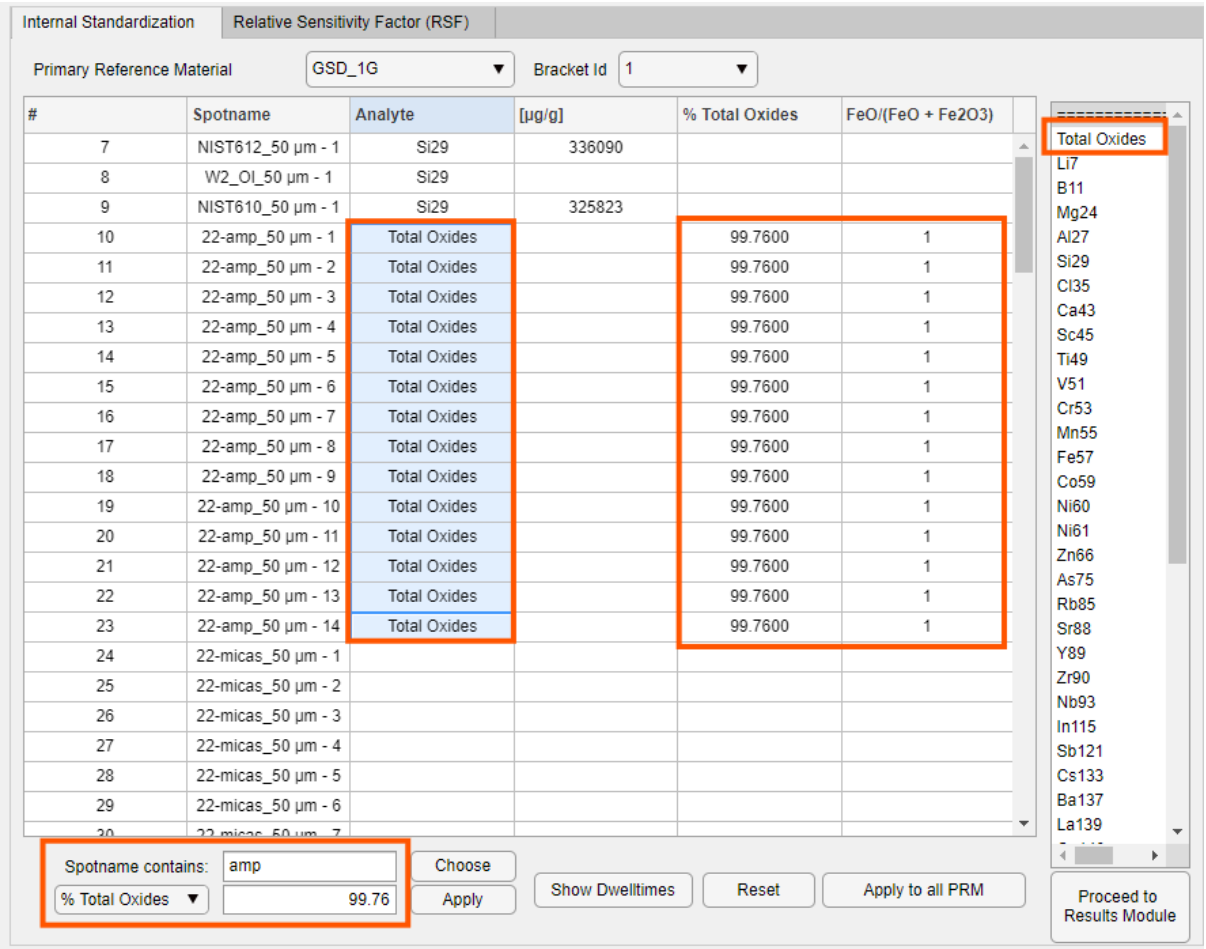


Once you filled in the internal standards, you can press the “Apply to all PRM” button to make the other PRMs have the same internal standards. This only works PRM-wise and not bracket-id wise.

<u>Make sure to fill in the internal standards for all PRM – bracket id pairs!</u> You can change them using the drop down menus above the table, similar to the Drift Module.

7.2.2. <u>Total Oxide</u>

If you don’t know an internal standard analyte mass fraction for your spot, you can also choose to quantify it using a normalization to a given amount of total oxides. You can either choose an analyte or Total Oxides. If Total Oxides is used, the “µg/g” column is emptied. You can navigate the GUI in the same manner as above. You can also use the string search and fill in the values by changing the dropdown menu to % Total Oxides.



#	Spotname	Analyte	[µg/g]	% Total Oxides	FeO/(FeO + Fe2O3)
7	NIST612_50 µm - 1	Si29	336090		
8	W2_OI_50 µm - 1	Si29			
9	NIST610_50 µm - 1	Si29	325823		
10	22-amp_50 µm - 1	Total Oxides		99.7600	1
11	22-amp_50 µm - 2	Total Oxides		99.7600	1
12	22-amp_50 µm - 3	Total Oxides		99.7600	1
13	22-amp_50 µm - 4	Total Oxides		99.7600	1
14	22-amp_50 µm - 5	Total Oxides		99.7600	1
15	22-amp_50 µm - 6	Total Oxides		99.7600	1
16	22-amp_50 µm - 7	Total Oxides		99.7600	1
17	22-amp_50 µm - 8	Total Oxides		99.7600	1
18	22-amp_50 µm - 9	Total Oxides		99.7600	1
19	22-amp_50 µm - 10	Total Oxides		99.7600	1
20	22-amp_50 µm - 11	Total Oxides		99.7600	1
21	22-amp_50 µm - 12	Total Oxides		99.7600	1
22	22-amp_50 µm - 13	Total Oxides		99.7600	1
23	22-amp_50 µm - 14	Total Oxides		99.7600	1
24	22-micas_50 µm - 1				
25	22-micas_50 µm - 2				
26	22-micas_50 µm - 3				
27	22-micas_50 µm - 4				
28	22-micas_50 µm - 5				
29	22-micas_50 µm - 6				

In the "% Total Oxides" column you can enter a value between 0 and 100, in the "FeO/(FeO + Fe2O3)" column you can enter values between 0 and 1. The total oxide value is 100 minus the volatile content of your mineral. If your mineral is volatile-free the total oxide is 100%. If your mineral has 4% volatiles, your % total oxide is 96. If you put 1 for the iron ratio, all Fe is considered to be ferric.

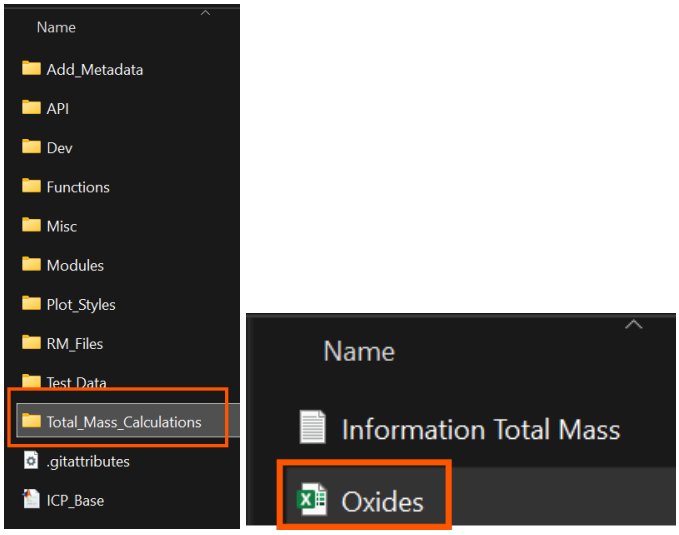
7.2.2.1. How it works

ICP-Base calculates preliminary mass fractions (µg/g) of all analytes for a spot. No internal standard was yet applied. Next, ICP-Base converts those preliminary mass fractions to wt % oxides (based on an oxide list that can be customized, see next chapter) and normalizes them to e.g. 99.76 % total oxide. After the normalization, it converts them back to element µg/g, which is the output you get.

Attention: If you use this feature with a reference material like Sca-17 which only contains 9 elements, only those 9 can be used for the normalization. The result will therefore not properly reflect the real composition (since so many elements are missing). E.g. If Sca-17 does not have Mg but you want to use the feature on an olivine, Mg is not considered in the normalization which will give you wrong results. **If you use the % total oxides feature, make sure to do it with a PRM that contains all the major elements of your sample and most of the minors and trace elements too!**

7.2.2.2. Customizing the oxide list

Similar to the RM_Files and the Plot_Styles, in the ICP-Base folder you find a subfolder called “Total_Mass_Calculations” in which is a Oxide.xlsx file.



ICP-Base will read in this Oxide.xlsx file automaticallly when the quantification module is launched (if it is named differently it is ignored).

This file holds a list of elements, their oxide names, conversion factor. These can be changed as needed (added/deleted/different speciation). There are more elements written in this file than there are probably measured. This is fine, ICP-Base ignores all the elements that were not analyzed.

The column Preferred_Isotope becomes important if you analyzed multiple isotopes of an element e.g. Ni61 and Ni62. If we would consider both for normalization, it would be wrong since the element is considered twice. Therefore, you can choose a preferred isotope which is then used for the total oxide calculation. If you don’t specify that, ICP-Base will take the first isotope it comes across e.g. Ni61 in this case.

The isotope that is not the preferred one will not be considered in the normalization but it will still be quantified (using dark magic).

In the last column, you can specify if an element should be considered as a “Major” or not. This has nothing to do with the quantification calculation, this is just for the Results Module where the majors can be displayed as oxides. In other words, this is just cosmetic.

	A	B	C	D	E
1	Name	Oxide_Name	Conversion_Factor	Preferred_Isotope	Major_Element
2	Ag	Ag2O	1.079		No
3	Al	Al2O3	1.889	Al27	Yes
4	As	As2O3	1.32		No
5	Au	Au2O3	1.271		No
6	B	B2O3	3.22		No
7	Ba	BaO	1.116		No
8	Be	BeO	2.785		No
9	Bi	Bi2O3	1.159		No
10	Br	Br	1	Br79	No
11	Ca	CaO	1.4	Ca43	Yes
12	Cd	CdO	1.143		No
13	Ce	CeO2	1.357		No
14	Cl	Cl	1	Cl35	No
15	Co	CoO	1.271		No
16	Cr	Cr2O3	1.462	Cr53	Yes

7.2.3. Sequential quantification

Up to this point, I only explained the case where all your PRM – bracket id pairs are quantified in parallel but as teased above, they can also be quantified in sequence. **This is only possible for spots that are bracketed by at least two different PRMs.**

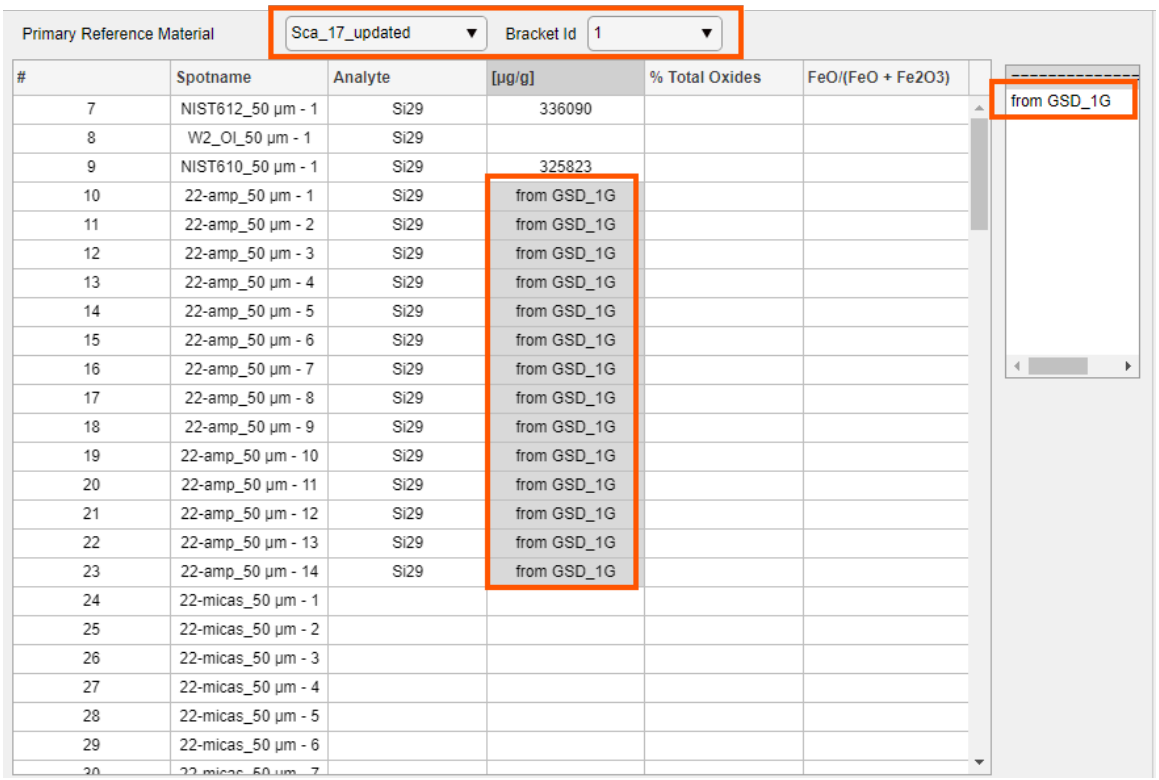
This is especially useful if your internal standard analyte that you want to use is only present in one of the PRMs. Example: The internal standard element that should be used (for some reason) is Ti49. This element is in the GSD-1G reference material but not in the Sca-17 PRM!

Here, the sequential quantification comes in. First set Ti49 for the GSD-1G as the internal standard.

Primary Reference Material: GSD_1G ▼ Bracket Id: 1 ▼

#	Spotname	Analyte	[µg/g]	% Total Oxides	FeO/(FeO + Fe2O3)
7	NIST612_50 µm - 1	Si29	336090		
8	W2_OI_50 µm - 1	Si29			
9	NIST610_50 µm - 1	Si29	325823		
10	22-amp_50 µm - 1	Ti49	123456		
11	22-amp_50 µm - 2	Ti49	123456		
12	22-amp_50 µm - 3	Ti49	123456		
13	22-amp_50 µm - 4	Ti49	123456		
14	22-amp_50 µm - 5	Ti49	123456		
15	22-amp_50 µm - 6	Ti49	123456		
16	22-amp_50 µm - 7	Ti49	123456		
17	22-amp_50 µm - 8	Ti49	123456		
18	22-amp_50 µm - 9	Ti49	123456		
19	22-amp_50 µm - 10	Ti49	123456		
20	22-amp_50 µm - 11	Ti49	123456		
21	22-amp_50 µm - 12	Ti49	123456		
22	22-amp_50 µm - 13	Ti49	123456		
23	22-amp_50 µm - 14	Ti49	123456		
24	22-micas_50 µm - 1				
25	22-micas_50 µm - 2				
26	22-micas_50 µm - 3				
27	22-micas_50 µm - 4				
28	22-micas_50 µm - 5				
29	22-micas_50 µm - 6				
30	22-micas_50 µm - 7				

Next, switch to the Sca_17 PRM with the dropdown menu above the table. For this PRM, choose one of the possible analytes as the internal standard. Then click in the “µg/g” column. On the right side of the table, a new list box will pop up. Since the selected spots are bracketed by both the GSD-1G and the Sca-17 PRM, you have the option to choose “from GSD_1G”.



#	Spotname	Analyte	[µg/g]	% Total Oxides	FeO/(FeO + Fe2O3)
7	NIST612_50 µm - 1	Si29	336090		
8	W2_OI_50 µm - 1	Si29			
9	NIST610_50 µm - 1	Si29	325823		
10	22-amp_50 µm - 1	Si29	from GSD_1G		
11	22-amp_50 µm - 2	Si29	from GSD_1G		
12	22-amp_50 µm - 3	Si29	from GSD_1G		
13	22-amp_50 µm - 4	Si29	from GSD_1G		
14	22-amp_50 µm - 5	Si29	from GSD_1G		
15	22-amp_50 µm - 6	Si29	from GSD_1G		
16	22-amp_50 µm - 7	Si29	from GSD_1G		
17	22-amp_50 µm - 8	Si29	from GSD_1G		
18	22-amp_50 µm - 9	Si29	from GSD_1G		
19	22-amp_50 µm - 10	Si29	from GSD_1G		
20	22-amp_50 µm - 11	Si29	from GSD_1G		
21	22-amp_50 µm - 12	Si29	from GSD_1G		
22	22-amp_50 µm - 13	Si29	from GSD_1G		
23	22-amp_50 µm - 14	Si29	from GSD_1G		
24	22-micas_50 µm - 1				
25	22-micas_50 µm - 2				
26	22-micas_50 µm - 3				
27	22-micas_50 µm - 4				
28	22-micas_50 µm - 5				
29	22-micas_50 µm - 6				

ICP-Base will first quantify the spots using the GSD-1G reference material and Ti49 as the internal standard. Once that is done, it will quantify the same spot using the Sca-17 reference material with Si29, as quantified by the GSD-1G, as the internal standard.

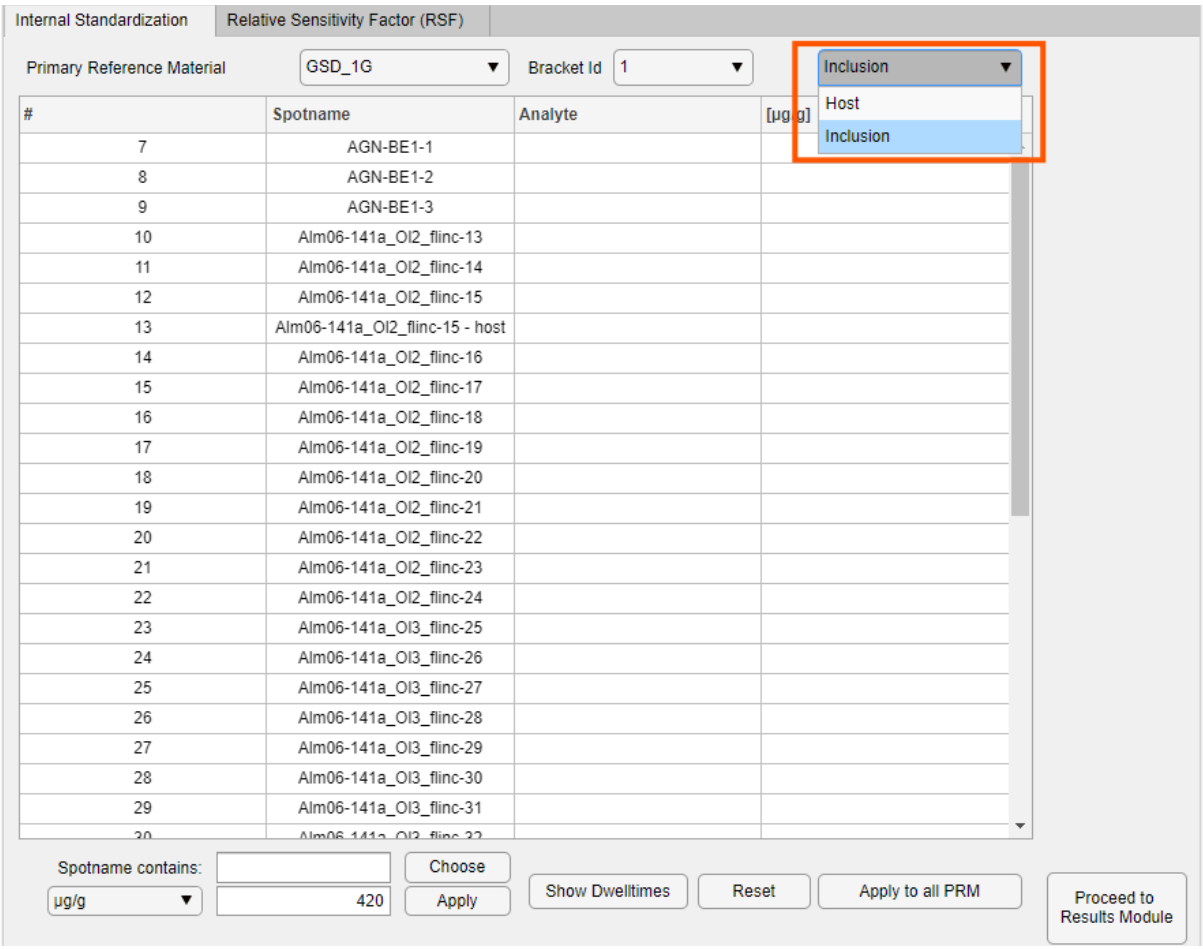
So we do the quantification for one chosen reference material first and then use one of the analytes that was quantified as the internal standard for another reference material. This can be done with as many PRMs as wished and in any order. There has to be a start somewhere of course (if you put in “from XY” for all of them its not gonna work, obviously). Similarly, if your first PRM does not quantify the analyte that the other PRM should use, it will, obviously, also not work.

If you have simply no clue about any internal standard element for a given spot, you can start this sequential quantification by using % Total Oxides. And then getting e.g. Si29 form that quantification for all other PRMs.

This feature also works PRM-wise and not bracket id-wise. You can do it separately for all bracket ids though.

7.2.4. Inclusion quantification

When it comes to the quantification of inclusion, everything works pretty much the same as described above. One of the few differences is that the % Total Oxide feature cannot be used for the quantification of inclusions (it would be wrong doing so for many reasons). To set the internal standard for the inclusions, use the dropdown menu and go to "Inclusion" (only visible in Inclusion Mode, that's why it is not there in the above explanations).



If you leave the dropdown menu on "Host" you will quantify the host, and not the inclusion.

You can also use the sequential quantification feature. This feature works within the "Host" and "Inclusion" – so it is not possible to use the host somehow to quantify the inclusion, they are separate.

7.2.4.1. NaCl salinity equivalent (wt%)

If you have fluid inclusion, you might only know the NaCl salinity equivalent. In sills, it was possible to input this directly but we decided against that since the conversion of NaCl eq. (wt%) to Na (µg/g) is pretty straightforward.

To convert sodium chloride (NaCl) weight percent (wt%) to sodium (Na) concentration in micrograms per gram (µg/g), we have to understand that 1 wt% means 1 gram of NaCl is present in 100 grams of the total sample. Since Na makes up only part of the NaCl molecule, we can use the ratio of the molar mass of sodium (22.99 g/mol) to that of NaCl (58.44 g/mol), which is approximately 0.3934. This means that for every gram of NaCl, there are about 0.3934 grams of Na. To find the Na content per gram of sample, divide the 0.3934 grams of Na by 100 grams of sample, giving 0.003934 grams of Na per gram of sample. Finally, convert grams to micrograms by multiplying by 10^6 resulting in 3934 µg of Na per gram of sample.

Therefore, 1 wt% NaCl is equivalent to 3934 µg/g Na. Equally, 2 wt% NaCl is equivalent to 2 x 3934 µg/g = 7868 µg/g Na and so on. Just take whatever wt% NaCl eq. you have and multiply it by 3934 to get the Na mass fraction in µg/g. You can use this value and Na as your internal standard.

Personal recommendation since this came up once at a conference: Imagine you have e.g. 20 inclusions of the same generation and you do Raman for all of them to get the NaCl eq. and later you analyze exactly the same inclusions with LA-ICP-MS. The NaCl equivalent is different for every inclusion and ranges from, lets say, 2 to 10 wt%. What you could do, and what I **do not** recommend, is to use the exact NaCl eq. that you have obtained for the inclusion as the internal standard. In this case, one inclusion would have the internal standard of 2 wt%, the other one of 4% and another one of 10 wt%. This means that that the second inclusion, even though from the same generation and probably of same composition, will yield results double of the first inclusion. The third inclusion of this example will yield 5x the results of the first inclusion. In other words, the spread of your data will be massive! Since we are talking about generations, in my limited knowledge about NaCl eq., I think it is fair to use the average of all your salinities. If you then use this average NaCl eq. for all inclusions (which I **do** recommend), you "shift" them equally and the spread of your data will (probably) be much lower since you are not introducing it yourself. As already stated, my experience with NaCl eq. is limited. This paragraph is written more out of a LA-ICP-MS point of view.

7.2.5. Ignoring and disregarding spots

If you leave a PRM – bracket id pair empty or just don't input any internal standard for the host, ICP-Base will simply ignore it so it is no problem. Generally, cells that are left empty or do not have all the information (e.g. Si in the "Analyte" column but no value in the "µg/g" column) are ignored.

A quick way of deleting multiple entries of the "µg/g" column is to simply type few letters and press "Enter" on your keyboard. ICP-Base will recognize the letters as wrong input and make the cell empty.

Once an analyte was chosen in the “Analyte” column you cannot get rid of it, just let it be and ignore it.

7.3. Relative Sensitivity Factor (RSF)

The relative sensitivity factor (RSF) can be used to assess the best internal standard for a secondary reference material and to visualize matrix effects. To use it, navigate to the corresponding Tab in the Quantification Module. This is an optional feature that is not mandatory to use, you can also just ignore it if you want.

Most of the times, it is desirable to have secondary reference materials that are very well reproduced (if e.g. my SRM has 44 µg/g Ti according to the reference material file and we measure 44 µg/g, we have absolutely perfectly reproduced it) since this is a sign that the measurement session was successful and my data is of good quality.

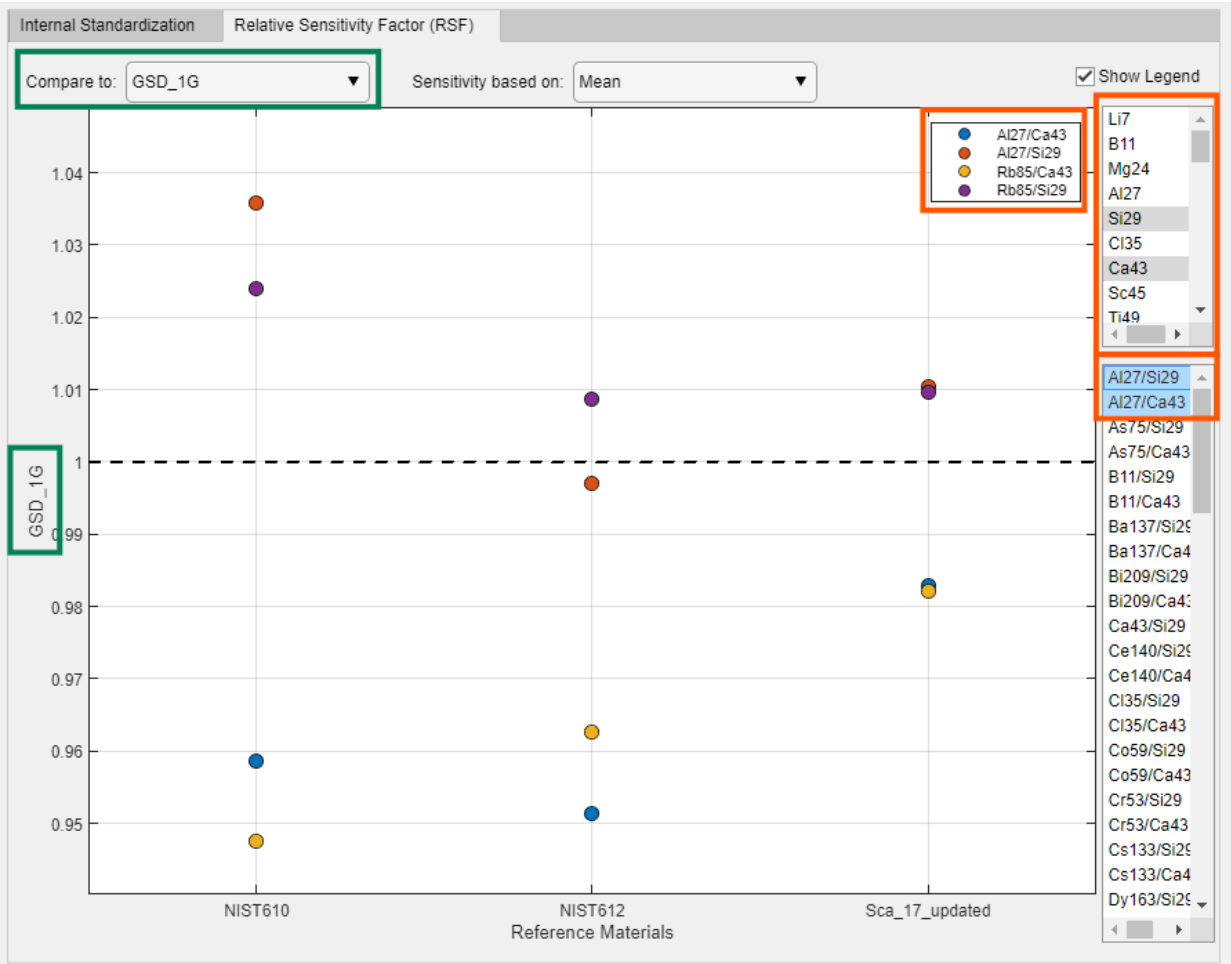
The RSF is a way of testing how well a PRM, using a specific internal standard, can reproduce a given SRM. This is all just purely based on sensitivities that are calculated from the reference material files and the measured cps. There is no sensical user input that can greatly influence this.

The RSF is also a measure of how well the ablated matrices are matching each other. If the matrices are matching, the RSF is probably very close to 1, if they are not matching at all or there is some weird matrix effect the RSF will be very high or very low but nowhere near 1. This can be used to explain (or at least get a hint) of the magnitude of the matrix effects that make your data bad.

If the reference material is very badly characterized, the RSF will also be worse and/or less meaningful!

7.3.1. How it works

In the upper list box on the right hand side, you can choose elements that you are interested in using as internal standards. In my example, I am want to use Si or Ca as the internal standard for quantification of my unknowns. I select those two analytes by holding “ctrl” on the keyboard and clicking on them. This will fill the lower list box with all the possible ratios. Now it is possible to select those ratios and they will be plotted. By changing the “Compare to” dropdown menu, you can adjust the vertical axis of the plot. It will always plot all possible reference materials.



7.3.2. *What it means*

Lets look at the example of Al/Si in the above example. First, the $Sensitivity_{Al27}/Sensitivity_{Si29}$ is calculated for each reference material. Next, the sensitivity ratio of the reference materials on the horizontal axis are ratioed by the sensitivity ratio of the reference material on the vertical axis. So we get e.g. $({}^{NIST610}Sensitivity_{Al27}/{}^{NIST610}Sensitivity_{Si29})$ / $({}^{GSD_1G}Sensitivity_{Al27}/{}^{GSD_1G}Sensitivity_{Si29})$. The closer this ratio is to 1 the better. If the ratio is e.g. 0.96 this PRM (vertical axis) will underestimate this SRM (horizontal axis) for 4%. Analogous, if the ratio is e.g. 1.10, the PRM (vertical axis) will overestimate the SRM (horizontal axis) for 10%.

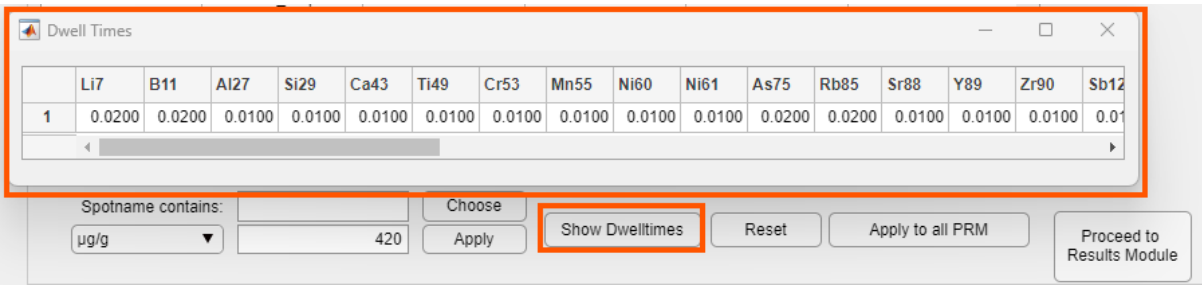
7.3.3. *How we can use it*

In the above example, I am interested in using Si or Ca as an internal standard and (for some reason) I am interested in getting good analyses of Al and Rb. So I selected those elements. What is now visible on the plot is that generally, if we use Si29 as the internal standard, we will overestimate the SRMs whereas with Ca43, we will underestimate. Since relatively, we overestimate less (below 4%) by using Si29, I would use Si29 as the internal standard.

You can just play around with this and explore your data. For some reference material pairs you will see systematic discrepancies and for some perfect matches. This is just a visualization at the end so don't be scared to play around with it.

7.4. Adjusting the dwell times

The dwell times are important for the calculation of the limits of detection. If you did not import ICP-MS metadata holding the dwell times, press the "Show Dwelltimes" button and change them accordingly. Now change the value (in seconds) to whatever you want and press "Enter" on your keyboard. Then close the Dwell Times table using the x and done.



7.5. Quick guide

- In this module the internal standards are assigned to the spots
- Either a single analyte can be chosen or % Total Oxides
- Assign them using a direct selection on the table or string search
    - String search can also do && (and) and || (or)
        - E.g. "amp && micas" will select all spots containing "amp" and "micas"
- Make sure to do this for all PRM – bracket id pairs
    - If you don't, nothing bad happens, ICP-Base will just not quantify it
- Inclusion Quantification
    - Choose "Inclusion" or "Host" in the dropdown menu top right and enter you internal standards.
    - No % Total Oxide possible
    - Everything else works pretty much the same
- Sequential quantification
    - First input the internal standards for the first PRM (let's call it XY) that should be used for quantification
    - If a spot is bracketed by multiple PRMs, you can go to another PRM (let's call it Z), select an analyte (let's say Si29) and then click on the "µg/g" column and select "from XY"
    - This will make ICP-Base first quantify the spots using PRM XY. Then the result of Si29 of that quantification will be used as the internal standard for the quantification of the same spots using PRM Z.

    - This can be done with multiple standards
    - It hard to explain this shortly, if this is unclear go read chapter 7.2.3
- Adjust dwell times if necessary
    - Open the dwell times using the “Show Dwelltimes” button
    - Change the value (in seconds)
    - Close table, done
- Relative Sensitivity Factor (RSF)
    - Not mandatory to use
    - Ratios the sensitivities of analytes of different PRMs and SRMs and compares them to each other by ratioing those ratios again.
        - If this ratio lands at 1, you have no matrix effect and the e.g. SRM will be perfectly reproduced
        - If the ratio lands at e.g. 1.10, you SRM will be about 10% off with that internal standard.
    - Gives you an idea about:
        - Which internal standard to use to best reproduce a secondary reference material
        - Matrix effects
        - Why your data is bad
        - Why your data is good
    - Hard to explain shortly, if this is unclear go read chapter 7.3

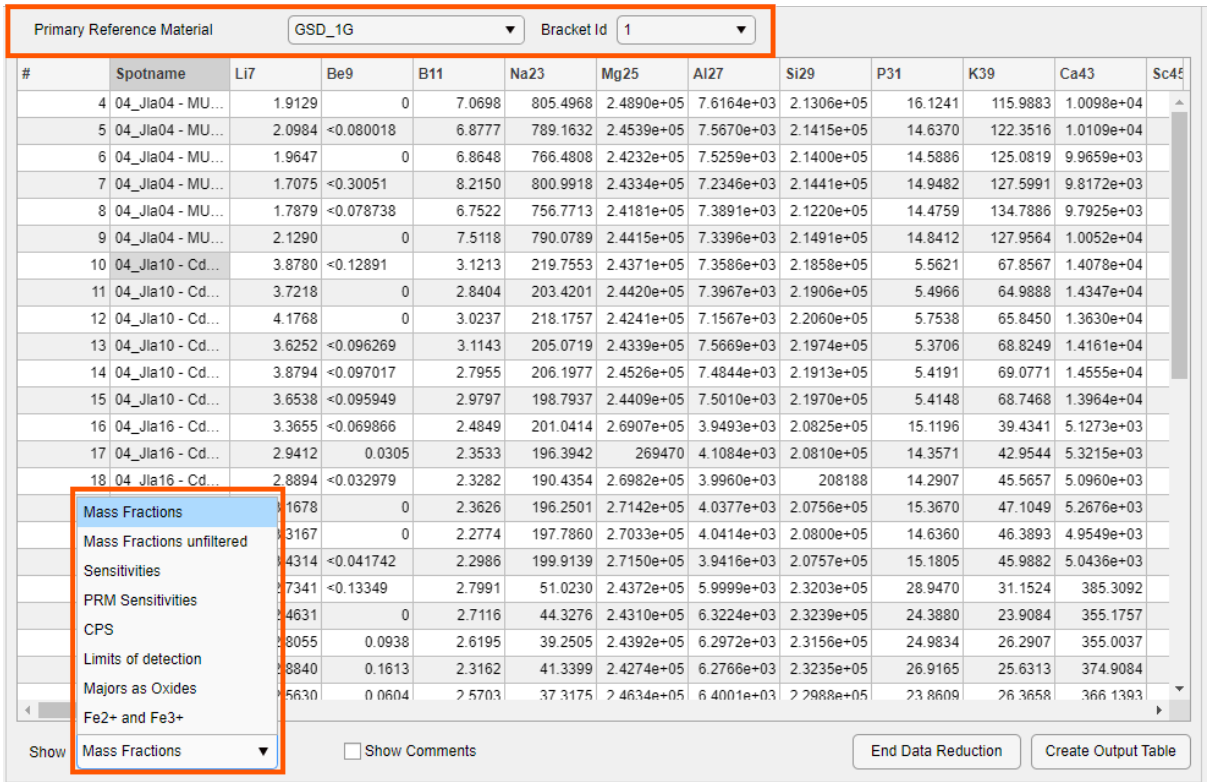
7.6. Proceeding to next module

Once you set all your internal standards, you are ready to proceed. The quantification will trigger once you press the “Proceed to Results Module” button. Depending on how much spots you want to quantify, this can take a few seconds.

8. Results Module

This module will now show your results in µg/g. Again, this will show the results of a given PRM – bracket id pair as selected in the dropdown menu.

8.1. Raw Results

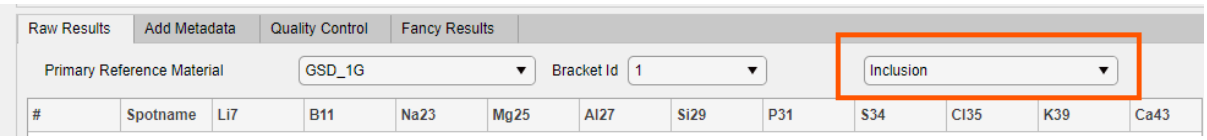


#	Spotname	Li7	Be9	B11	Na23	Mg25	Al27	Si29	P31	K39	Ca43	Sc45
4	04_Jla04 - MU...	1.9129	0	7.0698	805.4968	2.4890e+05	7.6164e+03	2.1306e+05	16.1241	115.9883	1.0098e+04	
5	04_Jla04 - MU...	2.0984	<0.080018	6.8777	789.1632	2.4539e+05	7.5670e+03	2.1415e+05	14.6370	122.3516	1.0109e+04	
6	04_Jla04 - MU...	1.9647	0	6.8648	766.4808	2.4232e+05	7.5259e+03	2.1400e+05	14.5886	125.0819	9.9659e+03	
7	04_Jla04 - MU...	1.7075	<0.30051	8.2150	800.9918	2.4334e+05	7.2346e+03	2.1441e+05	14.9482	127.5991	9.8172e+03	
8	04_Jla04 - MU...	1.7879	<0.078738	6.7522	756.7713	2.4181e+05	7.3891e+03	2.1220e+05	14.4759	134.7886	9.7925e+03	
9	04_Jla04 - MU...	2.1290	0	7.5118	790.0789	2.4415e+05	7.3396e+03	2.1491e+05	14.8412	127.9564	1.0052e+04	
10	04_Jla10 - Cd...	3.8780	<0.12891	3.1213	219.7553	2.4371e+05	7.3586e+03	2.1858e+05	5.5621	67.8567	1.4078e+04	
11	04_Jla10 - Cd...	3.7218	0	2.8404	203.4201	2.4420e+05	7.3967e+03	2.1906e+05	5.4966	64.9888	1.4347e+04	
12	04_Jla10 - Cd...	4.1768	0	3.0237	218.1757	2.4241e+05	7.1567e+03	2.2060e+05	5.7538	65.8450	1.3630e+04	
13	04_Jla10 - Cd...	3.6252	<0.096269	3.1143	205.0719	2.4339e+05	7.5669e+03	2.1974e+05	5.3706	68.8249	1.4161e+04	
14	04_Jla10 - Cd...	3.8794	<0.097017	2.7955	206.1977	2.4526e+05	7.4844e+03	2.1913e+05	5.4191	69.0771	1.4555e+04	
15	04_Jla10 - Cd...	3.6538	<0.095949	2.9797	198.7937	2.4409e+05	7.5010e+03	2.1970e+05	5.4148	68.7468	1.3964e+04	
16	04_Jla16 - Cd...	3.3655	<0.069866	2.4849	201.0414	2.6907e+05	3.9493e+03	2.0825e+05	15.1196	39.4341	5.1273e+03	
17	04_Jla16 - Cd...	2.9412	0.0305	2.3533	196.3942	269470	4.1084e+03	2.0810e+05	14.3571	42.9544	5.3215e+03	
18	04_Jla16 - Cd...	2.8894	<0.032979	2.3282	190.4354	2.6982e+05	3.9960e+03	208188	14.2907	45.5657	5.0960e+03	
		1678	0	2.3626	196.2501	2.7142e+05	4.0377e+03	2.0756e+05	15.3670	47.1049	5.2676e+03	
		3167	0	2.2774	197.7860	2.7033e+05	4.0414e+03	2.0800e+05	14.6360	46.3893	4.9549e+03	
		4314	<0.041742	2.2986	199.9139	2.7150e+05	3.9416e+03	2.0757e+05	15.1805	45.9882	5.0436e+03	
		7341	<0.13349	2.7991	51.0230	2.4372e+05	5.9999e+03	2.3203e+05	28.9470	31.1524	385.3092	
		4631	0	2.7116	44.3276	2.4310e+05	6.3224e+03	2.3239e+05	24.3880	23.9084	355.1757	
		8055	0.0938	2.6195	39.2505	2.4392e+05	6.2972e+03	2.3156e+05	24.9834	26.2907	355.0037	
		8840	0.1613	2.3162	41.3399	2.4274e+05	6.2766e+03	2.3235e+05	26.9165	25.6313	374.9084	
		5630	0.0604	2.5703	37.3175	2.4634e+05	6.4001e+03	2.2988e+05	23.8609	26.3658	366.1393	

It is also possible to show the unfiltered mass fractions where there were no detection limits applied and some other results. If you choose “Majors as Oxides” the majors, as defined in the Oxides.xlsx file (see chapter 7.2.2.2), will be converted to wt% oxides. If you used a certain FeO/(FeO+Fe2O3) ratios, you can also visualize the Fe2+ and Fe3+ contributions (µg/g) to the total Fe57.

Whenever you press the “Create Output Table” button, it will save the table that is visualized.

8.1.1. Inclusion data

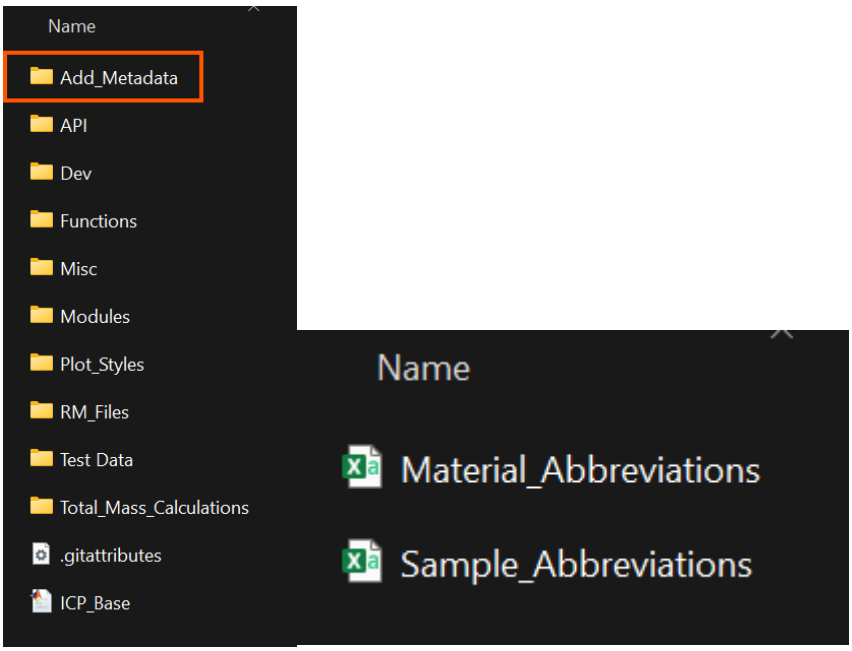


To visualize your quantified inclusion data, make sure to switch to “Inclusion” in the dropdown menu. This dropdown menu is hidden in “Matrix Mode” hence it is not visible in the example above.

8.2. Add Metadata

This is still a little bit work in progress but it generally works fine. This is needed if you want to visualize the “Session drift” in the “Quality Control” Tab.

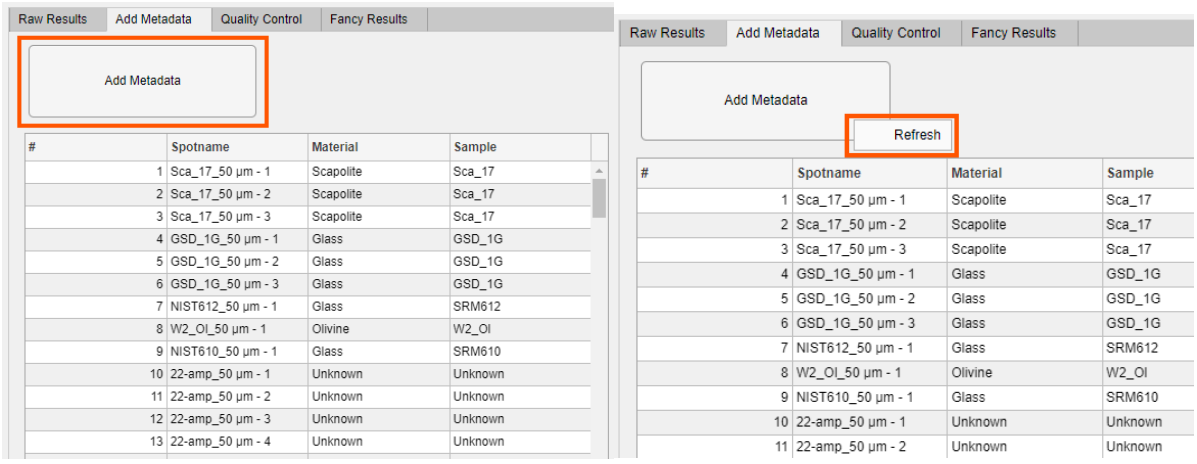
First, get the subfolder “Add_Metadata” in the ICP-Base folder and open Sample_Abbreviations (or Material_Abbreviations).



In those files, the first row holds the sample name as is should be stored in ICP-Base. All other rows hold possible strings that are part of the spot name. So if the spot contains e.g. NISTSRM610 ICP-Base will store it as “SRM610”.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	SRM610	SRM612	SRM614	SRM616	GSD_1G	GSD_2G	ATHO_G	BCR_2G	GSE_1G	AGN_BE11	Sca_17	Sc_1	Grt_1
2	NIST610	NIST612	NIST614	NIST616	GSD-1G	GSD-2G	ATHO	BCR-2G	GSE	BE11	Sca-17	Sc-1	Grt-1
3	SRM610	SRM612	SRM614	SRM616	gsd-1g	gsd-2g	atho	bcr-2g	gse	AGN-BE11	Sc-17	sc-1	Grt_1
4	NISTSRM6	NISTSRM6	NISTSRM6	NISTSRM6	GSD_1G	GSD_2G		BCR-2g			Scap-17	Sc_1	grt-1
5	SRM 610	SRM 612	SRM 614	SRM 616	gsd_1g	gsd_2g		BCR_2G			scap-17	sc_1	grt_1
6	NIST 610	NIST 612	NIST 614	NIST 616	gsd_1G	gsd_2G		BCR_2g			sca-17		
7	SRM_610	SRM_612	SRM_614	SRM_616	GSD_1g	GSD_2g		BCR_2G			Sca_17		
8	NIST_610	NIST_612	NIST_614	NIST_616	GSD						sca_17		
9	NIST_SRM	NIST_SRM	NIST_SRM	NIST_SRM_616							scap_17		

The same goes for the material. Once you did that save the file and close it. Next, press the “Add Metadata” button and if successful, you should see the result. “Unknown” is the default.



#	Spotname	Material	Sample
1	Sca_17_50 µm - 1	Scapolite	Sca_17
2	Sca_17_50 µm - 2	Scapolite	Sca_17
3	Sca_17_50 µm - 3	Scapolite	Sca_17
4	GSD_1G_50 µm - 1	Glass	GSD_1G
5	GSD_1G_50 µm - 2	Glass	GSD_1G
6	GSD_1G_50 µm - 3	Glass	GSD_1G
7	NIST612_50 µm - 1	Glass	SRM612
8	W2_Ol_50 µm - 1	Olivine	W2_Ol
9	NIST610_50 µm - 1	Glass	SRM610
10	22-amp_50 µm - 1	Unknown	Unknown
11	22-amp_50 µm - 2	Unknown	Unknown
12	22-amp_50 µm - 3	Unknown	Unknown
13	22-amp_50 µm - 4	Unknown	Unknown

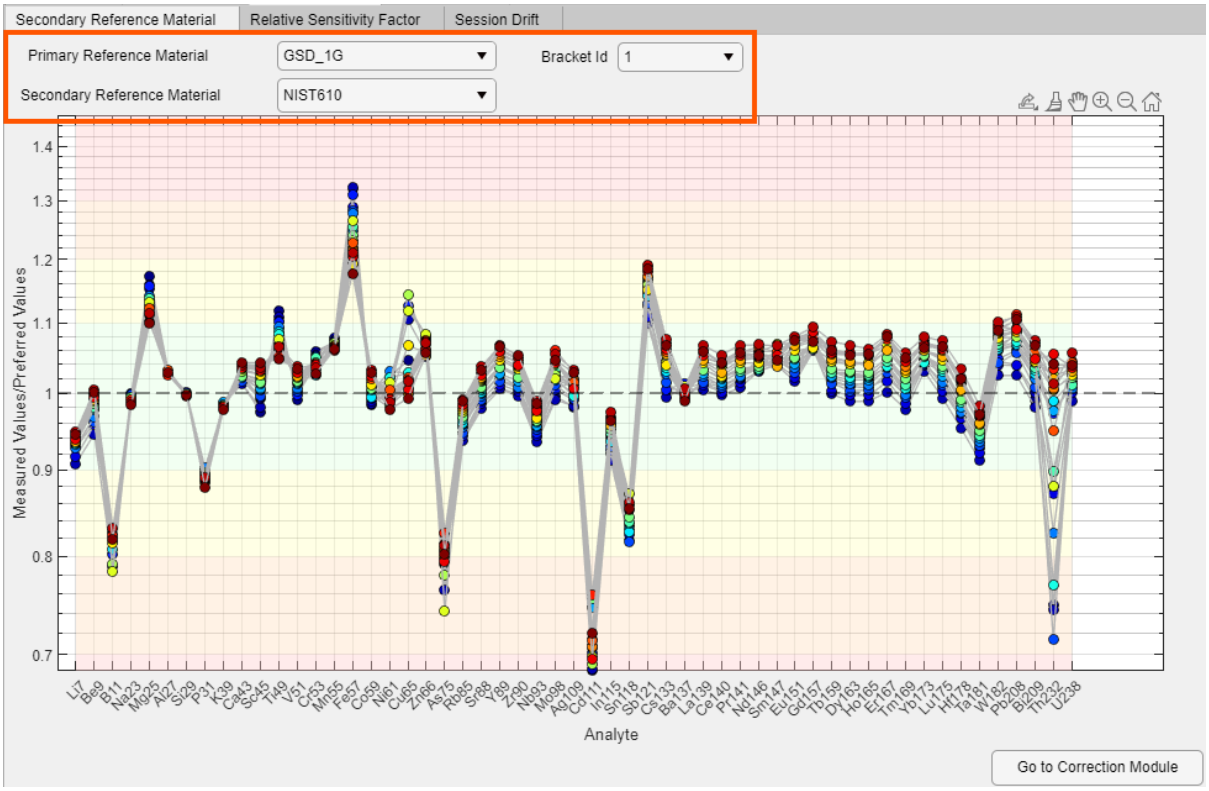
#	Spotname	Material	Sample
1	Sca_17_50 µm - 1	Scapolite	Sca_17
2	Sca_17_50 µm - 2	Scapolite	Sca_17
3	Sca_17_50 µm - 3	Scapolite	Sca_17
4	GSD_1G_50 µm - 1	Glass	GSD_1G
5	GSD_1G_50 µm - 2	Glass	GSD_1G
6	GSD_1G_50 µm - 3	Glass	GSD_1G
7	NIST612_50 µm - 1	Glass	SRM612
8	W2_Ol_50 µm - 1	Olivine	W2_Ol
9	NIST610_50 µm - 1	Glass	SRM610
10	22-amp_50 µm - 1	Unknown	Unknown
11	22-amp_50 µm - 2	Unknown	Unknown

If you want to change something, you can change the .xlsx files and the right-click the “Add Metadata” button and click “Refresh”.

8.3. Quality Control

8.3.1. Secondary Reference Material

In this Tab, you can compare how well the secondary standards were reproduced. The plot ratios the measured value with the “Preferred Value” which is the value from the reference material file. If the ratio is 1, it is perfectly reproduced. The green area is from 0.9 to 1.1 (so ± 10 %), the yellow area is from ± 10 to ± 20 %, the orange area is from ± 20 to ± 30 % and everything red is more than ± 30 % deviation from the perfectly reproduced value.



This example shows the NIST610 SRM that was quantified using the GSD-1G as PRM with bracket id 1. Most analytes were reproduced within 10 % (which is mostly good enough for LA-ICP-MS).

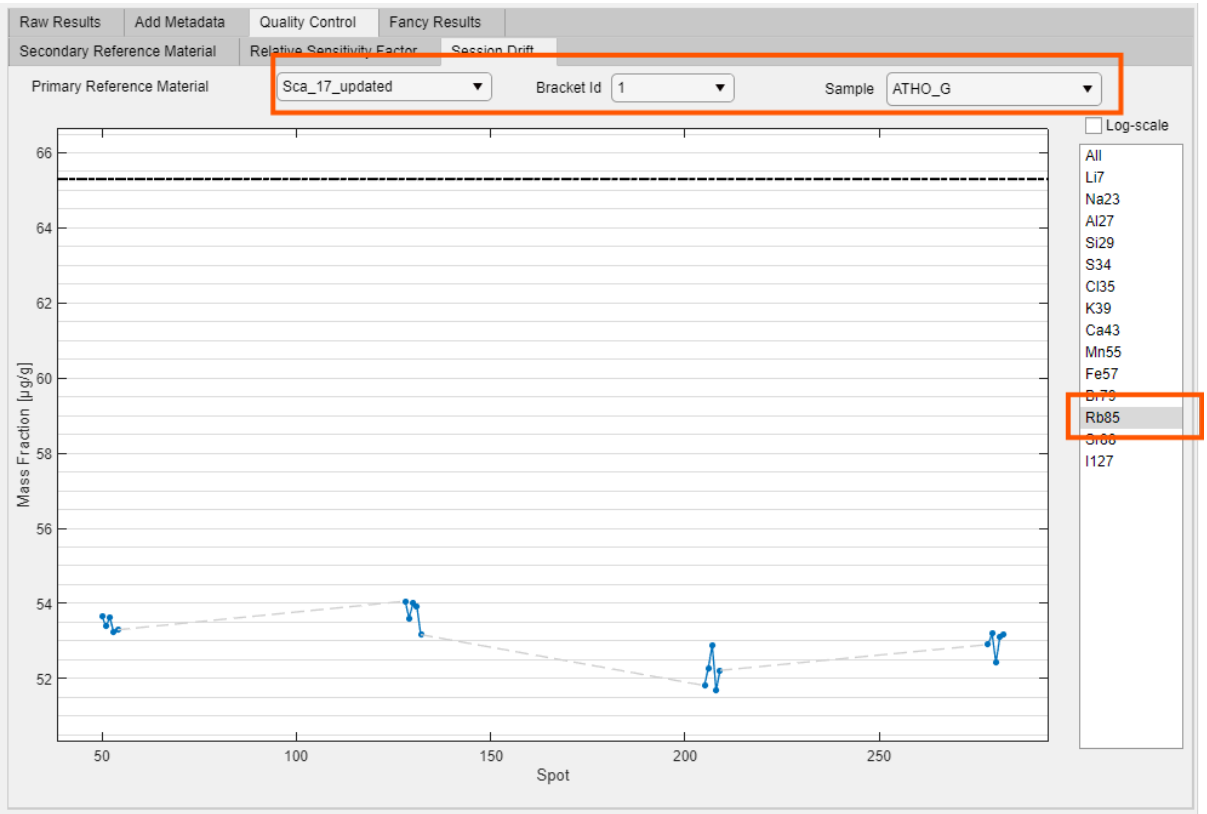
The colors of the points represent the time of the analysis (acquisition time; integer time points). If the spot was measured at the start of the session it will be dark blue and the spots measured at the end of the session are red (colormap "Jet", google it). In the example we can see that e.g. the REEs were reproduced better at the start of the session vs the end of the session. Fe57 is way off but behaves opposite.

8.3.2. Relative Sensitivity Factor

Same as in the Quantification Module (see chapter 7.3).

8.3.3. Session Drift

This plot only works after adding the sample name in the "Add Metadata" Tab (see chapter 8.2. Also, right now it only works for the matrix/host and not for inclusions.



In this plot, you can visualize the quantified analytes over the measurement session. This way you can see if there is a certain drift in the results. In the example above, we quantified the ATHO-G SRM with the Sca-17 PRM. We are seeing no weird drift as the blue dots scatter randomly. The grey dashed line depict the spots that were measured in between the ATHO-G spots. The thick dashed line on top shows the "Preferred Value" that is the value from the reference material file. So we can see that we quantified Rb85

consistently over the whole session with 52 to 54 µg/g which is about 12 µg/g under the preferred value.

8.4. Fancy Results

Work in progress, not finished yet.

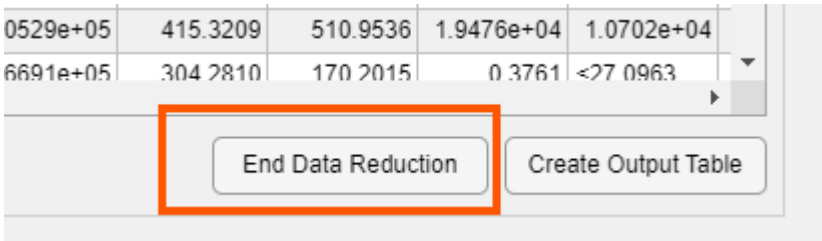
8.5. Quick guide

- Raw results
    - Show the results of a selected PRM – bracket id pair
        - Mass fractions
        - Mass fractions unfiltered (without detection limits)
        - Sensitivities
        - CPS
        - And more
    - Change to "Inclusion" to see results for inclusion
    - Create an output table
        - It will always output the table that you are currently looking at
- Add Metadata
    - Add sample names and materials
        - Add your sample name to the Sample_Abbreviations.xlsx in the Add_Metadata subfolder of the ICP-Base folder
        - Same goes for Materials
        - First row of the .xlsx holds the sample name that ICP-Base will take, the other rows hold the strings that are searched for in the spot name
        - Only necessary to do this for "Session Drift" in the "Quality Control" Tab
- Quality Control
    - Secondary Reference Material
        - Ratios the results of a secondary reference material by its "Real Value" which comes from the reference material file (.csv in RM_Files folder)
    - Relative Sensitivity Factor
        - Same as in the Quantification Module, read it up there
    - Session Drift
        - Shows the results over the time of the session
        - For secondary reference materials there is a black dashed line to show the "real value"
        - Grey dashed lines show that there were some other spots measured in between
    - Fancy Results

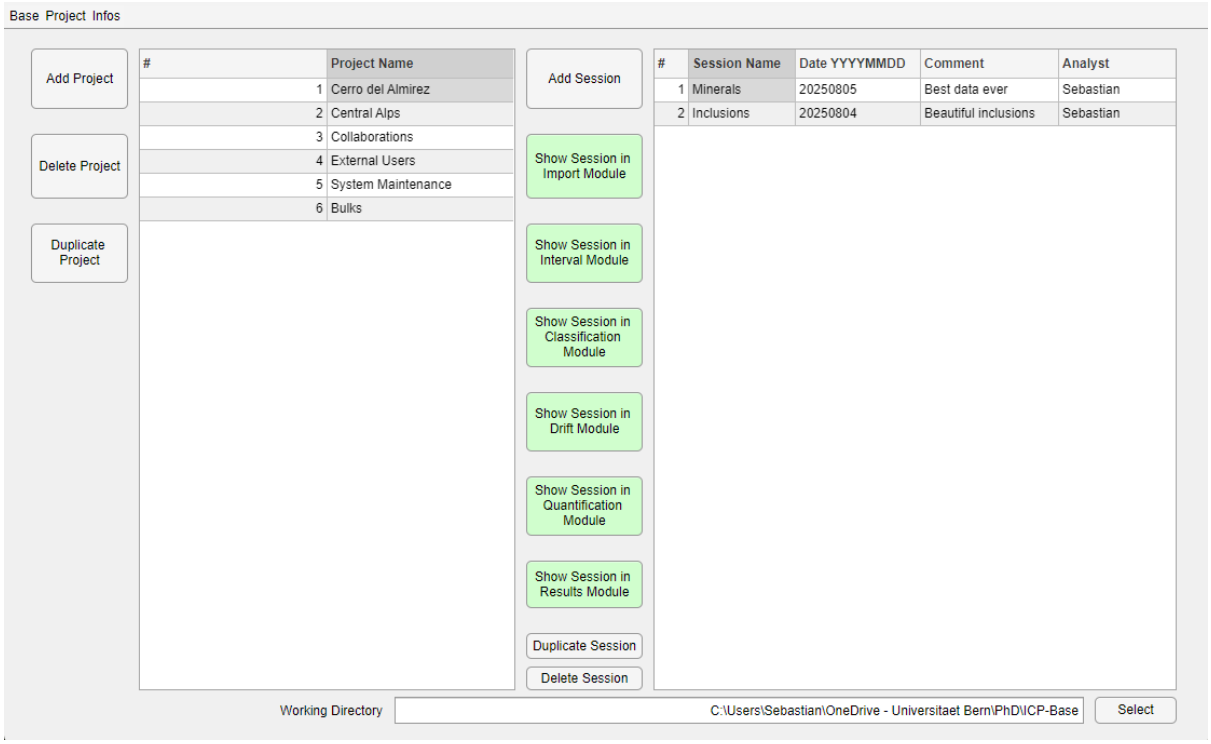
- Not implemented yet

8.6. End Data Reduction – Saving the Base

This button will end the data reduction. Congrats you did it!

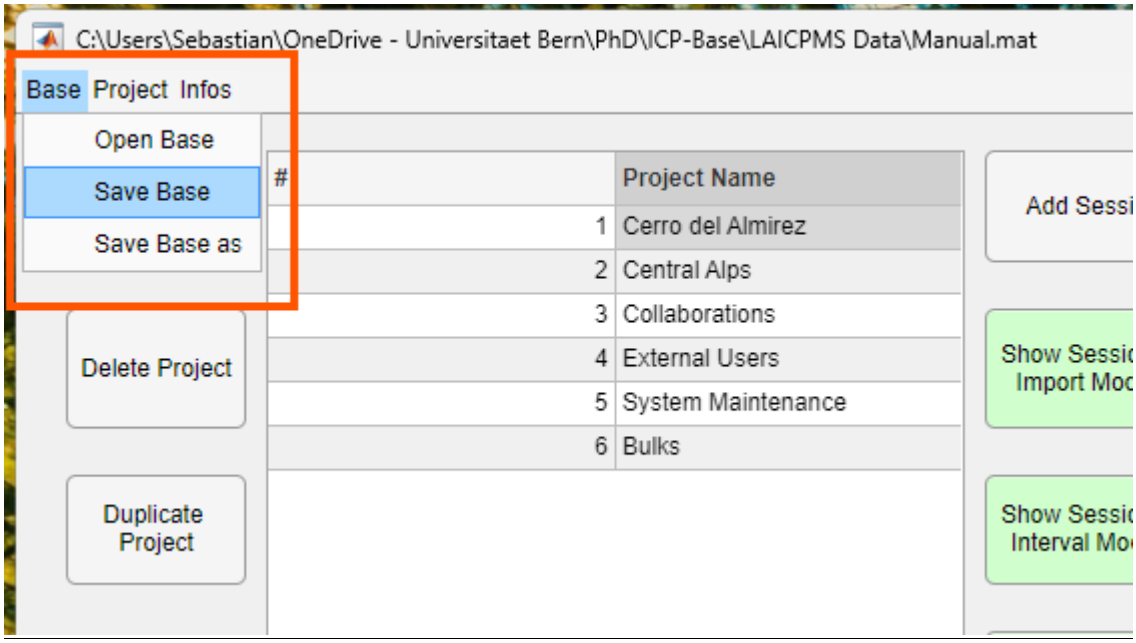


The Results Module will be closed and all modules will be shown in green in the Base.



You can now select a session and jump to any module. You can also duplicate or delete the session. If you want to test a few things or do method development, I recommend duplicating the session and then playing around with it.

<u>Now make sure to actually also save the Base itself to your computer!</u>



I recommend to not store the Base in the ICP-Base folder but somewhere else.

Next time you want to do something, open ICP-Base and click “Open Base” and load it back in. Then you can proceed with your new measurement sessions and everything will be stored at the same place.

Over time, as you continue to doing your data reduction with ICP-Base, you will build an extensive database with all your raw data, metadata and results. I plan to make a “Long-term monitoring” Module soon. Nobody will have more idea about what is going on with their machine than you! You are the best lab manager ever, give yourself a clap on the shoulder, you are doing well.